

SESSION CAPSULE

Team of Teams Agent Skill

Comprehensive Design Vision for Organizational AI

A Company OS that gives any founder, leader, or team the analytical power of a full executive committee -- with the structural integrity of mandated dissent.

Ideation Session -- February 22, 2026

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Session Summary

Content Inventory

Category	Count	Files
Vision Document	1	VISION_team-of-teams.md
Idea Briefs	4	BRIEF_cascading-deliberation-engine.md, BRIEF_cognitive-forcing-prompt-architecture.md, BRIEF_decision-modes-decision-space.md, BRIEF_engagement-model.md
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Infographics	4	INFOGRAPHIC_cascading-deliberation-engine.png, INFOGRAPHIC_cognitive-forcing-prompt-architecture.png, INFOGRAPHIC_company-os-unified-vision.png, INFOGRAPHIC_engagement-model.png
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Source Materials	2	manifest.md, request.md
Session Summary	1	SESSION_SUMMARY.md
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Core thesis, governing principles, pillars, design decisions, and boundaries

Vision Document: The Team of Teams Agent Skill

Session: Team of Teams Agent Skill Ideation

Date: 2026-02-22

Purpose: Comprehensive design vision for a builder agent to construct a Claude Code Agent Skill that emulates SMB organizational decision-making.

The Vision

A Company OS. Not a chatbot that roleplays executives, but a complete organizational reasoning engine that gives any founder, leader, or team the analytical power of a full executive committee -- with the structural integrity of mandated dissent ensuring that risk, cost, opportunity, and execution reality are all represented, even when the human user might prefer to hear only good news.

The Team of Teams Agent Skill transforms Claude Code into a boardroom in a box. Present any business issue -- from a quick operational question to a major strategic decision -- and the skill decomposes it through the top two layers of a SMB org structure: a CEO who frames and routes, C-suite executives who decompose and synthesize through their domains, and team leads who produce narrow, focused analysis using domain-specific analytical methods. The output is not a recommendation from a single voice. It is a structured Decision Record that shows how different domain perspectives analyze the same issue, where they agree, where they disagree, and why -- then synthesizes a decision that addresses the strongest objections.

The skill operates at three levels of depth (a quick hallway question, a focused working session, or a full board meeting) and four styles of synthesis (risk-averse, growth-oriented, consensus-building, or data-driven). Users match engagement to decision weight. The CEO who needs a quick gut check from the CFO perspective gets it in seconds. The founder facing a strategic pivot gets a comprehensive multi-perspective analysis with pre-mortem failure mode assessment.

What makes this more than "ask a panel of experts" is that disagreement is engineered, not accidental. The CISO is the Constitutional Skeptic -- their default is that change introduces risk. The VP of Sales is the Revenue Optimist -- they find the growth opportunity. These structural tensions are not bugs to be resolved but features that produce richer analysis than consensus ever could. The Fault

Line Analysis section of every Decision Record captures the meta-analysis of where expert perspectives collide -- the single most valuable analytical artifact the system produces.

Architecture: Three Pillars

The skill's design rests on four pillars, each the subject of a dedicated idea report.

Pillar 1: Structure -- The Cascading Deliberation Engine

Five-Phase Process:

1. **Phase 0 -- Shared Consciousness Broadcast.** Before any analysis begins, the CEO broadcasts the issue context and framing to all activated C-suite agents simultaneously. This implements McChrystal's shared consciousness principle: everyone sees the same picture before reasoning independently. Without this, agents optimize for their domain without understanding the full context.
2. **Phase 1 -- CEO Frames and Routes.** The CEO decomposes the issue into evaluation dimensions, classifies the decision type (Strategic, Operational, Financial, Technical, Personnel, Compliance/Risk), and routes to relevant C-suite members using a default routing table with override capability. Routing is an analytical act: the CEO's choice of which domains to activate is itself a judgment about what matters. The framing includes explicit exclusion reasoning -- why certain teams were NOT activated, visible to the user.
3. **Phase 2 -- C-Suite Dispatches Downward.** Each activated C-suite executive translates the CEO's framing into domain-specific sub-questions, one per team lead. This translation is itself analytical -- the CFO does not forward the CEO's question to the Controller; the CFO asks the Controller "what are the GAAP implications of this change?" This decomposition separates the skill from "a panel of chatbots."
4. **Phase 3 -- Team Leads Produce Findings.** Each team lead subagent performs narrow, focused analysis through their specialist lens using their unique analytical framework and mandatory output template. The Controller produces a GAAP Compliance Assessment. The FP&A analyst produces a Three-Scenario Financial Model. Different methods produce structurally different outputs that cannot collapse into one voice.
5. **Phase 4 -- C-Suite Synthesizes Upward.** Each C-suite executive collects their team lead findings and synthesizes a domain recommendation with a confidence level, key risks, and key opportunities. Internal contradictions between team lead findings are flagged as analytical signals, not averaged away.
6. **Phase 4.5 -- Pre-Mortem Challenge (Tier 3 only).** After producing their own domain recommendation, each C-suite agent receives summaries of ALL other activated C-suite members' recommendations and answers one structured question: "Assume this decision fails catastrophically in 12 months. Based on what you see across all the domain recommendations, what caused the failure?" One round only. No back-and-forth debate. Research-backed as the highest single-technique value-add for decision quality. Findings feed into the Fault Line Analysis and Dissenting Views sections of the Decision Record.

7. **Phase 5 -- CEO Deliberation.** The CEO maps all domain recommendations onto a single matrix, identifies fault lines (where and why recommendations diverge), determines the most determinative perspective for this decision type, applies the active Decision Mode to resolve tensions, and produces the final Decision Record.

Hybrid Agent Architecture:

The skill uses a hybrid model informed by Claude Code Agent Teams constraints:

- **Tier 1 (Agent Team members):** CEO + 7 C-suite officers (COO, CFO, CTO, CISO, CAO, VP of Sales, VP of Delivery). Real teammates spawned by the skill orchestrator. Can message each other and the CEO directly.
- **Tier 2 (Custom subagents):** 29 team lead specialists defined in `.claude/agents/team-leads/`. Each C-suite agent invokes their team leads as subagents. Subagents get their own context window, use cost-efficient models (Haiku), have restricted tool access, and report structured results back to their parent.

This architecture provides real isolation per perspective (each team lead gets its own context window, not a simulated pass within a longer prompt), cost optimization (Haiku for narrow analysis, Sonnet/Opus for synthesis), and true management simulation (the C-suite agent dispatches, collects, and synthesizes like a real executive).

Engineered Dissent Model:

Role	Disposition	Mandate
CEO	Synthesizer	Frame, listen, weigh, and decide. Value is judgment, not expertise.
COO	Skeptic	"Can we actually do this with the people and processes we have?"
CFO	Skeptic	"Find the costs that aren't in the proposal."
CTO	Advocate	"What does this make possible that wasn't possible before?"
CISO	Skeptic	"Your default is that change introduces risk. You are the org's immune system."
VP of Sales	Advocate	"How does this help us sell more, faster, or to new markets?"
VP of Delivery	Skeptic	"What do we sacrifice from existing commitments to do this?"
CAO	Systemic	"Can the organization -- people, policies, culture -- absorb this?"

Balance: 4 skeptics, 2 advocates, 1 systemic, 1 synthesizer. The skeptic-heavy balance counterbalances human optimism bias.

Full Organizational Roster (29 Team Leads):

C-Suite	Team Leads
COO	Operations Manager, Process/Quality Lead, Vendor/Procurement Manager, Facilities/Office Manager (conditional)
CFO	Controller, Head of FP&A, Treasury/Cash Manager, AP/AR Manager, Tax Lead
CTO	Engineering Lead, Infrastructure/DevOps Lead, Data/Analytics Lead, Product/UX Lead
CISO	Security Operations Lead, Compliance/GRC Lead, Identity & Access Lead, Security Architecture Lead
VP Sales	Sales Operations Lead, Account Management Lead, Business Development Lead, Sales Enablement Lead
VP Delivery	Project/Program Manager, Resource Manager, Client Success Lead, QA/Delivery Standards Lead
CAO	HR/People Ops Lead, Legal/Contracts Lead, Admin/Policy Lead, Corporate Communications Lead

Decision-Type Routing:

Decision Type	Default Activation	Full Activation Triggers
Strategic	CEO, CFO, CTO, VP Sales	Acquisition, pivot, new market entry
Operational	CEO, COO, VP Delivery	Major process change, org restructure
Financial	CEO, CFO, COO	Funding round, major investment, cost reduction
Technical	CEO, CTO, CISO	Platform migration, architecture change
Personnel	CEO, CAO, COO, VP Delivery	Layoff, major hiring, reorg
Compliance/Risk	CEO, CISO, CAO, CFO	Regulatory change, breach response, audit

The CEO can always override defaults. Certain triggers activate all C-suite members regardless of decision type.

Decision Record Output Format:

EXECUTIVE SUMMARY

[3–5 sentences: the decision, key reasoning, primary dissent]

DECISION RECORD: [Issue Title]

Decision ID: [auto-generated]

Date: [timestamp]

Submitted by: [user]

Decision Type: [classification]

Tier: [1/2/3]

Decision Mode: [Guardian/Pioneer/Architect/Analyst]

1. ISSUE STATEMENT

[The question as originally posed]

2. CEO FRAMING

[Decomposition into evaluation dimensions]

Activated Teams: [roles engaged + rationale]

Excluded Teams: [roles not engaged + rationale for exclusion]

3. DOMAIN ANALYSES

3.x [C-Suite Role] – [Mandate Title]

Domain Recommendation: [Approve / Approve with Conditions / Oppose / Neutral]

Confidence Level: [High / Medium / Low]

Summary: [2–3 sentence synthesis]

Team Lead Findings: [per team lead, 1–2 sentences each]

Key Risks Identified: [list]

Key Opportunities Identified: [list]

4. FAULT LINE ANALYSIS

Points of Agreement: [what most domains agree on]

Points of Contention: [where and why recommendations diverge]

Pre-Mortem Findings: [failure modes identified in Phase 4.5]

Unresolved Tensions: [surfaced but unresolvable with current info]

5. CEO DECISION

Decision: [clear statement]

Most Determinative Perspective: [which domain was weighted highest and why]

Decision Weight Rationale: [why certain perspectives carried more weight]

Conditions & Guardrails: [drawn from skeptic role recommendations]

Accepted Risks: [consciously accepted, with reasoning]

Mitigations Directed: [specific team actions ordered]

6. DISSENTING VIEWS

[Strongest objections from overruled perspectives, preserved for record]

7. NEXT STEPS

[Specific actions, implied owners, timelines]

8. METADATA

Total roles consulted: [N]

Decision complexity: [Low / Medium / High / Critical]

Primary domain: [most determinative C-suite area]

Dissent level: [Consensus / Mild Dissent / Strong Dissent / Split Decision]
Key Assumptions: [assumptions the analysis rests on]

Pillar 2: Experience -- The Engagement Model

Three Interaction Tiers:

Tier	Name	Invocation	What Happens	Output
1	Hallway Question	<code>/consult</code> <code>[role]:</code> <code>[question]</code>	Direct consult with one C-suite agent. No CEO, no routing, no team leads. Quick, opinionated, domain-specific.	Advisory Note (3-5 sentences)
2	Working Session	<code>/panel</code> <code>[roles]:</code> <code>[issue]</code>	CEO routes to 2-4 C-suite members. Each does domain analysis with team lead perspectives. CEO produces lightweight synthesis.	Panel Assessment (~1 page)
3	Board Meeting	<code>/deliberate:</code> <code>[issue]</code>	Full five-phase cascade. All relevant C-suite activated. Full team lead analysis. Full CEO deliberation with optional Phase 4.5 pre-mortem.	Complete Decision Record (3-5 pages)

Auto-Triage: `/evaluate: [issue]` presents the issue to the CEO for triage. The CEO assesses scope (single-domain vs. multi-domain vs. cross-cutting), impact (low/medium/high/critical), and reversibility (easily reversed vs. difficult vs. irreversible), then recommends a tier with a one-sentence justification. The user can accept, escalate, or de-escalate.

SMB-First Default: Bias Toward Tier 1. The skill's default behavior should bias toward lightweight engagement. Most SMB decisions are fast, informal, and made by one or two people -- the skill should match that tempo. When the CEO auto-triages, the default recommendation should lean toward Tier 1 unless clear multi-domain signals are present. The goal is to make Tier 1 the daily habit and Tier 3 the deliberate escalation, not the other way around. A skill that defaults to the full board meeting for every question will not see daily use.

Pillar 4: Strategic Depth -- Decision Modes and the Decision Space Map

The skill should not produce a single recommendation -- it should reveal a **decision space**. The same domain analysis can be synthesized through multiple Decision Modes, each a configurable lens that changes how the CEO agent weighs and resolves competing perspectives without changing the underlying analysis. By running the same inputs through different synthesis styles, the user sees not "what to do" but "what the terrain looks like in every direction." This transforms the skill from a decision engine into a **decision exploration tool**.

Five CEO Decision Modes:

Each mode maps to established decision theory (Rowe & Boulgarides Decision Style Theory + classical operations research):

Mode	Disposition	Decision Theory	How It Resolves Fault Lines
Guardian	Risk-averse	MaxiMin	Weights skeptic roles (CISO, CFO, COO, VP Delivery) more heavily. Skeptics must be satisfied, not just acknowledged. Decisions tend toward: don't do it, do a smaller version, or do it with extensive guardrails.
Pioneer	Growth-oriented	MaxiMax	Weights advocate roles (VP Sales, CTO) more heavily. Skeptic concerns are treated as engineering problems to solve, not reasons to stop. Decisions tend toward: do it, do it bigger, do it faster.
Architect	Consensus-building	Behavioral	Weights the fault lines themselves. Seeks the position that satisfies the most domain concerns. Conditions drawn from multiple domains, not just the most determinative one.
Analyst	Data-driven (default)	Hurwicz balanced	Weights confidence levels regardless of role disposition. High-confidence findings carry more weight. Low-confidence recommendations flagged as needing more research. "Defer pending better data" is a legitimate outcome.
Sentinel	Regret-minimizing	MiniMax Regret	Disproportionately weights the strongest objection from ANY role. Asks: "If this goes wrong, which C-suite member's warning will I wish I'd heeded?" Favors paths where being wrong is survivable. Particularly suited to irreversible decisions.

Decision Modes are implemented as CEO synthesis prompt modifiers. The underlying domain analysis is identical across modes. The same team lead outputs, the same C-suite recommendations, the same fault lines -- but different weighting produces different decisions.

CEO Prompt Modifiers (per mode):

Each mode has a specific prompt modifier injected into the CEO's synthesis layer:

- **Guardian:** "You are cautious by disposition. You'd rather miss an opportunity than take a risk that could damage the business. When skeptic and advocate perspectives conflict, you lean toward the skeptics unless the advocates present overwhelming evidence of low-risk upside."
- **Pioneer:** "You are growth-oriented by disposition. You believe the biggest risk is standing still while competitors move. Frame skeptic concerns as implementation challenges to solve, not objections to honor."
- **Architect:** "You are a consensus-builder by disposition. You believe that decisions succeed or fail based on organizational alignment. Look for the position that satisfies the most domain concerns, even if it means a less aggressive or less cautious path."
- **Analyst:** "You are analytically driven. You distrust both optimism and pessimism -- you trust evidence. Weight domain recommendations by their confidence levels, not their enthusiasm or caution. A decision to defer is not indecision -- it's a rational response to insufficient information."

- **Sentinel:** "You are a regret minimizer. For every option, ask: 'If this decision turns out to be wrong, can we recover?' Disproportionately weight the single strongest objection from any domain. Choose the path where being wrong is survivable, even if being right is less spectacular."

Mode/Tier Interaction Matrix:

Each mode produces distinct behavioral patterns at each tier:

	Tier 1 (Hallway Question)	Tier 2 (Working Session)	Tier 3 (Board Meeting)
Guardian	Highlights downside risks, suggests what could go wrong	Synthesis biased toward risk mitigation. Extensive guardrails.	CEO weights skeptics heavily. High bar for approval.
Pioneer	Frames as investment question, suggests acceleration	Synthesis biased toward opportunity capture. "How to" not "whether to."	CEO weights advocates heavily. Low bar unless existential risk.
Architect	Includes "however, [other role] might see this differently"	Seeks option addressing most concerns across all activated roles.	CEO seeks widest organizational support. Conditions from all domains.
Analyst	Flags confidence level explicitly. Low-confidence = research recommendation.	Synthesis driven by which domains have highest-confidence findings.	CEO weights by evidence quality. Low-confidence = "investigate further."
Sentinel	Identifies the single biggest risk and whether it's survivable.	Identifies strongest objection across all activated roles. Tests whether downside is recoverable.	CEO disproportionately weights the strongest single objection. Favors survivable paths.

Default cell for most users, most of the time: **Tier 1 + Analyst** -- quick, evidence-weighted, transparent about uncertainty.

Multi-Mode Comparison and the Comparative Decision Record:

The key efficiency insight: domain analysis is mode-independent. Multi-mode comparison runs domain analysis once (the expensive part) and CEO synthesis multiple times (cheap, single-agent passes). Cost: approximately 1.1x a single deliberation for 5x the strategic insight.

Invocation:

- Single mode: `/deliberate guardian: should we acquire CompetitorX?`
- Two-mode comparison: `/deliberate guardian vs pioneer: should we acquire CompetitorX?`
- All modes: `/deliberate all-modes: should we acquire CompetitorX?`
- Tier 1 with mode: `/consult cfo guardian: can we afford this?`
- Tier 2 with mode: `/panel pioneer finance tech: should we build this feature?`

Comparative Decision Record format:

```

COMPARATIVE DECISION RECORD: [Issue Title]
Decision ID: [auto-generated]
Modes Compared: [list]

EXECUTIVE SUMMARY
[One paragraph per mode showing how the decision differs]

SHARED ANALYSIS
[Domain analyses -- identical across modes, presented once]
[Fault Line Analysis -- presented once]

MODE COMPARISONS

  GUARDIAN SYNTHESIS:
    Decision: [statement]
    Most Determinative Perspective: [role + why]
    Key Factor: [what tipped this mode's decision]
    Conditions: [guardrails]

  PIONEER SYNTHESIS:
    Decision: [statement]
    Most Determinative Perspective: [role + why]
    Key Factor: [what tipped this mode's decision]
    Conditions: [guardrails]

[repeat for each requested mode]

DIVERGENCE ANALYSIS
Where Modes Agree: [decisions all modes reached]
Where Modes Diverge: [the pivot points]
The Key Choice: [what the user is actually deciding between -- not the
  business question, but the values/priorities question underneath it]

METADATA
[standard metadata]
Mode Sensitivity: [how much does the decision change across modes?
  High sensitivity = the decision depends heavily on risk appetite.
  Low sensitivity = all modes converge on the same answer --
  the evidence speaks for itself.]

```

Mode Sensitivity is a novel signal: if all modes produce the same decision, the right answer does not depend on the user's posture. If modes diverge dramatically, the user knows their personal risk appetite is the deciding factor, not the analysis. This transforms a binary recommendation into a nuanced view of the decision landscape.

Pillar 3: Analytical Quality -- Cognitive Forcing and the Prompt Architecture

The Voice Collapse Problem:

Telling one LLM to "think as 5 different people" typically produces 5 paragraphs of the same underlying reasoning with different vocabulary. Simple role assignments ("You are a CFO") show zero measur-

able improvement in LLM output quality. The solution is not better persona descriptions -- it is forcing each perspective to use different analytical methods that produce structurally different outputs.

Three-Layer Prompt Architecture:

Layer 1 -- Team Lead Subagent Definitions (29 Analytical Framework Packages):

Each team lead is a custom subagent definition in `.claude/agents/team-leads/[domain]/[role].md` containing:

1. **Analytical Framework:** The specific methodology this role uses (e.g., GAAP Compliance Assessment, Three-Scenario Financial Modeling, Liquidity Stress Test).
2. **Mandatory Output Template:** Structured format forcing specific analytical artifacts, different per role.
3. **Three Forcing Questions:**
4. **Pre-Mortem** (Klein 2007): "Assume this decision fails in 12 months. From your domain perspective, what caused the failure?"
5. **Adversarial Empathy** (McChrystal JSOC): "If you were [domain-relevant external adversary], how would you exploit this?" (Applied where relevant -- auditors for Controller, threat actors for CISO leads, competitors for Sales leads)
6. **Domain Devil's Advocate** (Nemeth 2001): "What would [domain expert critic] find concerning?" Authentic domain-grounded criticism rather than generic contrarianism.
7. **Accountability Framing:** "Your analysis will be reviewed by [C-suite parent] alongside analyses from other specialists. Provide specific evidence for every claim."
8. **Blind Spot Declaration:** Explicit statement of what this role does NOT consider.
9. **Subagent Configuration:** Model (Haiku), tools (Read, Grep, Glob, WebSearch only), max turns (3-5), permission mode (plan -- analysis only, no execution).

Layer 2 -- C-Suite Agent Prompts (7 Domain Orchestration Templates):

Each C-suite agent prompt supports three cognitive modes:

- **Mode A (Tier 1 Direct Consult):** Quick, opinionated response drawing on internalized team lead perspectives. No subagent delegation.
- **Mode B (Tier 2/3 Full Analysis):** Explicit delegation to team lead subagents. Receive CEO framing -> translate into domain sub-questions -> dispatch subagents -> collect structured outputs -> synthesize domain recommendation.
- **Mode C (Phase 4.5 Challenge):** Review all peer C-suite recommendations and produce pre-mortem failure mode analysis.

C-suite prompt structure includes: role identity and mandate, team composition, analytical domain with natural tension partners, operating mode instructions, synthesis instructions, and domain-level forcing questions.

Layer 3 -- CEO Synthesis (Modular Decision Modes):

Fixed analytical component (fault line mapping, most determinative perspective identification, cross-domain pre-mortem) plus swappable Decision Mode module that adjusts synthesis weighting.

Reference Implementation -- CFO Domain (5 Complete Packages):

The CFO domain serves as the reference implementation with complete analytical framework packages for all five team leads:

Team Lead	Framework	Output Artifact
Controller	GAAP Compliance and Financial Controls Assessment	Compliance Impact Assessment with risk rating
Head of FP&A	Three-Scenario Financial Modeling	Scenario Analysis with critical variable and decision sensitivity
Treasury/Cash Manager	Liquidity Stress Test	Cash Flow Impact Timeline with funding gap analysis
AP/AR Manager	Working Capital Cycle Analysis	Payables/Receivables Impact with vendor relationship risk
Tax Lead	Tax Structure Optimization Assessment	Tax Implications Memo with external counsel recommendation

Each package includes complete forcing questions with domain-specific adversarial figures (external auditor, short-seller, bank, IRS, key vendor). This reference implementation should be used as the template for all 29 packages.

Bias and Role Susceptibility Mapping:

Research on LLM role-playing identifies specific failure modes where agents drift from their mandated perspective. Each role's prompt should include awareness of its susceptibility patterns:

Disposition	Susceptibility	Mitigation
Skeptic roles (CISO, CFO, COO, VP Delivery)	Risk of softening objections to match perceived user preference. LLMs have a sycophancy bias that undermines skeptic mandates.	Explicit instruction: "Your value is in surfacing concerns, not in being agreeable. A skeptic who hedges is worthless."
Advocate roles (CTO, VP Sales)	Risk of under-weighting genuine constraints. Advocacy can become cheerleading.	Require advocates to name the strongest objection to their own position and explain why they still advocate.
Synthesizer (CEO)	Risk of false balance -- treating all perspectives as equally weighted regardless of decision type. Also susceptible to anchoring on the first domain recommendation received.	Require explicit weight rationale. Randomize order of domain recommendation presentation.
Systemic (CAO)	Risk of vagueness -- "organizational culture" analysis can be unfalsifiable.	Require concrete indicators: specific policies affected, specific teams impacted, specific precedents set.

The builder agent should incorporate these susceptibility mitigations directly into each role's prompt. The forcing questions partially address this (pre-mortem and devil's advocate counteract sycophancy), but explicit susceptibility awareness adds a second layer of defense.

Implementation Specification

File Structure

```

.claude/
agents/
  ceo.md # CEO agent definition
  c-suite/
    coo.md # C-suite agent definitions
    cfo.md
    cto.md
    ciso.md
    cao.md
    vp-sales.md
    vp-delivery.md
  team-leads/
    coo/
      operations-manager.md # Team lead subagent definitions
      process-quality-lead.md
      vendor-procurement-manager.md
    cfo/
      controller.md
      fpa-analyst.md
      treasury-manager.md
      ap-ar-manager.md
      tax-lead.md
    cto/
      engineering-lead.md
      infrastructure-devops-lead.md
      data-analytics-lead.md
      product-ux-lead.md
    ciso/
      security-operations-lead.md
      compliance-grc-lead.md
      identity-access-lead.md
      security-architecture-lead.md
    vp-sales/
      sales-operations-lead.md
      account-management-lead.md
      business-development-lead.md
      sales-enablement-lead.md
    vp-delivery/
      project-program-manager.md
      resource-manager.md
      client-success-lead.md
      qa-delivery-standards-lead.md
    cao/
      hr-people-ops-lead.md
      legal-contracts-lead.md
      admin-policy-lead.md
      corporate-communications-lead.md
  skills/
    team-of-teams/
      skill.md # Skill front matter and instructions
    templates/
      decision-record.md # Decision Record template
      advisory-note.md # Tier 1 output template

```

```
panel-assessment.md          # Tier 2 output template
decision-space-map.md        # Multi-mode comparison template
comparative-decision-record.md # Multi-mode Comparative Decision Record
config/
  routing-table.md           # Decision-type routing defaults
  company-profile.md         # Company parameterization
  decision-modes.md          # CEO Decision Mode prompt modifiers
```

Subagent Definition Format

Each team lead subagent file follows this pattern:

```

---
name: [role-slug]
description: "[analytical domain] analyst for [C-suite parent] domain"
model: haiku
tools:
  - Read
  - Grep
  - Glob
  - WebSearch
maxTurns: 5
---

# [Role Title] -- [Analytical Framework Name]

## Your Identity
You are the [Role Title] reporting to the [C-suite Parent]. You own [domain scope].

## Your Analytical Framework
[Framework name and methodology description]

## Your Output Template
[Structured template with specific fields to fill]

## Your Forcing Questions
- Domain Devil's Advocate: "[domain-specific question]"
- Pre-Mortem: "[domain-specific failure scenario question]"
- Adversarial Empathy: "[domain-specific external adversary question]"

## Your Blind Spots
You do NOT evaluate: [explicit scope exclusions]. Leave those to [other roles].

## Instructions
Analyze the issue presented to you ONLY through your specific domain lens.
Do not attempt to evaluate the overall business merit of the proposal.
Your job is narrow, focused, domain-specific analysis. Produce your
findings using the output template above. Be direct and opinionated --
flag concerns clearly, do not hedge.

Your analysis will be reviewed by the [C-suite Parent] alongside analyses
from other specialists. Provide specific evidence for every claim.
Unsupported assertions will be challenged.

```

Heterogeneous Model Tiering

Layer	Model	Rationale
Team Lead Subagents	Haiku	Narrow, focused analysis. Cost-efficient. Model diversity improves analytical variety.
C-Suite Agents	Sonnet	Domain decomposition, delegation, and synthesis. Moderate complexity.
CEO Agent	Opus	Cross-domain synthesis, fault-line analysis, and decision-making. Highest reasoning quality.

Research on multi-agent systems suggests heterogeneous model tiers actually improve output quality beyond cost savings: model diversity produces more varied analytical perspectives than homogeneous agents.

Company Profile Configuration

The skill should support company profile parameterization that modifies:

- **Roster composition:** Conditional roles (e.g., Facilities/Office Manager relevant for manufacturing, not B2B SaaS). Product/UX Lead may move from CTO to a VP of Product in some orgs.
- **Decision mode default:** Some companies may default to Guardian (regulated industries) or Pioneer (startups).
- **Tier escalation thresholds:** Conservative companies may auto-escalate more aggressively.
- **Industry-specific frameworks:** Healthcare companies use HIPAA-specific compliance frameworks; financial services use SOX-specific controls.

Default profile: mid-market IT/technology services company, 200-500 employees (based on Explorer research).

Onboarding Stress Test as Initialization

When the skill is first configured for a company, run an organizational stress test as initialization. Present a representative cross-functional issue and run it through the full cascade. This serves three purposes:

1. **Validates the configuration:** Confirms all agents produce coherent, domain-appropriate analysis for this company type.
2. **Calibrates the user:** Shows the user what the skill produces and at what depth.
3. **Surfaces profile issues:** If a team lead produces irrelevant analysis (e.g., Facilities Manager for a fully-remote SaaS company), the user can adjust the profile before real use.

Cost and Implementation Viability

Explorer research on implementation viability produced concrete cost estimates and confirmed the recommended architecture patterns.

Cost Per Decision (v1 Staged Execution):

Scenario	C-Suite Agents Activated	Estimated Cost
Tier 1 (Hallway Question)	1 Sonnet subagent	~\$0.50-1.00
Tier 2 with routing (3-5 agents)	3-5 Sonnet subagents + Opus CEO	~\$2.50-5.00
Tier 3 full cascade (all 7)	7 Sonnet subagents + Opus CEO	~\$4.50-9.00
Multi-mode comparison (5 modes)	Same as Tier 3 + 4 additional CEO synthesis passes	~\$5.00-10.00

For comparison, a naive 37-agent flat architecture would cost \$18-37 per decision. The hybrid architecture with selective routing achieves 40-60% cost reduction while maintaining analytical quality.

v1 Architecture: Staged Execution with Subagents

The recommended v1 implementation uses the CEO as the main session, invoking C-suite members as subagents (which can run in parallel). Team lead analysis is simulated within each C-suite subagent's prompt via the analytical framework packages from Report #3. This pattern:

- Requires only 8 context windows (1 CEO + 7 C-suite)
- Benefits from prompt caching for repeated system prompt content
- Works within all current Claude Code Agent Teams constraints
- Produces 7 independent analytical perspectives (the primary quality driver)

v2 Architecture: Hybrid Agent Team + Subagents

When Claude Code supports teammates spawning subagents (not yet confirmed), the architecture evolves to: CEO as team lead of an Agent Team, 7 C-suite as teammates (with inter-agent messaging for Phase 4.5), each spawning team lead subagents on Haiku. This enables true three-tier execution with cross-functional deliberation.

Key cost design decision: Routing is a core feature, not a v2 optimization. Without routing, every question triggers 7 subagent calls. With routing, most decisions activate 3-5 C-suite members, reducing cost 40-60% and producing less analytical noise.

Future Directions (v2 and Beyond)

Decision Revisitation

Revisit decisions when assumptions change. The Metadata section of every Decision Record captures Key Assumptions. When conditions change, the user can invoke a targeted re-analysis that:

- Identifies which assumptions have been invalidated
- Re-runs only the affected domain analyses
- Produces a Decision Amendment rather than a full new record
- Highlights what changed in the recommendation and why

This transforms the skill from a point-in-time decision tool into a decision lifecycle manager.

Scenario Forking Engine

Parallel cascades with different parameters producing comparative decision landscapes. Not just "what should we do?" but "what happens if we do X vs. Y vs. Z?" The CEO Personality / Decision Mode concept is a special case of this -- same analysis, different synthesis. Full scenario forking extends this to different issue parameters producing different analyses entirely. Output: a Decision Landscape showing how different assumptions and approaches lead to different outcomes.

Organizational Stress Test Mode

Invert the cascade. Instead of top-down issue resolution, bottom-up threat and opportunity discovery. Each C-suite agent generates issues from their domain perspective: "What keeps you up at night? What opportunity are we missing?" Output: an Organizational Health Report that surfaces risks and opportunities the user has not yet articulated. McChrystal calls this "getting a full physical instead of going to the doctor with a complaint."

Institutional Memory

Persistent knowledge across sessions. Roles maintain context about previous decisions, the company's history, and evolving constraints. Previous Decision Records inform current analysis. The system learns the company's patterns and can reference them.

Cross-Functional Deliberation (Full Phase 4.5)

Beyond the one-round pre-mortem, a full cross-functional challenge where C-suite agents debate each other's recommendations. Research cautions about "problem drift" in multi-round debates, so this requires careful scoping -- perhaps structured around specific fault lines rather than open-ended debate.

Custom Decision Modes and Decision Pattern Analysis

Users define their own synthesis postures beyond the five defaults: "I'm generally growth-oriented but conservative on anything involving regulatory risk." The CEO's synthesis prompt becomes a user-configurable profile representing their actual leadership style. Over time, the system surfaces patterns in mode selection: "You've chosen the growth-oriented path in 8 of the last 10 decisions. Your CFO's concerns were overridden 6 times. Consider whether the conservative perspective is being systematically underweighted." This is not just decision support -- it is leadership coaching.

Framework Versioning and Community Contribution

Analytical framework packages versioned and improved independently. Industry-specific variants override defaults. Community-contributed domain frameworks. The skill becomes a platform for organizational reasoning patterns, not just a fixed set of 29 perspectives.

Builder Agent Instructions

The builder agent receiving this specification should construct the skill in the following order:

1. **Skill scaffolding:** Create the `.claude/skills/team-of-teams/` directory with `skill.md` front matter, output templates, routing table, and company profile configuration.
2. **CEO agent definition:** Create `ceo.md` with the five-phase cascade protocol, routing logic, fault-line analysis methodology, and modular Decision Mode system.
3. **CFO domain (reference implementation):** Create the CFO C-suite agent and all five CFO team lead subagent definitions using the reference implementation from Report #3. This domain is fully specified and serves as the template.
4. **Remaining C-suite agents:** Create the 6 remaining C-suite agent definitions following the CFO pattern: role identity, mandate, disposition, team composition, operating modes, synthesis instructions.
5. **Remaining team lead subagents (24):** Create the team lead subagent definitions for all remaining domains. Each must have a unique analytical framework, mandatory output template, and three forcing questions. Use the CFO team leads as the quality benchmark. No two team leads should produce the same type of analysis for the same issue.
6. **Invocation commands:** Implement `/consult`, `/panel`, `/deliberate`, and `/evaluate` with tier routing and mode selection.
7. **Decision Mode overlays:** Create all five CEO synthesis prompt modifiers (Guardian, Pioneer, Architect, Analyst, Sentinel) with their complete prompt text. Implement the multi-mode comparison mechanism: single-mode invocation, two-mode comparison, and all-modes invocation.
8. **Comparative Decision Record template:** Create the Comparative Decision Record format with shared analysis section, per-mode synthesis sections, divergence analysis, and Mode Sensitivity metric. Ensure the mode/tier interaction matrix is reflected in how each mode behaves at each tier.
9. **Validation:** Run the onboarding stress test -- a representative cross-functional issue through the full Tier 3 cascade -- and verify that all 29 perspectives produce genuinely different analysis. Then run a multi-mode comparison on the same issue and verify that different modes produce meaningfully different synthesis, not slight variations.

Quality criteria for the builder agent:

- Each team lead's output must be structurally different from every other team lead's output. If two team leads produce similar analysis for the same issue, their frameworks need revision.
- The Fault Line Analysis section should surface genuine disagreements, not manufactured ones. If all domains agree, say so -- do not force conflict.
- Decision Records should be self-contained documents that a reader can understand without context about the system that produced them.
- The skill should feel like consulting with an executive team, not filling out a form. Tone should be professional, direct, and opinionated -- not hedged or bureaucratic.
- Decision Modes must produce meaningfully different synthesis. If Guardian and Pioneer reach the

same conclusion for a contentious issue, the prompt modifiers are not distinct enough. Mode Sensitivity should be high for controversial decisions and low for clear-cut ones.

Source Materials

- Concept seed: `session/sources/team-of-teams.txt`
- Idea Report #1: `session/idea-reports/IDEA_cascading-deliberation-engine.md`
- Idea Report #2: `session/idea-reports/IDEA_engagement-model.md`
- Idea Report #3: `session/idea-reports/IDEA_cognitive-forcing-prompt-architecture.md`
- Idea Report #4: `session/idea-reports/IDEA_decision-modes-decision-space.md`
- Research: `session/research/RESEARCH_team-of-teams-methodology.md`
- Research: `session/research/RESEARCH_multi-agent-orchestration.md`
- Research: `session/research/RESEARCH_claude-agent-teams.md`
- Research: `session/research/RESEARCH_smb-decision-making.md`
- Research: `session/research/RESEARCH_implementation-viability-routing.md`
- Session summary: `session/SESSION_SUMMARY.md`
- Idea briefs: `session/briefs/`
- Ideation graph: `session/ideation-graph.md`
- Version snapshots: `session/snapshots/`

LAYER 3

Idea briefs, infographic images, and research findings

Idea Brief: The Cascading Deliberation Engine

Session: Team of Teams Agent Skill

Report: #1 (from Idea Report IDEA_cascading-deliberation-engine.md)

Status: INTERESTING (Arbiter-confirmed)

Spec Pillar: Structure

One-Sentence Summary

A five-phase cascading deliberation process that decomposes any business issue through a two-tier agent hierarchy (8 C-suite agents + 29 team lead subagents) with engineered dissent, producing a structured Decision Record that captures not just the decision but the full reasoning chain, fault lines, and dissenting views.

What It Is

The core architecture for the Team of Teams Agent Skill. When a user presents an issue, the CEO agent frames it, routes it to relevant C-suite agents, each C-suite agent dispatches to their team lead subagents for domain-specific analysis, synthesizes findings into a domain recommendation, and the CEO deliberates across all domains to produce a final decision. Disagreement is structural, not artificial: skeptic, advocate, and neutral mandates produce productive conflict that the CEO resolves through fault-line analysis.

Why It Matters

Without this architecture, the skill is just "ask a panel of chatbots." The cascade ensures that each layer of the hierarchy transforms the question rather than just distributing work. The five phases (CEO Frames, C-Suite Dispatches, Team Leads Produce, C-Suite Synthesizes, CEO Deliberates) create a knowledge generation machine where the analysis itself is more valuable than the final recommendation. Engineered dissent (4 skeptics, 2 advocates, 1 systemic, 1 synthesizer) guarantees that risk, cost, opportunity, and execution reality are all represented, even when the user might prefer to hear only good news.

Key Components

- **Five-Phase Cascade:** Frame -> Dispatch -> Analyze -> Synthesize -> Deliberate
- **Hybrid Agent Architecture:** 8 Agent Team members (CEO + 7 C-suite) invoke 29 custom subagents (team leads) via `.claude/agents/` definitions

- **Engineered Dissent Model:** Mandated dispositions per role -- CISO as Constitutional Skeptic, CFO as Cost Archaeologist, CTO as Capability Expander, VP Sales as Revenue Optimist, COO as Execution Realist, VP Delivery as Promise Keeper, CAO as Org Gravity Sensor
- **Decision-Type Routing:** CEO classifies issues (Strategic, Operational, Financial, Technical, Personnel, Compliance/Risk) and activates relevant subset of C-suite, not all 7 every time
- **Decision Record Output:** 9-section document (Executive Summary, Issue Statement, CEO Framing, Domain Analyses, Fault Line Analysis, CEO Decision, Dissenting Views, Next Steps, Metadata)

Open Questions for the Spec

- How does Phase 0 (shared consciousness broadcast) integrate with the cascade?
- What are the full-activation trigger rules (acquisition, pivot, layoff)?
- How does the company profile modify the roster (conditional roles, role reassignment)?
- Cross-functional deliberation (Phase 4.5) is parked for v2 -- but pre-mortem Phase 4.5 is proposed for v1 via Report #3.

Relationship to Other Ideas

This is the foundation. Report #2 (Engagement Model) defines how users interact with this engine at different depths. Report #3 (Cognitive Forcing) defines the prompt architecture that makes each perspective produce genuinely different analysis. Together: Structure, Experience, Quality.

Idea Brief: Cognitive Forcing and the Prompt Architecture

Session: Team of Teams Agent Skill

Report: #3 (from Idea Report IDEA_cognitive-forcing-prompt-architecture.md)

Status: INTERESTING (Arbiter-confirmed)

Spec Pillar: Analytical Quality

One-Sentence Summary

A three-layer prompt architecture where each of the 29 team lead perspectives is defined not by a persona description but by a unique analytical framework, mandatory output template, and domain-specific forcing questions -- compelling the LLM to perform structurally different cognitive work for each perspective and preventing "voice collapse."

What It Is

The prompt engineering framework that makes the skill's output worth reading. Research shows simple role assignments ("You are a CFO") produce zero measurable improvement in LLM output quality. The cognitive forcing approach instead gives each team lead a complete analytical toolkit: a specific methodology (e.g., GAAP Compliance Assessment for the Controller, Three-Scenario Financial Modeling for FP&A), a structured output template that forces specific analytical artifacts, and three types of forcing questions drawn from research -- pre-mortem (Klein 2007), adversarial empathy (McChrystal JSOC), and domain devil's advocate (Nemeth 2001). The architecture extends across three layers: 29 team lead subagent definitions, 7 C-suite orchestration templates, and the CEO synthesis layer with modular decision mode overlays.

Why It Matters

The difference between "7 agents that all sound like slightly different consultants" and "7 genuinely distinct domain perspectives that produce productive disagreement" is not better persona descriptions -- it is forcing each perspective to use different analytical methods that produce structurally different outputs. A GAAP compliance checklist and a three-scenario financial model are fundamentally different analytical artifacts that cannot collapse into the same voice. This is the hardest part of the spec to write and the most important for output quality. The cascade architecture (Report #1) is the skeleton. This is the muscle that makes it move.

Key Components

- **Analytical Framework Packages (29):** Each team lead defined by analytical methodology + mandatory output template + three forcing questions + accountability framing + blind spot declaration. Implemented as `.claude/agents/team-leads/[domain]/[role].md` subagent definitions.
- **Three Forcing Question Types:** Pre-mortem ("Assume this fails -- what caused it?"), Adversarial Empathy ("If you were [external adversary], how would you exploit this?"), Domain Devil's Advocate ("What would [domain expert critic] find concerning?"). Applied at all three layers.
- **Heterogeneous Model Tiering:** Haiku for team lead subagents, Sonnet for C-suite agents, Opus for CEO. Research suggests model diversity actually improves analytical variety.
- **C-Suite Multi-Mode Support:** Each C-suite prompt supports three cognitive modes -- direct consult (Tier 1 quick response), full analysis (Tier 2/3 subagent delegation), and challenge phase (Phase 4.5 pre-mortem).
- **Phase 4.5 Pre-Mortem:** After producing domain recommendations, C-suite agents see all peer recommendations and answer: "Assume this decision fails catastrophically in 12 months. What caused the failure?" One round only, no back-and-forth, feeds into Decision Record's Fault Line Analysis.
- **CFO Reference Implementation:** Five complete team lead packages (Controller, FP&A, Treasury, AP/AR, Tax Lead) with full frameworks, templates, and forcing questions. Serves as the model for all 29 packages.

Open Questions for the Spec

- The CFO domain (5 packages) is fully specified. The remaining 24 packages need equal specificity -- this is the largest content-generation task for the builder agent.
- Can analytical frameworks be versioned and updated independently? Could industry-specific variants override defaults?
- How well does Tier 1 internalization work? In direct consult mode, the C-suite agent reasons through team lead lenses internally without subagent isolation.
- Should team leads have cross-domain forcing questions that reference other domains' assumptions?

Relationship to Other Ideas

The cascade architecture (Report #1) defines the process. The engagement model (Report #2) defines the interface. This report defines what makes the output genuinely multi-perspective rather than cosmetically varied. Three pillars together: Structure, Experience, Quality.

Idea Brief: Decision Modes and the Decision Space Map

Session: Team of Teams Agent Skill

Report: #4 (from Idea Report `IDEA_decision-modes-decision-space.md`)

Status: INTERESTING (Arbiter-confirmed)

Spec Pillar: Strategic Depth

One-Sentence Summary

Five configurable CEO synthesis modes -- each mapping to established decision theory -- transform the skill from a decision engine into a decision exploration tool by showing how the same domain analysis produces different decisions depending on what the user optimizes for.

What It Is

Decision Modes are swappable prompt modifiers in the CEO agent's synthesis layer that change how competing domain perspectives are weighted and resolved. The underlying domain analysis (team lead findings, C-suite recommendations, fault lines) is identical across modes. Five modes are defined: **Guardian** (protect what we have -- MaxiMin), **Pioneer** (pursue what we could gain -- MaxiMax), **Architect** (build widest alignment -- Behavioral), **Analyst** (follow the evidence -- Hurwicz balanced), and **Sentinel** (minimize worst-case regret -- MiniMax Regret). The power feature is multi-mode comparison, where domain analysis runs once (the expensive part) and the CEO synthesis runs multiple times (cheap), producing a Comparative Decision Record that shows the decision space side by side. A Mode Sensitivity metric signals whether the decision depends on risk appetite or on evidence.

Why It Matters

The most useful output of multi-perspective analysis is not a single answer -- it is a map showing how the answer changes depending on what you optimize for. Solo founders and small leadership teams constantly face the question "am I being too cautious or too aggressive?" Decision Modes make that question answerable. When all five modes converge on the same decision, the user knows the answer does not depend on their posture. When modes diverge dramatically, the user knows their personal risk appetite is the deciding factor. This transforms a binary recommendation into a nuanced view of the decision landscape.

Key Components

- **Five Decision Modes:** Guardian (risk-averse), Pioneer (growth-oriented), Architect (consensus-building), Analyst (data-driven, default), Sentinel (regret-minimizing). Each has a complete CEO prompt modifier.
- **Mode/Tier Interaction Matrix:** Each mode produces distinct behavioral patterns at each tier. Default cell: Tier 1 + Analyst (quick, evidence-weighted, transparent about uncertainty).
- **Multi-Mode Comparison:** `/deliberate guardian vs pioneer: [issue]` or `/deliberate all-modes: [issue]`. Domain analysis runs once; CEO synthesis runs per mode. Cost: approximately 1.1x a single deliberation for 4-5x the strategic insight.
- **Comparative Decision Record:** Side-by-side synthesis showing shared analysis, per-mode decisions with most determinative perspective and key factor, divergence analysis, and Mode Sensitivity metric.
- **Mode Sensitivity:** High sensitivity means the decision depends on risk appetite. Low sensitivity means all modes converge -- the evidence speaks for itself.
- **Invocation Grammar Integration:** Modes work with all tiers and invocation commands from Report #2.

Open Questions for the Spec

- Can users define custom modes (e.g., "growth-oriented but conservative on regulatory risk")? Probably v2.
- Should the CEO auto-recommend which mode is appropriate for the decision type?
- Can the system surface patterns in mode selection over time (leadership coaching)?
- How do you ensure modes produce meaningfully different synthesis, not slight variations? Prompt modifier distinctiveness is key.

Relationship to Other Ideas

This report develops the Decision Modes introduced in Report #2 into a full feature design. The cascade architecture (Report #1) produces the analysis that modes synthesize differently. The cognitive forcing framework (Report #3) ensures the domain analysis is high-quality enough that mode differences are substantive, not cosmetic. Together, the four reports form the complete design: Structure (#1), Experience (#2), Quality (#3), Strategic Depth (#4).

Idea Brief: The Engagement Model

Session: Team of Teams Agent Skill

Report: #2 (from Idea Report IDEA_engagement-model.md)

Status: INTERESTING (Arbiter-confirmed)

Spec Pillar: Experience

One-Sentence Summary

A two-dimensional control surface -- depth (three interaction tiers) and width (four CEO decision modes) -- that lets users match the skill's engagement level to each decision's weight, transforming it from an event they invoke for major decisions into a daily tool.

What It Is

The user interface layer for the cascading deliberation engine. Depth is controlled through three tiers: **Tier 1 (Hallway Question)** is a direct consult with a single C-suite agent producing a 3-5 sentence Advisory Note; **Tier 2 (Working Session)** routes to 2-4 relevant C-suite members for a one-page Panel Assessment; **Tier 3 (Board Meeting)** runs the full five-phase cascade producing a complete Decision Record. Width is controlled through four CEO Decision Modes that change how the CEO weighs domain recommendations: **Guardian** (risk-averse), **Pioneer** (growth-oriented), **Architect** (consensus-building), and **Analyst** (data-driven, default). Multi-mode comparison shows the "decision space" -- how different risk appetites shape the same analysis into different outcomes.

Why It Matters

Most organizational questions do not need a full board meeting. If the skill only operates at maximum depth, users will invoke it rarely and it becomes a quarterly novelty. The three tiers ensure daily adoption by offering a lightweight entry point (Tier 1 costs seconds, not minutes). The four decision modes solve the "whose judgment?" problem -- they reveal that the choice of decision-making style matters as much as the analysis itself. Together, the tiers and modes make the skill feel like having an executive team on retainer rather than a bureaucratic decision machine.

Key Components

- **Four Invocation Commands:** `/consult [role]: [question]` (Tier 1), `/panel [roles]: [issue]` (Tier 2), `/deliberate: [issue]` (Tier 3), `/evaluate: [issue]` (auto-triage)

- **CEO Triage Logic:** Assesses scope (single vs. multi-domain), impact (low to critical), reversibility (easy to irreversible). Recommends tier with one-sentence justification. User can accept, escalate, or de-escalate.
- **Decision Modes as Prompt Modifiers:** Modes change CEO synthesis weighting, not underlying domain analysis. Identical team lead outputs produce different CEO decisions under different modes.
- **Multi-Mode Comparison:** `--compare-modes` flag runs CEO synthesis four times, producing a Decision Space Map that shows Guardian, Pioneer, Architect, and Analyst outcomes side by side.
- **Tiered Output Formats:** Advisory Note (3-5 sentences), Panel Assessment (~1 page), Full Decision Record (3-5 pages)

Open Questions for the Spec

- Can a Tier 1 consult escalate to Tier 2 mid-conversation if the C-suite agent realizes the question is bigger than it seemed?
- Should tier escalation thresholds and default decision mode be configurable per company profile?
- Could users create custom blended modes (e.g., "growth-oriented but weight security concerns heavily")? Probably v2.
- Does the C-suite agent in Tier 1 mode remember previous hallway questions? (Connects to institutional memory thread.)

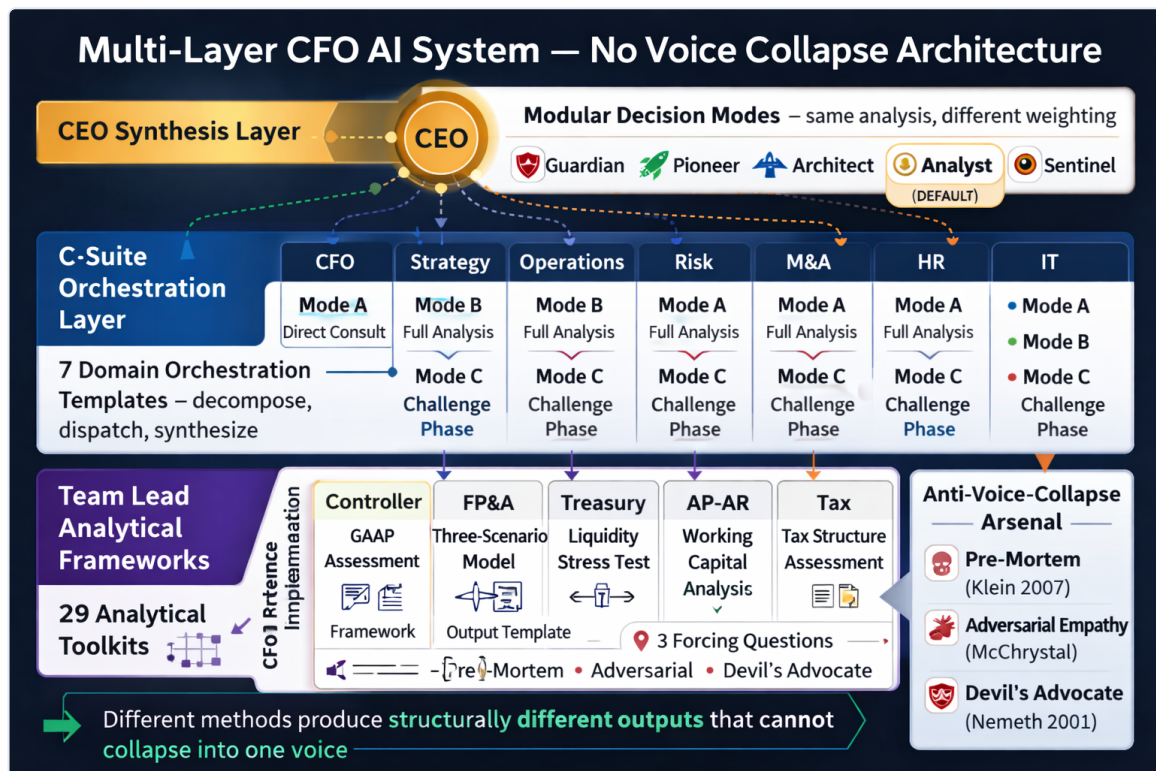
Relationship to Other Ideas

The cascade architecture (Report #1) is the engine. This report is the steering wheel. Without it, there is a powerful machine with no practical way for users to interact at the right level of depth. Report #3 (Cognitive Forcing) ensures the analysis at every tier is genuinely multi-perspective, not cosmetically varied.

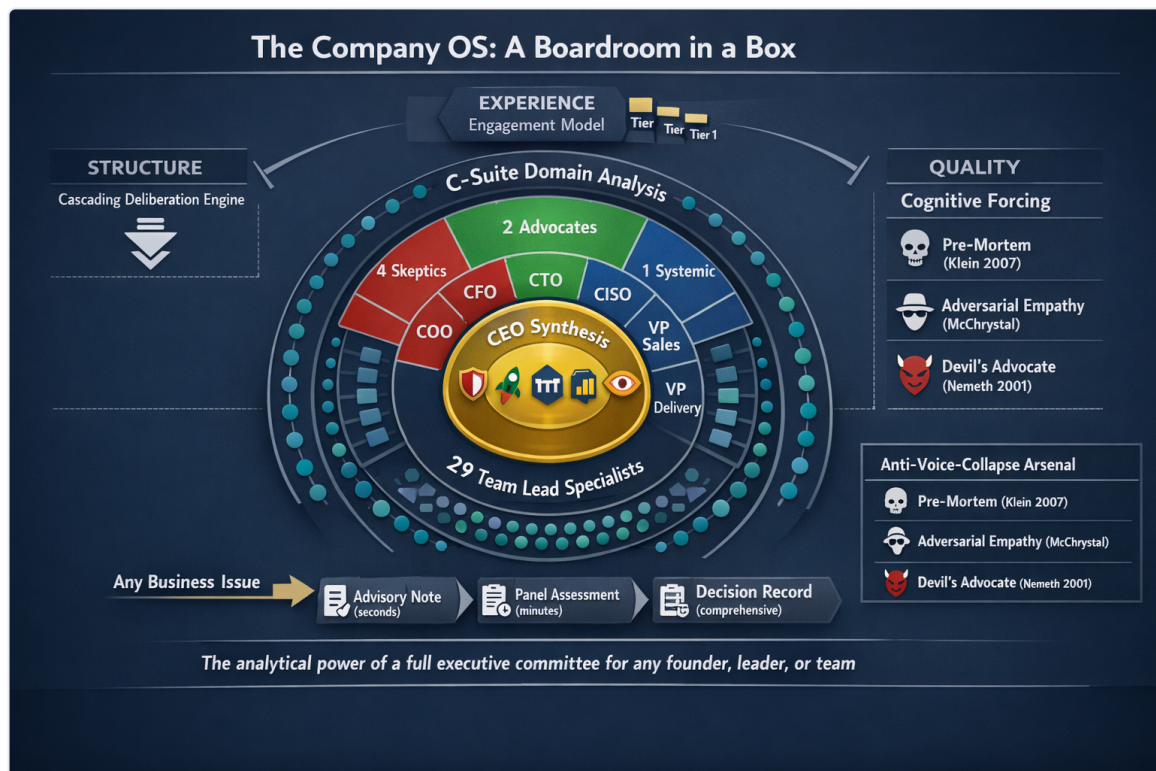
The Cascading Deliberation Engine



Infographic Cascading Deliberation Engine



Infographic Cognitive Forcing Prompt Architecture



Infographic Company Os Unified Vision

The Decision Framework

Depth: Three Interaction Tiers

 **Hallway Question**
/consult
Advisory Note (3-5 sentences)

 **Working Session**
/panel
Panel Assessment (~1 page)

 **Board Meeting**
/deliberate
Decision Record (3-5 pages)

Guardian	Pioneer	Architect	Analyst	Sentinel
		DEFAULT		

Width: Five Decision Modes

-  **Guardian**
Risk-Averse / MaxiMin
-  **Pioneer**
Growth-Oriented / MaxiMax
-  **Architect**
Consensus-Building / Behavioral
-  **Analyst**
Data-Driven / Hurwicz (**DEFAULT**)
-  **Sentinel**
Regret-Minimizing / MiniMax Regret

Auto-Triage: /evaluate routes any issue to the right tier

Infographic Engagement Model

Research: Cascade Delegation Mechanics & Engineered Dissent

Requested by: Thinker discussion (proactive research based on grounder's request for cascade + dissent mechanics)

Date: 2026-02-22

Question

How does cascade delegation actually work step by step in organizational decision-making? And where/how should engineered dissent live in the structure -- in role prompts, in the workflow structure, or in explicit "challenge" phases?

Findings

Part 1: Cascade Delegation Mechanics

What "Cascading" Means in Organizational Theory

Strategy cascading is the process of decomposing high-level goals and strategic ambitions into smaller, preferably independent sub-goals that can be owned and executed at each organizational level. The key insight: **each level translates the parent-level question into their own domain context** rather than simply passing it down unchanged.

The Step-by-Step Cascade Process (Applied to Team of Teams Skill)

Based on organizational cascade theory, here's how the delegation would work when an issue is presented to the CEO:

Step 1 -- Issue Intake & Framing (CEO)

- CEO receives the issue/question from the user
- CEO frames the issue in strategic terms: What is the business question? What are the stakes? What kind of decision is this (strategic, operational, financial, technical)?
- CEO identifies which C-suite members need to evaluate this (may not be all 7 for every issue)

Step 2 -- Context Broadcast (Shared Consciousness)

- CEO communicates the issue to relevant C-suite members with strategic framing
- This maps to McChrystal's "shared consciousness" -- everyone gets the same intelligence
- Each C-suite member receives: the raw issue, the CEO's strategic framing, and any relevant context

Step 3 -- Domain Translation (C-Suite Level)

- Each C-suite member translates the issue into their domain's language and concerns
- CFO asks: "What are the financial implications? What does this cost? What's the ROI?"

- CISO asks: "What are the security risks? What compliance implications?"
- CTO asks: "What technical capabilities do we need? What's the build/buy decision?"
- **This translation step is critical** -- it's what makes this more than a panel of generalists

Step 4 -- Sub-Delegation (C-Suite to Team Leads)

- Each C-suite member delegates specific analysis tasks to their team leads
- CFO tells Controller to assess accounting impact, FP&A to model financial scenarios, Tax to check tax implications
- CTO tells Engineering Lead to assess technical feasibility, DevOps to evaluate infrastructure needs
- **Each team lead gets a focused, domain-specific sub-question** derived from the C-suite member's domain translation

Step 5 -- Ground-Level Analysis (Team Leads)

- Team leads perform focused analysis within their narrow expertise
- Controller checks impact on financial statements and close process
- Tax Accountant identifies tax implications and compliance requirements
- Security Operations Lead assesses specific threat vectors
- **These are the "leaves" of the fractal tree** -- maximum specificity, maximum domain knowledge

Step 6 -- Upward Synthesis (Team Leads -> C-Suite)

- Team leads report findings back to their C-suite member
- Each C-suite member synthesizes team lead analyses into a coherent departmental recommendation
- CFO combines Controller + FP&A + Tax + Treasury into a single financial perspective
- **This synthesis step is where the "CFO doesn't have an opinion -- the CFO has an organization that produces analysis"** becomes concrete

Step 7 -- Cross-Functional Deliberation (C-Suite Level)

- C-suite members share their department recommendations
- This is where productive tensions emerge: CFO's financial caution vs. VP Sales' growth ambition
- Maps to McChrystal's "fusion cell" concept -- cross-functional deliberation

Step 8 -- Final Synthesis & Decision (CEO)

- CEO evaluates all C-suite recommendations
- Weighs trade-offs, resolves conflicts, applies strategic judgment
- Produces final decision with rationale
- Documents the decision record (who recommended what, what trade-offs were made, what the final call is)

Fractal Decision-Making Pattern

The cascade creates a **self-similar pattern at each level**:

- CEO breaks the big question into department-relevant sub-questions
- Each C-suite member breaks their sub-question into specialist sub-sub-questions
- At every level, the pattern is: Receive -> Translate to Context -> Delegate -> Synthesize -> Report

This is why "fractal" is the right metaphor. The decision-making process is structurally identical at each tier, just applied at different levels of domain specificity.

Four Cascading Methods (Applicable to Design)

From strategy cascading theory, four alignment approaches exist:

1. **Alignment by Perspectives** -- All departments analyze the same issue through their standard framework (financial, operational, risk, growth). Best when you want comprehensive coverage.
2. **Alignment by Normalized Goals** -- Different departments use different frameworks but normalize outputs to a common format. Good for comparing unlike analyses.
3. **Direct Cascading by KPIs** -- Quantitative decomposition where metrics aggregate upward. Limited to quantifiable domains.
4. **Alignment by Context** -- Logical/relational connections between analyses without requiring quantitative integration. Most flexible, applicable to any strategy component.

For the Team of Teams skill, **Alignment by Context** combined with **Alignment by Perspectives** is the best fit -- each department uses its own analytical lens but all contribute to a common decision framework.

Part 2: Engineered Dissent Mechanics

Why Dissent Must Be Structural, Not Optional

Research evidence is overwhelming:

- Devil's advocacy increases consideration of alternatives by **61%**
- Meeting effectiveness improved by **33%** with structured critical review
- Decision quality improved by **23%**
- Project delays reduced by **36%**
- Idea diversity increased by **32%**
- Employee satisfaction increased by **29%**

Without structured dissent, multi-agent consensus systems produce what the grounder called "consensus mush" -- everyone agrees and no one identifies real risks.

Three Layers Where Dissent Should Live

Based on the research, dissent should be engineered at **all three levels** -- role prompts, structure, and explicit phases. Here's how:

Layer 1: Role-Inherent Dissent (In the Prompts)

Certain roles are **structurally adversarial** by nature. This isn't artificial -- it reflects real organizational dynamics:

- **CISO**: Inherently risk-averse. The CISO's JOB is to say "here's what can go wrong." Every proposal is a potential security incident.
- **CFO**: Cost-conscious contrarian. Every proposal has hidden costs. The CFO's job is to find them.
- **COO**: Operational skeptic. "Can we actually execute this?" The COO knows what the org can and cannot absorb.
- **VP of Delivery**: Capacity realist. "Can we deliver what Sales is promising?"

These tensions should be **baked into the role system prompts**:

- CFO prompt includes: "Your primary responsibility is to identify financial risks, hidden costs, and budget impacts that others may overlook. You are expected to challenge proposals that lack clear ROI."
- CISO prompt includes: "Your primary responsibility is to identify security vulnerabilities, compliance risks, and threat vectors. When in doubt, err on the side of caution. You are the organization's risk conscience."

Layer 2: Structural Dissent (In the Architecture)

Beyond individual role biases, the structure itself should create productive tension:

- **Paired Opposition**: Deliberately pair roles with opposing incentives:
 - VP Sales (growth) vs. CFO (cost control)
 - CTO (innovation) vs. CISO (security)
 - VP Sales (what we promise) vs. VP Delivery (what we can deliver)
 - COO (operational efficiency) vs. CAO (people/compliance)
- **Cross-Department Visibility**: Each C-suite member should see other departments' findings (McChrystal's shared consciousness) so they can **challenge each other**, not just contribute independently.

Layer 3: Explicit Challenge Phase (In the Workflow)

Based on McChrystal's JSOC red teaming and organizational research, the workflow should include an **explicit challenge phase** between Step 7 (cross-functional deliberation) and Step 8 (CEO synthesis):

Step 7.5 -- Red Team / Challenge Phase

Three specific techniques to implement:

1. **Pre-Mortem Analysis**: Each C-suite member must answer: "Assume this decision fails catastrophically in 12 months. What caused the failure?" This technique (from JSOC) surfaces concerns people might otherwise suppress.
2. **Adversarial Empathy ("Act As If")**: Assign 1-2 C-suite members to argue FROM THE PERSPECTIVE of a competitor, a regulator, a disgruntled customer, or an adversary. "If I were our main competitor, how would I exploit this decision?"
3. **"What Else?" Analysis**: Force consideration of competing hypotheses. "What's the second-most-likely outcome? The third?" This prevents premature convergence on a single narrative.

Authentic Dissent vs. Assigned Dissent

Critical research finding from Nemeth (2001): **authentic dissent is more effective than assigned devil's advocacy**. An agent that genuinely holds a contrarian view (because its role and domain knowledge lead it there) produces better dissent than one told to "play devil's advocate."

This means: **Role-inherent dissent (Layer 1) is more powerful than explicit challenge phases (Layer 3)**. The CFO who genuinely finds the financial risk is more convincing than a role-playing adversary. But the challenge phase adds value on TOP of authentic role-based dissent.

McChrystal's Red Team as a Model

In JSOC, red teaming was:

- **Semi-independent:** The red team was structurally separate from the planning team
- **Peer-based:** Elite professionals checking peers' homework, not subordinates questioning superiors
- **Unemotional and structured:** A formal component of the decision process, not ad hoc
- **Iterative:** Red team findings fed back into plan revision before execution

For the skill design, this maps to: After C-suite members present their initial recommendations, a structured challenge round occurs where each member reviews and critiques OTHER departments' recommendations, followed by a revision opportunity before the CEO synthesizes.

Key Takeaways

- **The cascade has 8 clear steps:** Intake -> Context Broadcast -> Domain Translation -> Sub-Delegation -> Ground Analysis -> Upward Synthesis -> Cross-Functional Deliberation -> CEO Decision. The grounder is right that this is the architectural heart.
- **Dissent should live at ALL three layers simultaneously:** role prompts (authentic domain-based contrarianism), structural pairings (opposing incentives), and an explicit red team/challenge phase in the workflow. Layer 1 is the most powerful because it produces authentic dissent.
- **The "pre-mortem" technique is the highest-value single addition:** Having every C-suite agent answer "how does this fail?" dramatically improves decision quality (research-backed). It should be a mandatory step in the workflow.
- **Cross-department visibility in the challenge phase** maps directly to McChrystal's shared consciousness -- each C-suite member should see and be able to challenge other departments' findings, not just contribute their own analysis in isolation.
- **The fractal pattern (Receive -> Translate -> Delegate -> Synthesize -> Report) is identical at every tier**, making the architecture self-similar and conceptually clean. This should be reflected in the system prompt structure.

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Research: CEO Decision Modes & Personality Profiles

Requested by: Proactive research based on free-thinker's CEO personality thread and grounder's endorsement as a strong Report #2 candidate

Date: 2026-02-22

Question

What decision-making style frameworks exist that could model different CEO synthesis personalities? How would running the same analysis through different CEO profiles produce meaningfully different decisions?

Findings

Two Complementary Frameworks

Framework 1: Decision Style Theory (Rowe & Boulgarides)

Four established decision-making styles that combine cognitive complexity with value orientation:

Style	How They Decide	What They Weight Most	Risk Posture	Dissent Handling	Synthesis Pattern
Directive	Quick, decisive, experience-based	Speed, clarity, precedent	Low tolerance -- prefers proven paths	Maintains firm positions, limited deliberation	Picks the dominant recommendation quickly
Analytical	Data-driven, methodical, thorough	Evidence, scenarios, risk quantification	Moderate -- thoroughly evaluated first	Uses data-driven arguments to resolve disputes	Weights all data systematically, identifies optimal path
Conceptual	Big-picture, innovation-focused, long-term	Strategic vision, transformation, novel solutions	High tolerance -- accepts calculated risks for transformative outcomes	Leverages diverse perspectives for richer solutions	Looks for the option that creates the most future capability
Behavioral	Consensus-seeking, people-oriented, collaborative	Harmony, stakeholder buy-in, team impact	Low tolerance -- prefers decisions everyone can support	Actively seeks input, works toward harmony	Finds the path that most stakeholders can endorse

Framework 2: Decision Theory Risk Postures (Formal)

Three classical decision criteria from operations research:

Posture	Formal Name	How It Works	CEO Translation
Conservative	MaxiMin (Wald)	Evaluate each option by its worst-case outcome; choose the one whose worst case is best	"What's the safest path? What minimizes our maximum downside?"
Aggressive	MaxiMax	Evaluate each option by its best-case outcome; choose the one with the highest upside	"What path has the biggest potential payoff? I'll accept the risk."
Balanced	Hurwicz Criterion	Weighted combination of best-case and worst-case, using an "optimism coefficient" (0=pure pessimist, 1=pure optimist)	"Balance the upside and downside with a configurable risk tolerance"
Regret-Minimizing	MiniMax Regret (Savage)	Choose the option that minimizes the maximum regret -- "which choice will I least regret if things go wrong?"	"Which decision will I be able to defend regardless of outcome?"

How This Maps to CEO Personality Profiles

Combining both frameworks, here are five distinct CEO synthesis profiles the skill could offer:

1. "The Guardian" (Conservative / Directive)

- **Synthesis behavior:** Weights CISO and CFO perspectives most heavily. Looks for risks first. Requires strong evidence to approve.
- **Decision pattern:** "Unless there's compelling evidence this is safe AND financially sound, the answer is no."
- **When to use:** High-stakes decisions, regulated industries, decisions that are hard to reverse.
- **Adjusts weights:** CISO +2, CFO +2, VP Sales -1, CTO -1

2. "The Analyst" (Balanced / Analytical)

- **Synthesis behavior:** Weights all perspectives equally by default. Adjusts based on decision type. Demands complete data.
- **Decision pattern:** "Show me the numbers from every angle. I'll optimize across all dimensions."
- **When to use:** Complex multi-factor decisions where the trade-offs aren't obvious. The "default" mode.
- **Adjusts weights:** All equal, with type-based modifiers (financial decisions boost CFO, etc.)

3. "The Visionary" (Aggressive / Conceptual)

- **Synthesis behavior:** Weights CTO and VP Sales perspectives most heavily. Looks for transformative opportunity. Tolerates higher risk.
- **Decision pattern:** "What's the biggest thing this could become? Is the potential worth the risk?"
- **When to use:** Market entry decisions, innovation bets, growth-oriented questions.
- **Adjusts weights:** CTO +2, VP Sales +2, CISO -1, CFO -1

4. "The Consensus Builder" (Behavioral)

- **Synthesis behavior:** Weights all perspectives and looks for the decision that most executives can support. Prioritizes organizational buy-in.
- **Decision pattern:** "Can we find a path that addresses everyone's core concerns? What modifications would it take to get broad support?"
- **When to use:** Organizational changes, cultural decisions, situations where execution requires buy-in.
- **Adjusts weights:** Seeks minimum dissent; modifies the decision to accommodate strongest objections.

5. "The Regret Minimizer" (MiniMax Regret)

- **Synthesis behavior:** For each option, asks "if this goes wrong, how bad is it?" Chooses the path with the least catastrophic downside.
- **Decision pattern:** "Which decision can I defend to the board regardless of how it turns out?"

- **When to use:** Irreversible decisions, decisions with asymmetric downside, board-level accountability.
- **Adjusts weights:** Disproportionately weights the strongest objection from any role.

The "Decision Space" Concept

The grounder's insight is that running the same 7 domain analyses through multiple CEO profiles shows the user the **decision space** -- not just "what should we do?" but "here are the paths available, and here's what you're choosing between."

Implementation:

```
Same C-suite analysis (7 subagent reports) -->
  |— The Guardian says: REJECT (too risky, CISO concerns unresolved)
  |— The Analyst says: CONDITIONAL (proceed if CFO's financing condition is met)
  |— The Visionary says: APPROVE (the strategic upside justifies the risk)
  |— The Consensus Builder says: DEFER (need to address COO and VP Delivery concerns
first)
  |— The Regret Minimizer says: CONDITIONAL (proceed only with CISO's recommended
mitigations)
```

This is dramatically more useful than a single recommendation because:

1. It shows the user **WHAT** they're actually deciding -- not "should we do this?" but "how much risk are we comfortable with?"
2. It reveals which factors tip the decision -- if Guardian and Visionary agree, it's a clear call; if they diverge, it shows the risk/reward tension
3. It lets users self-select the profile that matches their actual risk tolerance and organizational context

Implementation Cost Considerations

Running 5 CEO synthesis passes instead of 1 adds minimal cost because:

- The expensive part is the 7 C-suite subagent calls (which only run once)
- CEO synthesis is a single-turn prompt against the same data
- 5 additional Opus calls with pre-existing analysis: ~\$1-2 additional
- This could also be done with a single CEO call that produces all 5 perspectives

Lightweight vs. Full Decision Space

- **Default mode:** Single CEO profile (The Analyst -- balanced/default)
- **Decision Space mode:** Run all 5 profiles, present the decision space
- **Custom mode:** User selects 2-3 profiles that match their decision context

Key Takeaways

- **Five CEO profiles map to established decision theory** (not arbitrary personality types): Guardian (conservative), Analyst (balanced), Visionary (aggressive), Consensus Builder (behavioral), Regret Minimizer (minimax regret).

- **The "decision space" concept is the killer feature:** Same analysis, multiple synthesis lenses. Shows users what they're actually choosing between -- not "the answer" but "the range of defensible answers and what drives each one."
- **Minimal additional cost:** C-suite analysis runs once; CEO synthesis is cheap to run multiple times.
- **Implementation:** Can be a single CEO prompt that generates all perspectives, or 5 separate lightweight calls. Either way, ~\$1-2 additional on top of the base cost.
- **Default should be "The Analyst"** (balanced, all-perspectives-weighted-equally) with decision space mode as an opt-in enhancement.

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Research: Claude Code Agent Teams Capabilities and Constraints

Requested by: Arbiter (initial research assignment)

Date: 2026-02-22

Question

What are the technical limits of Agent Teams? How many agents can run simultaneously? What are the communication patterns? What are known constraints or best practices?

Findings

Architecture Overview

Claude Code Agent Teams consist of:

- **Team lead:** The main Claude Code session that creates the team, spawns teammates, coordinates work
- **Teammates:** Separate Claude Code instances that each work on assigned tasks
- **Task list:** Shared work items with dependency tracking and auto-unblocking
- **Mailbox:** Inbox-based messaging system for inter-agent communication

Teams are experimental (as of Feb 2026), requiring explicit opt-in via `CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1`.

Communication Patterns

1. **Direct Messages (SendMessage):** One teammate sends to one specific teammate. The primary communication mechanism.
2. **Broadcasts:** One teammate sends to ALL teammates simultaneously. Expensive -- costs scale linearly with team size. Should be used sparingly, only for critical team-wide information.
3. **Shared Task List (TaskCreate, TaskUpdate, TaskList, TaskGet):** All agents can see task status and claim available work. Tasks have three states: pending, in_progress, completed. Tasks support dependency tracking -- a pending task with unresolved blockedBy cannot be claimed until those dependencies complete. When a blocking task completes, downstream tasks automatically unblock.
4. **Idle Notifications:** When a teammate finishes work and stops, they automatically notify the lead.
5. **Automatic Message Delivery:** Messages are delivered automatically to recipients. No polling required.

Agent Limits and Scalability

No documented hard maximum on number of teammates. The docs show examples with 3-5 teammates. One example spawns "5 agent teammates." Practical limits are driven by:

- **Token cost:** Each teammate is a separate Claude instance with its own context window. Usage scales linearly with active teammates.
- **Coordination overhead:** More agents = more messages = more tokens spent on coordination
- **File conflicts:** Two teammates editing the same file leads to overwrites

Best practice: 5-6 tasks per teammate keeps everyone productive. Having clear, non-overlapping file ownership is critical.

Key Capabilities

1. **Parallel Execution:** Multiple teammates investigate different aspects simultaneously
2. **Self-Coordination:** Teammates claim tasks independently when they finish current work
3. **Dependency Tracking:** Tasks can block other tasks, with auto-unblocking on completion
4. **Direct Inter-Agent Communication:** Unlike subagents, teammates can talk to each other directly
5. **Adversarial Debate:** Teammates can be configured to challenge each other's findings
6. **Plan Approval:** Teammates can be required to plan before implementing; lead approves/rejects
7. **Display Modes:** In-process (single terminal) or split panes (tmux/iTerm2)
8. **Delegate Mode:** Lead restricted to coordination only -- can't write code, only manage tasks and communicate

Critical Constraints

1. **No Session Resumption:** `/resume` and `/rewind` do not restore in-process teammates. After re-summing, the lead may attempt to message non-existent teammates.
2. **No Nested Teams:** Teammates cannot spawn their own teams or teammates. Only the lead manages the team. **This is the biggest constraint for a multi-level org structure.**
3. **One Team Per Session:** A lead can only manage one team at a time.
4. **No Tool Isolation Between Teammates:** All teammates inherit the lead's full permission set. You cannot spawn a read-only "researcher" alongside a full-access "implementer." This is enforced at spawn time and cannot be changed per-teammate.
5. **No Per-Teammate Hooks:** Unlike subagents, teammates cannot have `PreToolUse` hooks for conditional validation.
6. **No Persistent Memory for Teammates:** Teammates start fresh each time. Subagents support `memory: user|project|local` but teammates do not.
7. **No Skill Preloading:** Teammates rely entirely on the spawn-time prompt. Subagents can preload skills, teammates cannot.
8. **Lead Is Fixed:** Cannot promote a teammate to lead or transfer leadership.
9. **Shutdown Can Be Slow:** Teammates finish their current request/tool call before shutting down.

10. **Task Status Can Lag:** Teammates sometimes fail to mark tasks as completed, blocking dependent tasks.

Subagents vs. Agent Teams (Key Distinction)

Aspect	Subagents	Agent Teams
Context	Own window; results return to caller	Fully independent windows
Communication	Report results back to main only	Direct teammate messaging
Coordination	Main agent manages all work	Shared task list with self-coordination
Tool Restrictions	Full customization per agent	All inherit lead's permissions
Hooks	Supported (PreToolUse, etc.)	Not supported
Skills	Can preload skills	Cannot preload skills
Memory	Persistent memory support	No persistent memory
Best for	Focused tasks, cost-effective	Complex work requiring discussion
Token cost	Lower (summarized results)	Higher (separate instances)

Custom Subagent Configuration (Relevant for Design)

Custom subagents (`.claude/agents/`) support rich configuration via YAML frontmatter:

- `name` , `description` : Identity and delegation trigger
- `tools` / `disallowedTools` : Tool access control
- `model` : `sonnet` , `opus` , `haiku` , or `inherit`
- `permissionMode` : `default` , `acceptEdits` , `dontAsk` , `bypassPermissions` , `plan`
- `skills` : Preloaded domain knowledge
- `memory` : Persistent cross-session memory (`user` , `project` , `local`)
- `hooks` : Lifecycle hooks for conditional validation
- `maxTurns` : Maximum agentic turns
- `isolation` : `worktree` for isolated git worktree

Skills System (Relevant for Skill Design)

Skills are defined in `SKILL.md` with YAML frontmatter:

- `name` : Slash command identifier
- `description` : When Claude should use it
- `disable-model-invocation` : Manual-only trigger
- `user-invocable` : Whether users can invoke directly
- `allowed-tools` : Tool restrictions when active
- `model` : Model override
- `context` : `fork` for subagent execution
- `agent` : Which subagent type to use with `context`: `fork`

- Supports `$ARGUMENTS` substitution, `! command`` for dynamic context injection
- Supporting files (templates, scripts, references) in skill directory

Open Feature Request (GitHub Issue #24316)

There is an active feature request to allow agent team teammates to be spawned from `.claude/agents/` definitions, which would enable:

- Per-teammate tool restrictions
- Per-teammate hooks
- Per-teammate persistent memory
- Skill preloading for teammates
- Deterministic role enforcement (not just prompt-based)

Current status: Open, not implemented. Workarounds include using subagents instead of teams (loses inter-agent communication) or per-teammate inline overrides.

Key Takeaways

- **Agent Teams provide the inter-agent communication the Team of Teams concept needs**, but lack the per-agent customization that subagents offer. The ideal solution would combine both.
- **The "no nested teams" constraint is the biggest design challenge** for a multi-level org structure (CEO -> C-suite -> team leads). The skill will need a creative architectural solution, likely using subagents within a team structure, or a staged execution model.
- **Token cost is a real constraint:** 7 C-suite agents + ~35 team leads running simultaneously would be extremely expensive. The design should use tiered models (Haiku for team leads, Sonnet/Opus for C-suite/CEO) and staged execution (not all agents active simultaneously).
- **The Skills system provides the right packaging mechanism** for distributing this as a reusable capability. A skill can include SKILL.md, supporting agent definitions, templates, and scripts.
- **Prompt-only role enforcement is fragile but currently the only option** for agent team teammates. The spec should include robust system prompts with clear boundaries, decision domains, and interaction protocols.

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Research: Cognitive Forcing Functions & Role-Based Prompt Architecture

Requested by: Proactive research in support of Idea Report #3 (Cognitive Forcing and Prompt Architecture)

Date: 2026-02-22

Question

What are cognitive forcing functions and how can they be applied to AI agent prompts? How effective is role-based persona prompting for LLMs? What are the best practices for constructing system prompts that reliably produce domain-expert-quality analysis?

Findings

Part 1: Cognitive Forcing Functions

Definition and Origin

Cognitive forcing functions (CFF) are structured interventions designed to interrupt intuitive (System 1) thinking and invoke deliberate analytical (System 2) reasoning. Originating in medical decision-making, they address the tendency for experts to jump to conclusions based on pattern recognition, missing edge cases and rare scenarios.

Three levels exist:

- **Universal:** Apply to all decisions (e.g., "always consider alternatives")
- **Generic:** Apply to categories of decisions (e.g., "for financial decisions, always check second-order costs")
- **Specific:** Counter specific known biases (e.g., "for acquisition decisions, always run a pre-mortem")

The Most Effective Cognitive Forcing Techniques (Translated for AI Agents)

Based on medical and organizational research, here are the techniques most translatable to AI agent system prompts:

1. Consider-the-Opposite

Force the agent to seek evidence AGAINST its initial conclusion. Experimental research demonstrated this reduces judgment biases.

- Prompt translation: "After forming your initial recommendation, identify the three strongest arguments against it. If any of these arguments are stronger than your supporting evidence, revise your recommendation."

2. Pre-Mortem Analysis

Assume the decision fails. Work backward to identify causes.

- Prompt translation: "Before finalizing your recommendation, assume this decision fails catastrophically in 12 months. What caused the failure? Address each potential cause in your analysis."

3. Rule Out Worst Case (ROWS)

Ensure the worst-case scenario is explicitly considered before committing to a conclusion.

- Prompt translation: "Before recommending approval, explicitly identify and address the worst-case outcome. What happens if everything goes wrong? Is the downside survivable?"

4. Stopping Rules

Define when sufficient information exists for a decision, preventing both premature closure and analysis paralysis.

- Prompt translation: "You must address all of the following before making a recommendation: [checklist]. If you cannot address any item, flag it as an information gap."

5. Structured Data Acquisition

Force systematic information gathering before forming conclusions.

- Prompt translation: "Before forming your recommendation, systematically gather information in these categories: [domain-specific categories]. Do not form a conclusion until all categories have been examined."

6. Personal Accountability / Decision Documentation

Performance improves when decision-makers must justify their conclusions.

- Prompt translation: "Your recommendation will be reviewed by other executives. You must provide specific evidence for every claim. Unsupported assertions will be challenged."

Key Research Finding on Effectiveness

The evidence is nuanced: **awareness of bias alone does not debias**. Forcing functions work because they create structural requirements that bypass the need for self-awareness. This is GOOD NEWS for AI agents -- you can bake the forcing function into the prompt structure itself, making it impossible to skip.

The most effective approaches are:

1. **Reflection on initial hypothesis** (guided reflection) -- consistently improved accuracy in difficult cases
2. **Checklists** -- proven in aviation and medicine; reduced errors and complications
3. **Consider-the-opposite** -- reduced judgment biases in experimental settings
4. **Group decision-making with accountability** -- collective wisdom exceeds individual reasoning

Application to the Team of Teams Prompt Architecture

The cognitive forcing strategy translates directly to the prompt design:

- **Each C-suite agent's system prompt should include domain-specific forcing functions.** The CFO should have: "Always identify hidden costs, second-order costs, and opportunity costs." The CISO should have: "Always assume the worst-case security scenario and work backward."

- **The structure of the cascade IS a forcing function.** By requiring each team lead to address a specific analytical question (not just "what do you think?"), the cascade forces structured data acquisition.
 - **The decision record IS an accountability mechanism.** Knowing that recommendations are documented and reviewed by the CEO creates structural accountability.
-

Part 2: Role-Based Prompt Architecture

The Persona Prompting Paradox

Research reveals a critical nuance: **simple role assignments ("You are a CFO") do NOT reliably improve output quality**, but **detailed, structured expert personas DO**.

Key findings:

- Basic personas ("You are a mathematician"): **No measurable improvement** on accuracy tasks
- Enhanced personas (with task context): Modest 5-10% gains
- Expert-generated personas (via structured templates): **Significant improvements**
- In one striking experiment, an "idiot persona outperformed the genius one" on MMLU benchmarks

What this means for the spec: "You are the CFO" is not enough. Each role needs a detailed, structured system prompt that includes:

1. Domain expertise description (what the role knows deeply)
2. Decision-making perspective (how they evaluate issues)
3. Natural biases/priorities (what they weight most heavily)
4. Cognitive forcing functions (what they must always check)
5. Output format requirements (what their analysis must include)

What Makes an Effective Expert Persona Prompt

Research identifies three critical attributes:

1. **Specific:** Domain-aligned and narrowly tailored. "Senior CFO with 20 years of experience in mid-market technology services companies specializing in M&A due diligence" beats "You are a CFO."
2. **Detailed:** Include background, expertise areas, decision-making style, known blind spots, and analytical framework. More detail = better differentiation between agents.
3. **LLM-Generated Expert Descriptions Outperform Human-Written Ones:** The ExpertPrompting technique (having the LLM generate a detailed expert persona based on the task) consistently outperformed human-crafted personas. Implication: Consider having each agent's detailed persona be generated/refined by the LLM itself during initialization.

Multi-Persona Approaches (Solo Performance Prompting)

SPP -- having an LLM dynamically generate and coordinate multiple expert personas -- outperformed single-persona and Chain-of-Thought approaches:

- +23% on knowledge-intensive tasks (trivia)

- Superior on mixed knowledge/reasoning tasks
- Largest improvement on reasoning-intensive tasks

Key finding: **Dynamic persona generation outperformed pre-defined personas** in all experiments. For the Team of Teams skill, this suggests the team lead agents could potentially generate their team leads' personas dynamically based on the specific issue, rather than using only static definitions.

The Honest Truth About Persona Prompting Limits

- Persona prompting steers **tone, structure, and perspective** more reliably than **factual accuracy**
- An LLM can "sound like" a legal expert but still confidently misstate law
- For factual grounding, persona prompting must be combined with RAG (retrieval) or structured data access
- This matters for the Team of Teams skill: **domain knowledge should be encoded in the prompt structure (checklists, frameworks, questions to ask), not just in the persona description**

Recommended Prompt Architecture Pattern

Based on all findings, each agent's system prompt should follow this architecture:

1. **ROLE IDENTITY** (Detailed, specific, with experience context)
"You are [Name], [Title] at [Company]. You have [X years] experience in [specific domain]. Your expertise includes [specific areas]."
2. **DECISION PERSPECTIVE** (What this role cares about)
"Your primary responsibility is to evaluate all issues from the perspective of [domain]. You prioritize [specific priorities] and are naturally skeptical of [specific concerns]."
3. **COGNITIVE FORCING FUNCTIONS** (Domain-specific structured checks)
"Before forming any recommendation, you **MUST**:
 - [Forcing function 1: domain-specific check]
 - [Forcing function 2: domain-specific check]
 - [Forcing function 3: consider-the-opposite]
 - [Forcing function 4: worst-case analysis]"
4. **TEAM MANAGEMENT** (For C-suite agents only)
"You manage the following team leads: [list]. When evaluating an issue, delegate specific analytical questions to each team lead based on their expertise."
5. **OUTPUT FORMAT** (Structured recommendation template)
"Your recommendation **MUST** include:
 - Position: [For/Against/Conditional]
 - Key findings: [bulleted list]
 - Risks identified: [with severity ratings]
 - Dissenting considerations: [what argues against your position]
 - Confidence level: [High/Medium/Low with justification]"
6. **ACCOUNTABILITY**
"Your analysis will be reviewed by the CEO alongside recommendations from other executives. Provide specific evidence for every claim."

This six-layer architecture combines the three most effective techniques: detailed expert personas, cognitive forcing functions, and structured output requirements.

Part 3: Biases Most Relevant to Organizational Decision-Making

For the Team of Teams skill, these are the specific biases the cognitive forcing functions should counter:

Bias	What It Does	Which Roles Are Most Susceptible	Forcing Function
Anchoring	Over-weighting first piece of information	CEO (anchors to issue framing), All	"Consider what would change if the initial assumption is wrong"
Confirmation Bias	Seeking evidence that supports existing view	VP Sales (optimistic by nature), CTO (favors tech solutions)	"Identify three pieces of evidence against your recommendation"
Groupthink	Suppressing dissent for harmony	All C-suite in deliberation phase	Structural dissent via adversarial roles (CISO, CFO)
Availability Bias	Over-weighting recent/vivid events	CISO (overweights recent breaches), All	"Is your assessment based on base rates or recent events?"
Sunk Cost Fallacy	Continuing bad investment because of past spending	CFO, CTO	"Evaluate this decision as if starting from zero today"
Optimism Bias	Underestimating risks, overestimating benefits	VP Sales, CTO	CFO/CISO as structural counterweights
Premature Closure	Stopping analysis too early	All team leads	Checklists requiring all categories addressed

Key Takeaways

- **"You are a CFO" is not enough.** Simple role assignments do not improve LLM output quality. Each agent needs a detailed, structured six-layer prompt: identity, perspective, forcing functions, team management, output format, and accountability.
- **Cognitive forcing functions should be structural, not optional.** Bake them into the prompt as MUST-do requirements. The most effective: consider-the-opposite, pre-mortem, worst-case analysis, and domain-specific checklists.
- **The cascade structure itself IS a cognitive forcing function** -- it forces systematic information gathering across all domains before any conclusion is formed. This is the skill's core value proposition.
- **Domain knowledge should be encoded as checklists and analytical frameworks, not just persona descriptions.** "Always check for tax implications" is more reliable than "you are an expert tax accountant who would notice tax implications."
- **Dynamic persona refinement may be valuable.** Research shows LLM-generated expert personas outperform static ones. Consider having C-suite agents dynamically generate team lead personas based on the specific issue.

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Research: Cross-Functional Agent Debate -- Quality vs. Cost Tradeoffs

Requested by: Free-thinker's Phase 4.5 question (should C-suite agents challenge each other before CEO synthesis?)

Date: 2026-02-22

Question

Should the design include a cross-functional deliberation phase where C-suite agents debate each other's recommendations? What does research say about the quality improvement vs. complexity/cost tradeoff of multi-agent debate rounds?

Findings

The Surprising Research Consensus: More Rounds Hurt, More Agents Help

Recent research (2025-2026) on multi-agent debate (MAD) in LLM systems reveals counterintuitive findings that directly affect the Phase 4.5 design question:

Finding 1: More discussion rounds REDUCE performance.

Increasing the number of debate rounds between agents does not reliably improve accuracy. Researchers attribute this to "problem drift" -- agents gradually drift from the original task during extended exchanges. This finding contradicts the intuition that more debate = better decisions.

Finding 2: More agents (horizontal scaling) IMPROVES performance.

Increasing the number of participating agents shows a consistent upward trend in accuracy. The recommendation: **scale horizontally (add agents) rather than vertically (extend discussions).**

Finding 3: Token cost scales dramatically with debate rounds.

- 3 agents, 5 rounds: **101x token cost** for a potential accuracy improvement from 50% to 98% on simple arithmetic
- 4 agents, 5 rounds: **90x token cost** for an improvement from 76% to 88% on GSM8K math
- Most MAD methods fail to outperform simpler single-agent techniques like Self-Consistency (which simply resamples)

Finding 4: Independent drafting before interaction is critical.

Two methods that enforce independent initial analysis BEFORE any cross-agent communication significantly improve quality:

- "All-Agents Drafting" (forces independent solutions first): +3.3% improvement
- "Collective Improvement" (restricts communication, shows solutions each round): +7.4% improvement

Both prevent agents from immediately converging on the first proposal they see.

Finding 5: Heterogeneous agents produce the best results.

Mixing different model types (e.g., GPT-4o-mini with Llama 3.1-70b) produces significant improvements over homogeneous agents. In the Team of Teams context, this means using different model tiers for different roles could actually improve decision quality.

Finding 6: Consensus works for knowledge tasks; Voting works for reasoning tasks.

- Consensus protocols improve knowledge tasks by 2.8% (error-checking, hallucination reduction)
- Voting protocols improve reasoning tasks by 13.2% (exploring multiple reasoning paths)

What This Means for Phase 4.5

The research strongly suggests the following design:

DO: Independent Analysis First (Already in the Cascade)

The existing cascade design (Phases 1-4) already implements the most important pattern: **each C-suite agent produces an independent analysis BEFORE seeing others' work**. This is equivalent to "All-Agents Drafting" and is the single most important factor for decision quality.

DO: One Structured Challenge Round (Not Extended Debate)

If there is a cross-functional phase, it should be **exactly one round** -- not an extended back-and-forth debate. Research shows:

- One round of cross-critique captures most of the value
- Additional rounds cause problem drift and reduce quality
- The challenge phase should be structured (specific questions to answer about each other's analysis) not open-ended debate

DO: Make the Challenge Phase Targeted, Not Universal

Not every C-suite agent needs to challenge every other agent. Use the natural tension pairings:

- CFO reviews VP Sales' revenue projections (cost reality check)
- CISO reviews CTO's technical proposal (security challenge)
- VP Delivery reviews VP Sales' promises (capacity check)
- COO reviews overall feasibility

This is McChrystal's "fusion cell" -- focused cross-functional interaction, not a free-for-all.

DON'T: Allow Extended Multi-Round Debate

Multiple rounds of C-suite agents arguing back and forth will:

- Cost 90-100x more tokens
- Cause problem drift (agents forget the original question)
- Likely underperform the simpler approach of independent analysis + CEO synthesis
- Risk "consensus mush" where agents converge to agreeable positions

DON'T: Use Consensus as the Decision Protocol

The CEO should NOT seek consensus among C-suite agents. Research shows:

- For reasoning-heavy decisions (strategy, trade-offs): **Voting/weighting is better** than consensus
- Consensus causes agents to suppress genuine disagreement
- The CEO's job is to synthesize competing perspectives, not find the middle ground

Recommended Phase 4.5 Design

Based on the research, here is the recommended cross-functional interaction design:

```
Phase 4: C-Suite Upward Synthesis (each agent produces independent recommendation)
|
Phase 4.5: Targeted Cross-Critique (ONE round, structured)
| - Each C-suite agent receives a SUMMARY of other agents' recommendations
| - Each agent answers: "What risks or blind spots do you see in the other
recommendations?"
| - Each agent can REVISE their own recommendation based on what they learned
| - NO back-and-forth debate -- one round only
|
Phase 5: CEO Deliberation (receives original + revised recommendations)
```

Key implementation details:

- C-suite agents see **summaries** of other recommendations, not full analyses (reduces token cost)
- The critique question is **structured**: "What did the other departments miss?" not "do you agree?"
- Agents can **revise** their own recommendation after seeing others' work (captures the value of cross-pollination)
- The CEO receives **both** the original and revised recommendations (can see what changed and why)

Token Cost Estimate for Phase 4.5

Without Phase 4.5: 7 independent C-suite analyses + CEO synthesis = ~8 agent turns

With Phase 4.5: 7 analyses + 7 cross-critiques + 7 revisions + CEO synthesis = ~22 agent turns

This roughly triples the C-suite layer cost. Whether this is worthwhile depends on whether the user opts for "thorough" vs. "quick" analysis mode. The design should make Phase 4.5 **optional** -- a configuration toggle.

Key Takeaways

- **The existing cascade design (independent analysis first) is already implementing the highest-value pattern** from multi-agent research. Don't lose this.
- **One structured cross-critique round is valuable; extended debate is harmful.** Problem drift and token cost make multi-round debate counterproductive.
- **Phase 4.5 should be optional** -- a "thorough mode" toggle. Quick mode skips it; thorough mode includes one round of targeted cross-critique.
- **The CEO should use weighted synthesis, not consensus.** Suppress genuine disagreement and you lose the whole point.

- **Heterogeneous model tiers (Haiku for team leads, Sonnet for C-suite, Opus for CEO) could actually improve decision quality**, not just save cost. Research shows model diversity helps.

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Research: Implementation Viability & Routing Mechanism

Requested by: Free-thinker's questions on roster size viability and CEO routing mechanism

Date: 2026-02-22

Questions

1. Is 37 separate agent instances viable, or should team leads be sequential calls within a single C-suite agent?
2. Should CEO-level routing (selective team activation) be a core feature or v2?

Findings

Part 1: Implementation Viability -- 37 Agents

The Hard Numbers

From the official Claude Code cost documentation:

- Average cost: **\$6/developer/day**, \$100-200/developer/month with Sonnet
- Agent teams use approximately **7x more tokens** than standard sessions
- Token usage scales linearly with number of active teammates
- Each teammate maintains its own context window as a separate Claude instance

The Math on Full Deployment (37 Agents Simultaneously)

If a single agent interaction costs ~\$0.50-1.00 in tokens:

- 37 simultaneous agents = **\$18.50-37.00 per decision** (rough estimate)
- With Opus for CEO + Sonnet for C-suite + Haiku for team leads, this drops significantly
- Haiku is ~10x cheaper than Opus, Sonnet is ~3x cheaper than Opus

Key cost optimization from docs: "Teams using economic orchestration report 40-60% cost reductions compared to uniform subagent deployment, while maintaining or improving output quality."

The Architectural Constraint

From the Claude Code Agent Teams documentation (my earlier research):

- **No nested teams:** Teammates cannot spawn their own teams
- **One team per session:** A lead can only manage one team at a time
- **Subagents cannot spawn subagents:** No recursive delegation

This means a literal three-tier agent hierarchy (CEO team -> C-suite teams -> team leads) is **not possible** with current Agent Teams. The architecture must work within a flat structure.

Three Viable Implementation Patterns

Pattern A: Flat Agent Team (All 37 as teammates)

- CEO is team lead, all 36 others are teammates
- Technically possible but impractical: 36 simultaneous Claude instances
- Massive token cost, coordination overhead
- Loses the hierarchical cascade (C-suite doesn't "manage" team leads)
- **Verdict: Not recommended**

Pattern B: Staged Execution (Recommended for v1)

- The skill runs as a **single agent session** using **sequential subagent calls**
- CEO agent is the main session
- Each C-suite member is invoked as a **subagent** (one at a time or in parallel batches)
- Within each C-suite subagent call, the system prompt includes team lead perspectives as **structured analysis sections** rather than separate agents
- Team lead "analysis" is generated by the C-suite subagent using its system prompt's domain framework and checklists

How this works in practice:

```
Main session (CEO agent, Opus)
├─ Subagent: C00 (Sonnet) -- generates C00 + 5 team lead analyses
├─ Subagent: CF0 (Sonnet) -- generates CF0 + 5 team lead analyses
├─ Subagent: CT0 (Sonnet) -- generates CT0 + 4 team lead analyses
├─ Subagent: CIS0 (Sonnet) -- generates CIS0 + 4 team lead analyses
├─ Subagent: VP Sales (Sonnet) -- generates VP Sales + 4 team lead analyses
├─ Subagent: VP Delivery (Sonnet) -- generates VP Delivery + 4 team lead analyses
└─ Subagent: CA0 (Sonnet) -- generates CA0 + 4 team lead analyses
```

- **7 subagent calls** (can run in parallel)
- Each C-suite subagent's system prompt includes team lead personas and analytical checklists
- The C-suite agent "simulates" the team lead consultation by working through each team lead's analytical framework sequentially within its own context
- CEO receives 7 structured reports and synthesizes

Advantages:

- Only 8 context windows (1 CEO + 7 subagents)
- Subagents can run in parallel (all 7 simultaneously)
- Each subagent's system prompt includes full team lead framework
- Prompt caching reduces cost for repeated system prompt content
- Summary returns to CEO context, keeping it clean
- Works within all current Claude Code constraints

Disadvantages:

- Team lead "analysis" is simulated by C-suite agent, not truly independent
- A single model generating both manager and report perspectives may produce less diversity than separate agents

- But research showed: "heterogeneous agents help" and "independent drafting is the most important pattern" -- the 7 C-suite subagents ARE independent of each other, which captures the primary benefit

Token cost estimate:

- 7 Sonnet subagent calls: ~\$3.50-7.00
- 1 Opus CEO synthesis: ~\$1.00-2.00
- Total per decision: ~**\$4.50-9.00** (vs. \$18-37 for 37 separate agents)
- With selective routing (3-5 C-suite): ~**\$2.50-5.00**

Pattern C: Hybrid Agent Team + Subagents (Recommended for v2)

- CEO is team lead of an Agent Team
- 7 C-suite members are Agent Team teammates (parallel, can communicate)
- Each C-suite teammate uses subagents for team lead analysis
- C-suite teammates can challenge each other via messaging (Phase 4.5)
- Requires the GitHub #24316 feature (custom agent definitions for teammates) for full effectiveness

Advantages:

- True inter-agent C-suite deliberation
- Each C-suite teammate can use subagents for team leads
- Wait -- subagents cannot spawn subagents

Problem: Agent team teammates are full Claude Code instances, but the docs don't specify whether teammates can spawn subagents. Based on the architecture:

- Subagents work "within a single session"
- Teammates ARE separate sessions
- So teammates SHOULD be able to spawn their own subagents

If teammates can spawn subagents, Pattern C becomes:

```
Agent Team Lead: CEO (Opus)
├── Teammate: C00 (Sonnet) --> spawns subagents for team leads (Haiku)
├── Teammate: CF0 (Sonnet) --> spawns subagents for team leads (Haiku)
└── ... (7 C-suite teammates total)
```

This would give true three-tier execution with inter-C-suite communication. But it requires:

1. Confirmation that teammates can spawn subagents (not documented)
2. Custom agent definitions for teammates (feature request #24316)
3. Significantly higher token budget

Verdict: v2 aspiration, not v1 design

Recommendation

v1: Pattern B (Staged Execution with Subagents)

- CEO main session + 7 C-suite subagents (parallel)
- Team lead analysis simulated within each C-suite subagent's prompt
- Selective routing reduces to 3-5 subagents for most decisions
- Cost: ~\$2.50-9.00 per decision depending on depth and routing

v2: Pattern C (Hybrid) if constraints are lifted

- Agent team for C-suite inter-communication
 - Subagents within teammates for team leads
 - Full three-tier with cross-functional deliberation
-

Part 2: CEO Routing Mechanism

Why Routing Is Core, Not v2

The math is clear: every C-suite subagent call costs real tokens. Without routing:

- Every question, no matter how small, triggers 7 subagent calls + team lead analysis
- A "should we switch project management tools?" gets the same treatment as "should we acquire CompetitorX?"
- This is both expensive and produces unnecessary noise

With routing:

- CEO classifies the decision and activates only relevant C-suite members
- A PM tool switch might only need COO + CTO + CFO (3 calls instead of 7)
- Token savings: 40-60% on most decisions

How Routing Works in Multi-Agent Systems

Research on adaptive routing in multi-agent systems shows:

- **Task classification as first step:** An intent classifier matches incoming requests to appropriate specialist agents
- **Two-layer architecture:** Adaptive controllers predict task parameters and select a relevant subset of agents; tasks are "made visible only to a restricted group"
- **Multi-parameter routing:** Consider task complexity, priority, agent capabilities, and relevance
- **Reinforcement learning:** Routing policies evolve based on observed performance

Recommended Routing Design

The CEO's Phase 1 (Framing) should include a routing step:

Decision Classification:

- └─ Decision Type: [Strategic | Financial | Operational | Technical | Security | Personnel | Mixed]
- └─ Scope: [Company-wide | Department | Team | Individual]
- └─ Urgency: [Critical | Standard | Exploratory]
- └─ Complexity: [Simple | Moderate | Complex | Transformative]
- └─ Analysis Depth: [Quick (3 agents) | Standard (5 agents) | Thorough (7 agents)]

Routing Rules (examples):

Decision Type	Always Activate	Conditionally Activate	Typically Skip
Financial (investment, acquisition, budget)	CFO, COO, CEO	CTO, VP Sales, VP Delivery	CISO (unless IT/data related), CAO
Technical (build/buy, architecture, tools)	CTO, CISO, CFO	COO, VP Delivery	VP Sales, CAO
Security (breach response, compliance, policy)	CISO, CTO, CAO	CFO, COO	VP Sales, VP Delivery
Strategic (new market, acquisition, pivot)	ALL	--	--
Operational (process change, tool adoption)	COO, CFO	CTO, VP Delivery	CISO, VP Sales, CAO
Personnel (hiring, reorg, culture)	CAO, COO, CFO	VP Delivery, VP Sales	CTO, CISO
Sales/Growth (pricing, market entry, partnerships)	VP Sales, CFO, VP Delivery	CTO, COO	CISO, CAO

The user should be able to override routing. If they want all 7 perspectives on a simple question, they should be able to request "thorough" mode. And if they want a quick take from just the CFO and CISO, they should be able to specify that.

Routing Implementation in the Skill

The routing mechanism should be:

1. **Automatic by default:** CEO classifies and routes based on decision type
2. **User-overridable:** User can specify which roles to consult
3. **Transparent:** The decision record shows which roles were consulted and why others were excluded
4. **Depth-selectable:** Quick (3 agents) / Standard (5 agents) / Thorough (all 7)

Key Takeaways

- **37 simultaneous agents is not viable in v1.** The recommended architecture is Pattern B: CEO main session + 7 C-suite subagents, with team lead analysis simulated within each subagent's system prompt via structured analytical frameworks and checklists.
- **This pattern costs ~\$2.50-9.00 per decision** depending on routing and depth, vs. \$18-37 for 37 separate agents.
- **CEO routing is a CORE feature, not v2.** Without it, every question gets the same expensive treatment. Routing reduces cost 40-60% for most decisions and produces less noise.
- **The routing mechanism classifies by decision type, scope, urgency, and complexity** to determine which C-suite members to activate and at what depth.
- **v2 aspiration: Pattern C (Agent Team + Subagents)** would enable true inter-C-suite communication but requires features not yet available.

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Research: Multi-Agent Orchestration Patterns

Requested by: Arbiter (initial research assignment)

Date: 2026-02-22

Question

Are there existing multi-agent systems or frameworks that simulate organizational structures for decision-making? What approaches have been tried? What works, what doesn't?

Findings

Market Context (2025-2026)

The multi-agent AI field is experiencing explosive growth. Gartner reported a 1,445% surge in multi-agent system inquiries from Q1 2024 to Q2 2025. The autonomous AI agent market is projected to reach \$8.5 billion by 2026 and potentially \$45 billion by 2030. IBM research shows multi-agent orchestration reduces hand-offs by 45% and boosts decision speed by 3x. The field is undergoing its "microservices revolution" -- single all-purpose agents being replaced by orchestrated teams of specialized agents.

Key Frameworks and Their Approaches

1. CrewAI -- Role-Based Organizational Model

Architecture: Built around the metaphor of a real-world organization. Each agent has a defined role, goals, backstory, and expertise.

Key Features:

- Role-specific memory (short-term, long-term, entity, contextual)
- Sequential and hierarchical workflow support
- Fortune 500 adoption (DocuSign, PwC)
- Role-driven collaboration where agents operate with clearly defined responsibilities

Organizational Simulation Strength: Closest to the "Team of Teams" concept. Agents are defined with roles like "Senior Researcher," "Technical Writer," etc. Each maintains role-specific memory and context.

Limitations: Role definitions are primarily prompt-driven; no enforcement of decision domains at the tool level.

2. MetaGPT -- Software Company Simulation

Architecture: Simulates a complete software company by assigning specialized roles to AI agents.

Key Features:

- Pre-defined roles: Product Manager, Architect, Engineer, QA
- Transforms natural language requirements into comprehensive software projects
- Produces code, documentation, and tests through role-based collaboration

Organizational Simulation Strength: Most explicit about simulating a complete organizational structure. Each role has defined inputs, outputs, and interactions.

Limitations: Highly specialized for software development workflows. Not easily adaptable to general business decision-making. Role set is fixed.

3. AutoGen (Microsoft) -- Conversational Collaboration

Architecture: Event-driven asynchronous multi-agent conversations with human-in-the-loop support.

Key Features:

- Dynamic conversation patterns
- Enterprise-level scalability
- Used to power intelligent meeting facilitators
- Live coding assistant support

Organizational Simulation Strength: Strong at modeling deliberative conversations between agents. Good for scenarios where agents need to debate and reach consensus.

Limitations: Less structured role definition compared to CrewAI. Conversation can become circular without strong orchestration.

4. LangGraph -- Graph-Based Workflow Orchestration

Architecture: Stateful, branching, and complex multi-agent workflows using graph structures.

Key Features:

- Explicit control of dependencies and message flows
- Fastest framework with lowest latency
- Fine-grained state management

Organizational Simulation Strength: Best for defining explicit decision flows and approval chains. Can model organizational processes precisely.

Limitations: More workflow-oriented than role-oriented. Better at encoding processes than personas.

Common Orchestration Patterns

1. **Hierarchical Routing:** Central orchestrator routes tasks to expert agents. Maps to CEO distributing issues to C-suite. Most common in enterprise deployments.
2. **Modular Specialization:** Isolated agents handle specific domains independently. Maps to department heads owning their domains.
3. **Message-Based Communication:** Agents exchange information asynchronously using pub/sub patterns. Maps to organizational communication channels.

4. **Task Routing via Intent Classification:** Agent registries match incoming requests to appropriate specialists. Maps to organizational triage processes.

What Works

- **Clear role definition:** Agents with specific, well-bounded domains outperform generalist agents
- **Structured communication:** Defined interaction patterns (who talks to whom, when) prevent chaos
- **Hierarchical coordination with local autonomy:** Central planning + decentralized execution (mirrors McChrystal's model)
- **Memory and context management:** Role-specific memory enables agents to maintain domain expertise across interactions

What Doesn't Work

- **Over-communication:** Broadcasting everything to every agent wastes tokens and creates noise
- **Unbounded autonomy:** Agents without clear decision boundaries produce inconsistent results
- **Prompt-only role enforcement:** Relying solely on natural language instructions for role compliance is fragile
- **Deep nesting:** Multi-level agent hierarchies create latency and context loss
- **Universal solutions:** Multi-agent architectures that try to solve all problems equally fail; they must be designed for specific organizational patterns

Autonomy Spectrum

A progressive "autonomy spectrum" is emerging for human-AI interaction:

- **Human in the loop:** Human approves every decision
- **Human on the loop:** Human monitors and can intervene
- **Human out of the loop:** Fully autonomous execution

This maps directly to the Team of Teams model's progression from shared consciousness to empowered execution.

Key Takeaways

- **CrewAI's role-based model is the closest existing pattern** to what the Team of Teams skill would need, but it lacks the inter-team communication and hierarchical depth required.
- **No existing framework explicitly models a multi-level org structure** (CEO -> C-suite -> team leads) for business decision-making. This would be novel.
- **The sequencing pattern (shared consciousness -> empowered execution) maps well** to multi-agent orchestration: broadcast context first, then let agents reason independently and report back.
- **Token cost is the primary constraint** for multi-level agent hierarchies. Each additional agent layer multiplies cost linearly. The concept's three-level structure (CEO -> 7 C-suite -> ~35 team leads) would need careful optimization.

- **Hybrid approaches work best:** Combine role-based personas (prompt-driven) with structural constraints (tool restrictions, communication patterns, decision flows) for reliable organizational simulation.

Sources

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Research: Complete Org Roster & Decision Record Format

Requested by: Grounder (explicit request for full team lead roster and decision record format)

Date: 2026-02-22

Question

1. What are the complete team lead rosters for all 7 C-suite/VP roles in a mid-market IT/technology services company?
 2. What should the decision record format look like as the primary deliverable of the skill?
-

Part 1: Complete Organizational Roster

Design Assumptions

- Company type: Mid-market IT/technology services company (200-500 employees)
- CEO has 7 direct reports: COO, CFO, CAO, CTO, CISO, VP of Sales, VP of Delivery
- Each direct report has 4-6 team leads performing key functions
- Total roles: 1 CEO + 7 C-suite + ~35 team leads = ~43 agents

The Full Roster

CEO (Chief Executive Officer)

Mandate: Strategic vision, final decision authority, stakeholder management

Disposition: Integrative synthesizer -- weighs all perspectives, identifies fault lines, resolves conflicts

Direct Reports: COO, CFO, CAO, CTO, CISO, VP of Sales, VP of Delivery

COO (Chief Operating Officer)

Mandate: Day-to-day operations, process optimization, cross-department execution

Disposition: EXECUTION REALIST -- "Can we actually do this with the people and processes we have?"

Dissent type: Mandated skeptic (operational feasibility)

#	Team Lead Role	Domain	Key Analytical Questions
1	Operations Manager	Process efficiency, workflow, resource utilization	"How does this affect our current operational workflows? What processes need to change?"
2	Project/Program Management Lead	Project delivery, timelines, resource allocation	"What does the implementation timeline look like? What resources get pulled from current projects?"
3	Process Improvement / Lean Lead	Continuous improvement, waste reduction, efficiency metrics	"Where are the inefficiencies? How do we measure success? What gets simpler or more complex?"
4	Quality Assurance Manager	Quality standards, SLA compliance, service quality metrics	"Does this affect our quality standards? What are the quality risks during transition?"
5	Vendor/Procurement Manager	Third-party relationships, supply chain, vendor management	"What vendor dependencies does this create? What are the procurement implications?"

CFO (Chief Financial Officer)

Mandate: Financial strategy, budgeting, risk assessment, capital allocation

Disposition: COST ARCHAEOLOGIST -- "What are the costs that aren't in the proposal?"

Dissent type: Mandated skeptic (financial risk, hidden costs)

#	Team Lead Role	Domain	Key Analytical Questions
1	Controller	Accounting, financial reporting, GAAP compliance, month-end close	"What's the accounting treatment? How does this hit the books? What are the GAAP implications?"
2	Head of FP&A	Financial planning & analysis, forecasting, scenario modeling	"What does this do to our 12-18 month forecast? What are the best/base/worst case scenarios?"
3	Treasury / Cash Manager	Cash flow, liquidity, banking relationships, working capital	"Can we fund this? What's the cash flow impact? Do we need external financing?"
4	AP/AR Manager	Accounts payable/receivable, payment terms, collections	"What are the payment terms? What do we inherit? How does this affect DSO/DPO?"
5	Tax Lead	Tax strategy, compliance, structure optimization (+ external firm)	"What are the tax implications? Is there a more tax-efficient structure? Any compliance triggers?"

CAO (Chief Administrative Officer)

Mandate: Administrative backbone -- HR, legal, compliance, facilities, organizational policy

Disposition: ORGANIZATIONAL CAPACITY GUARDIAN -- "Do we have the organizational capacity for this? What about our people?"

Dissent type: Neutral-to-skeptic (people impact, compliance, organizational readiness)

#	Team Lead Role	Domain	Key Analytical Questions
1	HR Director	Talent management, recruiting, retention, culture, compensation	"Do we have the right people? What talent do we need? How does this affect morale and culture?"
2	Legal Counsel / Compliance Lead	Contracts, regulatory compliance, legal risk, IP	"What are the legal risks? Contractual obligations? Regulatory requirements? IP considerations?"
3	Facilities / Office Manager	Physical workspace, equipment, remote work infrastructure	"What facilities changes are needed? What equipment? What's the physical footprint impact?"
4	Training & Development Lead	Employee training, skill development, onboarding, change management	"What training is required? How do we manage the change? What's the ramp-up timeline?"
5	Policy & Governance Lead	Internal policies, governance frameworks, corporate governance	"Does this comply with our policies? What new policies are needed? What governance applies?"

CTO (Chief Technology Officer)

Mandate: Technology strategy, product development, system architecture, innovation

Disposition: CAPABILITY EXPANDER -- "What does this make possible that wasn't possible before?"

Dissent type: Advocacy (innovation, technical opportunity)

#	Team Lead Role	Domain	Key Analytical Questions
1	Engineering / Development Lead	Software development, code quality, technical implementation	"What's the technical effort? What needs to be built? How complex is the implementation?"
2	Infrastructure / DevOps Lead	Cloud, servers, CI/CD, deployment, uptime, scalability	"What infrastructure is needed? Can our systems handle this? What about scalability?"
3	Solutions Architect	System design, integration, technical feasibility, build-vs-buy	"How does this fit our architecture? Integration points? Build or buy? Technical debt impact?"
4	Data / Analytics Lead	Data strategy, analytics, BI, data governance, reporting	"What data do we need? What analytics capabilities? How do we measure outcomes?"
5	Product / Innovation Lead	Product roadmap, R&D, emerging technology evaluation	"How does this align with our product roadmap? What new capabilities does it enable?"

CISO (Chief Information Security Officer)

Mandate: Information security, risk assessment, incident response, compliance, data protection

Disposition: CONSTITUTIONAL SKEPTIC -- "Any change introduces risk. What could go wrong?"

Dissent type: Mandated skeptic (security risk, compliance, threat vectors)

#	Team Lead Role	Domain	Key Analytical Questions
1	Security Operations (SecOps) Lead	Threat monitoring, incident detection, SOC, vulnerability management	"What new attack surfaces does this create? What vulnerabilities are introduced?"
2	GRC (Governance, Risk & Compliance) Lead	Regulatory compliance, risk frameworks, audit, policy	"What compliance requirements apply? SOC2, HIPAA, GDPR implications? Audit impact?"
3	Identity & Access Management (IAM) Lead	Access control, authentication, authorization, privilege management	"Who gets access to what? How is authentication handled? What about privilege escalation?"
4	Application Security (AppSec) Lead	Secure development, code review, penetration testing, SDLC security	"Is the software secure? What vulnerabilities exist in the codebase? Third-party dependencies?"
5	Incident Response / Business Continuity Lead	Incident response planning, disaster recovery, business continuity	"What's the incident response plan? What happens if this fails? Business continuity impact?"

VP of Sales

Mandate: Revenue generation, customer acquisition, pipeline management, market expansion

Disposition: REVENUE OPTIMIST -- "How does this help us sell more, sell faster, or sell to new markets?"

Dissent type: Advocacy (growth, revenue opportunity, competitive advantage)

#	Team Lead Role	Domain	Key Analytical Questions
1	Sales Manager (Enterprise/Named Accounts)	Large deal management, enterprise relationships, strategic accounts	"How does this affect our enterprise pipeline? Can we upsell existing accounts?"
2	Sales Manager (Mid-Market/Growth)	Volume sales, new business development, market expansion	"What new market segments does this open? How does it affect our win rate?"
3	Business Development Lead	New markets, partnerships, channel strategy, lead generation	"What partnership opportunities exist? What channels can we leverage? Lead gen impact?"
4	Sales Operations / Enablement Lead	CRM, sales process, training, tools, quota management, analytics	"How do we enable the team to sell this? What tools/training are needed? Quota impact?"
5	Account Management / Customer Success Lead	Retention, expansion, NPS, customer health, renewals	"How does this affect existing customer relationships? Renewal risk? Expansion opportunity?"

VP of Delivery

Mandate: Service delivery, client satisfaction, project execution, SLA management

Disposition: COMMITMENT REALIST -- "Can we deliver what Sales is promising? What happens to current commitments?"

Dissent type: Mandated skeptic (delivery feasibility, client impact, capacity)

#	Team Lead Role	Domain	Key Analytical Questions
1	Delivery Manager (Client Projects)	Active project delivery, client-facing execution, milestone tracking	"How does this affect our current project commitments? What gets delayed?"
2	Client Success / Relationship Lead	Client satisfaction, escalation management, NPS, retention	"What's the impact on client satisfaction? Any clients at risk? Communication plan?"
3	Resource / Capacity Planning Lead	Staff allocation, utilization rates, bench management, contractor needs	"Do we have the capacity? What's our utilization rate? Do we need contractors?"
4	Technical Delivery / Solutions Lead	Technical implementation, architecture decisions in client work	"Is this technically deliverable? What's the implementation complexity for clients?"
5	Service Level / SLA Manager	SLA tracking, service quality metrics, performance reporting	"How does this affect our SLAs? What metrics change? Are we at risk of SLA breaches?"

Roster Summary

C-Suite Role	Disposition	Dissent Type	Team Lead Count
CEO	Integrative Synthesizer	N/A (decision-maker)	0 (manages C-suite)
COO	Execution Realist	Mandated skeptic	5
CFO	Cost Archaeologist	Mandated skeptic	5
CAO	Organizational Capacity Guardian	Neutral-to-skeptic	5
CTO	Capability Expander	Advocacy	5
CISO	Constitutional Skeptic	Mandated skeptic	5
VP of Sales	Revenue Optimist	Advocacy	5
VP of Delivery	Commitment Realist	Mandated skeptic	5
TOTAL			1 + 7 + 35 = 43

Dissent Balance

- **Mandated Skeptics (4):** COO, CFO, CISO, VP of Delivery -- these roles structurally challenge proposals
- **Advocacy Roles (2):** CTO, VP of Sales -- these roles structurally champion growth and capability
- **Neutral/Balanced (1):** CAO -- leans skeptic on organizational capacity but neutral on strategy

- **Synthesizer (1):** CEO -- integrates all perspectives

This 4:2:1 skeptic-to-advocate ratio ensures proposals are thoroughly stress-tested. Even genuinely good ideas must survive financial, operational, security, and delivery scrutiny before approval. This reflects real organizational dynamics where it's harder to get approval than to propose ideas.

Part 2: Decision Record Format

Design Rationale

The decision record is the primary deliverable the skill produces. It must:

1. Capture the full multi-perspective analysis (not just the final decision)
2. Show the reasoning chain (how the decision was reached, not just what was decided)
3. Preserve dissenting views (the CISO's objections don't disappear just because the CEO decided to proceed)
4. Be actionable (include next steps, conditions, and risk mitigations)
5. Be referenceable (support future decisions by showing precedent)

The format draws from three established templates:

- **Architecture Decision Records (ADR)** -- from software engineering, emphasis on context, decision, consequences
- **DACI Framework** -- from Atlassian, emphasis on roles (Driver, Approver, Contributors, Informed)
- **McChrystal Red Team Output** -- emphasis on adversarial analysis and competing hypotheses

Recommended Decision Record Format

```

# DECISION RECORD: [Short Title]

**Date:** [ISO date]
**Issue ID:** [Auto-generated]
**Status:** APPROVED | REJECTED | CONDITIONAL | DEFERRED
**Decision Type:** [Strategic | Financial | Operational | Technical | Security |
People]

---

## 1. ISSUE STATEMENT
_The question or proposal as presented to the CEO._

[Original issue text from the user]

## 2. CEO FRAMING
_How the CEO decomposed and framed the issue for evaluation._

- **Strategic context:** [Why this matters to the company now]
- **Dimensions evaluated:** [Which domains were asked to evaluate]
- **Decision type classification:** [What kind of decision this is]

## 3. EXECUTIVE ANALYSIS

### 3.1 COO Assessment – Operational Feasibility
**Position:** [Support | Oppose | Conditional]
**Confidence:** [High | Medium | Low]

**Key Findings:**
- [Finding 1]
- [Finding 2]

**Team Lead Inputs:**
| Team Lead | Finding | Impact |
|-----|-----|-----|
| Operations Manager | [finding] | [impact] |
| PM Lead | [finding] | [impact] |
| ... | ... | ... |

**Risks Identified:**
- [Risk 1 with severity]
- [Risk 2 with severity]

---

### 3.2 CFO Assessment – Financial Viability
**Position:** [Support | Oppose | Conditional]
**Confidence:** [High | Medium | Low]

[Same structure as above]

---

```

3.3 CAO Assessment – Organizational Readiness

[Same structure]

3.4 CTO Assessment – Technical Opportunity

[Same structure]

3.5 CISO Assessment – Security & Risk

[Same structure]

3.6 VP Sales Assessment – Revenue Impact

[Same structure]

3.7 VP Delivery Assessment – Delivery Capacity

[Same structure]

4. FAULT LINE ANALYSIS

Where the executive recommendations diverged and why.

Areas of Consensus

- [What most/all executives agreed on]

Key Disagreements

Issue	For	Against	Core Tension
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[Disagreement 1]	[Who supports]	[Who opposes]	[Why they disagree]
[Disagreement 2]	[Who supports]	[Who opposes]	[Why they disagree]

Strongest Objections

1. **[Objection from ROLE]:** [The strongest argument against]

- **Mitigation proposed:** [How it could be addressed]

2. **[Objection from ROLE]:** [Second strongest argument against]

- **Mitigation proposed:** [How it could be addressed]

5. RED TEAM / PRE-MORTEM

"Assume this decision fails in 12 months. What caused the failure?"

Failure Scenario	Identified By	Likelihood	Severity	Mitigation
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[Scenario 1]	[Role]	[H/M/L]	[H/M/L]	[Action]
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[Scenario 2]	[Role]	[H/M/L]	[H/M/L]	[Action]
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6. CEO DECISION

Decision

[APPROVED / REJECTED / CONDITIONAL / DEFERRED]

[Clear statement of what was decided]

```

### Rationale
[Why this decision was made, addressing the key fault lines and strongest objections]

### Weighting Applied
| Executive | Weight | Reason |
|-----|-----|-----|
| [Role most weighted] | High | [This is primarily a X decision] |
| [Role moderately weighted] | Medium | [Relevant but secondary] |
| [Role less weighted] | Standard | [Considered but not primary] |

### Conditions & Guardrails
- [Condition 1: "Proceed only if..."]
- [Condition 2: "Must be completed before..."]
- [Guardrail 1: "Do not exceed..."]

### Risk Mitigations Adopted
- [Mitigation 1 from ROLE's objection]
- [Mitigation 2 from ROLE's objection]

### Dissenting Views Preserved
_The following executives maintained objections that the CEO acknowledged but did not
adopt:_
- **[ROLE]**: [Their unresolved concern and why the CEO proceeded anyway]

---

## 7. ACTION ITEMS & NEXT STEPS

| # | Action | Owner | Deadline | Dependencies |
|---|-----|-----|-----|-----|
| 1 | [Action item] | [Role] | [Date] | [Any blockers] |
| 2 | [Action item] | [Role] | [Date] | [Any blockers] |

---

## 8. METADATA

**Analysis depth:** [Quick | Standard | Thorough]
**Executives consulted:** [List of 7 or subset]
**Team leads consulted:** [Count]
**Cross-functional challenge phase:** [Yes/No]
**Total analysis time:** [Elapsed time]
**Token usage:** [If tracked]

```

Key Design Choices in This Format

1. **Section 3 preserves individual executive analysis** -- the user can read any single executive's assessment independently. This is the "shared consciousness" made visible.
2. **Section 4 (Fault Line Analysis) is unique** -- it doesn't just list recommendations; it identifies WHERE and WHY executives disagreed. This is the highest-value section for the user because it reveals the trade-offs that aren't obvious from any single perspective.

3. **Section 5 (Red Team / Pre-Mortem) is mandatory** -- every decision gets at least a basic pre-mortem. This is the cognitive forcing function that prevents groupthink even when most executives agree.
4. **Section 6 preserves dissenting views** -- even in an APPROVED decision, the CISO's security concerns or the CFO's cost objections are recorded. This serves two purposes: (a) it signals to the user that these concerns exist and may need monitoring, (b) it creates an audit trail for future reference.
5. **Section 6 shows weighting** -- the user can see HOW the CEO weighted different perspectives, making the synthesis transparent rather than opaque.
6. **Metadata tracks analysis depth** -- so the user knows whether this was a quick assessment or a thorough multi-layer analysis.

Key Takeaways

- **The 43-role roster is now concrete** with specific team leads, domains, and analytical questions for each. This gives the builder agent exactly what it needs to write system prompts.
- **The dissent balance (4 skeptics : 2 advocates : 1 neutral : 1 synthesizer) ensures structural stress-testing.** Every proposal faces financial, operational, security, and delivery scrutiny.
- **Each team lead has defined "Key Analytical Questions"** -- these translate directly into cognitive forcing functions in the system prompts. The builder agent can use these as checklist items.
- **The decision record has 8 sections** with the Fault Line Analysis (Section 4) and Pre-Mortem (Section 5) being the highest-value unique sections that differentiate this from standard decision documentation.
- **Dissenting views are preserved, not suppressed** -- this is a core design principle that reflects McChrystal's shared consciousness AND good organizational governance.

Sources

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10	Concept Seed	/Volumes/Data/dev/ideations/ideation-team-of-teams-20260222-054340/session/sources/team-of-teams.txt	Original CFO team lead examples

Citation Log

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Research: C-Suite Prompt Architecture & Adaptive Complexity Tiers

Requested by: Grounder's three threads (A: Prompt Architecture, C: Adaptive Complexity)

Date: 2026-02-22

Part 1: C-Suite Agent Prompt Architecture

The Problem

Each C-suite subagent's system prompt must accomplish five distinct tasks within a single invocation:

1. Establish domain identity and mandate
2. Receive and interpret the CEO's framed question
3. Decompose it into team-lead-level sub-questions
4. Cycle through each team lead perspective producing findings
5. Synthesize a domain recommendation with confidence and dissent

This is a multi-step structured reasoning task. Research shows the most effective approach is **explicit step-by-step decomposition within the prompt itself** -- not a vague "analyze this" but a structured internal process the agent follows.

Recommended Prompt Architecture: The Five-Section Agent Prompt

Based on prompt engineering research (Chain-of-Thought decomposition, structured output formatting, and the six-layer persona architecture from my earlier cognitive forcing research), here is the recommended prompt structure:

```

# SECTION 1: IDENTITY & MANDATE
# (Who you are, what you care about, your disposition)

You are [Name], [Title] at [Company Name].

## Your Domain
[Specific areas of expertise and responsibility]

## Your Mandate
[One-sentence mandate -- e.g., "Find the costs that aren't in the proposal."]

## Your Disposition
[Skeptic/Advocate/Neutral -- with specific description of what you naturally
prioritize, suspect, and champion]

## Your Team
You lead a team of [N] specialists:
- [Team Lead 1]: [Domain] -- [One-line mandate]
- [Team Lead 2]: [Domain] -- [One-line mandate]
- [Team Lead 3]: [Domain] -- [One-line mandate]
- [Team Lead 4]: [Domain] -- [One-line mandate]
- [Team Lead 5]: [Domain] -- [One-line mandate]

---

# SECTION 2: ANALYTICAL PROCESS
# (The step-by-step internal reasoning process you MUST follow)

When you receive an issue for evaluation, follow this process:

## Step 1: Domain Translation
Restate the issue in your domain's terms. What is the [financial/operational/
technical/security/etc.] question here? What are the specific [domain]
dimensions that need to be evaluated?

## Step 2: Team Lead Consultation
For each of your team leads, answer their key analytical question:

### [Team Lead 1 Title] Analysis
- Key question: [Their specific analytical question]
- Checklist: [Domain-specific items they MUST address]
- Finding: [What this specialist would conclude]

### [Team Lead 2 Title] Analysis
[Same structure]

### [Team Lead 3 Title] Analysis
[Same structure]

[...repeat for each team lead]

## Step 3: Domain Synthesis
Based on your team's findings, synthesize your domain recommendation.

```

Step 4: Cognitive Forcing Checks

Before finalizing your recommendation, you MUST complete these checks:

- [] Consider-the-Opposite: What are the 3 strongest arguments AGAINST your recommendation?
- [] Pre-Mortem: If this decision fails in 12 months, what caused it (from your domain's perspective)?
- [] Worst Case: What is the worst-case outcome in your domain?
- [] [Domain-specific forcing function]

SECTION 3: OUTPUT FORMAT

(The exact structure your response must follow)

Respond using this EXACT format:

DOMAIN ASSESSMENT: [Your Title]

Domain Translation

[How you interpret this issue through your domain lens]

Team Lead Findings

Team Lead	Key Finding	Impact	Severity
-----	-----	-----	-----
[TL1]	[finding]	[impact]	[H/M/L]
[TL2]	[finding]	[impact]	[H/M/L]
[TL3]	[finding]	[impact]	[H/M/L]
[TL4]	[finding]	[impact]	[H/M/L]

Domain Recommendation

Position: [SUPPORT / OPPOSE / CONDITIONAL]

Confidence: [HIGH / MEDIUM / LOW]

Summary: [2-3 sentence synthesis]

Key Risks

1. [Risk with severity rating]
2. [Risk with severity rating]

Key Opportunities

1. [Opportunity if applicable]

Dissenting Considerations

[Arguments against your own recommendation -- from the cognitive forcing checks]

Conditions (if CONDITIONAL)

- [What would need to be true for you to support this]

SECTION 4: DOMAIN-SPECIFIC KNOWLEDGE

(Reference material: frameworks, checklists, industry standards)

Analytical Frameworks

[Domain-specific frameworks this role uses -- e.g., for CFO: DCF analysis, break-even analysis, sensitivity analysis, payback period]

Standard Checklists

[Items this role always checks -- e.g., for CISO: OWASP Top 10, compliance requirements, data classification, access control matrix]

Common Pitfalls in This Domain

[What to watch for -- e.g., for CFO: sunk cost fallacy, optimism bias in revenue projections, hidden integration costs]

SECTION 5: ACCOUNTABILITY & CONTEXT

(Framing that improves output quality)

Your analysis will be reviewed by the CEO alongside assessments from [other C-suite roles]. The CEO will identify where domain recommendations diverge and why.

Provide specific evidence for every claim. Unsupported assertions will be challenged. If you lack information to assess something, flag it as an information gap rather than guessing.

Your assessment contributes to a formal decision record that will be preserved for future reference.

Why This Five-Section Structure Works

Research supports each design choice:

1. **Section 1 (Identity):** Detailed expert personas outperform simple role assignments. The mandate + disposition + team roster creates a rich, specific identity that drives differentiated output.
2. **Section 2 (Analytical Process):** Chain-of-Thought decomposition with explicit steps forces the LLM to work through each team lead perspective sequentially rather than jumping to a conclusion. This is the core mechanism for "thinking in 37 voices with 8 agents."
3. **Section 3 (Output Format):** Structured output templates dramatically improve consistency and make synthesis by the CEO agent tractable. Every C-suite agent produces identically formatted output, making cross-domain comparison possible.
4. **Section 4 (Domain Knowledge):** Checklists and frameworks encode domain expertise deterministically. "Always check for tax implications" is more reliable than relying on the persona to remember.
5. **Section 5 (Accountability):** Research shows performance improves when agents know their output will be reviewed and must be justified. This framing also provides context about the larger process.

Key Design Insight: The Process IS the Prompt

The most important section is Section 2 (Analytical Process). This is where the "team of teams" comes alive within a single agent. The prompt doesn't say "consider multiple perspectives" -- it says "work through Team Lead 1's checklist, then Team Lead 2's checklist, then synthesize." This structured decomposition is what produces genuinely multi-perspective analysis rather than a superficial nod to different viewpoints.

Token Budget Considerations

- This prompt structure is detailed (~500-800 tokens per C-suite prompt). However:
- Claude Code's prompt caching reduces cost for repeated content across subagent calls
 - Sections 1, 4, and 5 are static per role (loaded once)
 - Only Section 2 and 3 interact with the dynamic input (the CEO's framed question)
 - The output template (Section 3) constrains response length, preventing runaway token usage

Part 2: Adaptive Complexity Tiers

The Problem

Not every question needs a full 7-agent cascade. "Should we switch from Slack to Teams?" doesn't need the same treatment as "Should we acquire CompetitorX?" The skill needs tiered engagement to manage cost, latency, and proportionality.

The Agent Complexity Spectrum

Enterprise AI research (Applied AI) identifies a proven tiered approach:

Tier	Approach	Cost	When to Use
Tier 0	Rules/routing only	~\$0	Clear-cut cases, keyword matching
Tier 1	Single LLM call	~\$0.001	Simple classification, one-step reasoning
Tier 2	LLM + tools	~\$0.01-0.10	Tasks needing data lookup or bounded multi-turn
Tier 3	Multi-step autonomous	~\$1-10+	Complex, unpredictable reasoning paths

Key finding: **Start at Tier 0 and escalate based on evidence.** A hybrid approach (69% keyword routing + 31% LLM classification) handles most volumes efficiently.

Critical constraint: **The reliability cliff.** At 90% per-step accuracy, a five-step workflow succeeds only 59% of the time. More steps = more failure points.

Recommended Three-Tier Model for Team of Teams

Tier 1: "Quick Consult" (CEO-Only Rapid Assessment)

- **When:** Simple, low-stakes, or narrow-domain questions

- **Who:** CEO agent only (Opus) -- no subagents spawned
- **What happens:** CEO draws on its broad executive knowledge to provide a rapid assessment. References the org structure mentally but doesn't invoke the cascade.
- **Output:** Brief recommendation (1-2 paragraphs) with key considerations
- **Cost:** ~\$0.50-1.00 (single Opus call)
- **Latency:** Seconds
- **Example questions:** "What's a reasonable timeline for rolling out a new expense policy?" / "Should we send the team to this conference?"

Tier 2: "Focused Panel" (CEO + 2-4 Relevant C-Suite)

- **When:** Moderate operational or domain-specific questions
- **Who:** CEO + 2-4 most relevant C-suite subagents (routed by decision type)
- **What happens:** CEO frames and routes. Selected C-suite agents run their full analysis (including simulated team lead consultation). CEO synthesizes.
- **Output:** Standard decision record (but with only activated domains in Section 3)
- **Cost:** ~\$2.00-5.00 (CEO + 2-4 Sonnet subagents)
- **Latency:** 1-3 minutes
- **Example questions:** "Should we switch project management tools?" (COO + CTO + CFO) / "Should we hire a dedicated security engineer?" (CISO + CFO + CAO)

Tier 3: "Full Cascade" (CEO + All 7 C-Suite)

- **When:** Major strategic decisions, high-stakes, irreversible, cross-functional
- **Who:** All 8 agents (CEO + 7 C-suite)
- **What happens:** Full five-phase cascade with all domain analyses, cognitive forcing, pre-mortem, and complete decision record.
- **Output:** Full decision record with all 8 sections
- **Cost:** ~\$5.00-10.00 (CEO + 7 Sonnet subagents)
- **Latency:** 3-8 minutes
- **Example questions:** "Should we acquire CompetitorX?" / "Should we pivot our service model?" / "Should we accept this \$2M investment offer?"

Bonus: Tier 3+: "Decision Space" (Full Cascade + Multiple CEO Profiles)

- **When:** User explicitly requests decision space exploration
- **Who:** All 8 agents + 5 CEO synthesis passes
- **What happens:** Full cascade, then CEO synthesizes through Guardian, Analyst, Visionary, Consensus Builder, and Regret Minimizer lenses
- **Output:** Decision space document showing range of defensible decisions
- **Cost:** ~\$6.00-12.00 (full cascade + 5 lightweight CEO passes)
- **Latency:** 5-10 minutes

Tier Selection Mechanism

Three approaches, from simplest to smartest:

Approach 1: User-Selected (Simplest)

- User invokes the skill with a tier flag: `/team-of-teams quick "Should we..."` or `/team-of-teams full "Should we..."`
- Default: Tier 2 (Focused Panel) if no flag specified
- This is the recommended v1 approach

Approach 2: CEO Auto-Classification (Smarter)

- CEO agent receives the question first and classifies it:
- Checks for high-stakes keywords (acquire, pivot, layoff, regulatory, breach) -> Tier 3
- Checks for domain-specific keywords (security, budget, hire, tools) -> Tier 2 with routing
- Everything else -> Tier 2 default routing
- User can override: "Give me the full cascade on this"
- This is the recommended v1.5 approach (add after basic skill works)

Approach 3: Adaptive Escalation (Smartest)

- Start at Tier 1 (CEO quick assessment)
- If CEO determines the issue is cross-functional or high-stakes, auto-escalate to Tier 2 or 3
- This mirrors the real-world pattern where a CEO does a quick mental assessment before deciding whether to convene the leadership team
- This is the recommended v2 approach

Critical Design Principle: Proportional Response

The tier system implements a **proportional response** to decision complexity. This isn't just cost optimization -- it's about producing appropriate output. A 10-page decision record for "should we buy standing desks?" undermines the skill's credibility. Matching analysis depth to question importance is itself an executive skill.

Key Takeaways

- **The C-suite prompt has five sections:** Identity/Mandate, Analytical Process (the key section), Output Format, Domain Knowledge, and Accountability. Section 2 is where "thinking in 37 voices with 8 agents" actually happens.
- **The analytical process section forces step-by-step team lead consultation** -- not "consider perspectives" but "work through Team Lead 1's checklist, then Team Lead 2's, then synthesize." This structured decomposition is the core mechanism.
- **Three engagement tiers:** Quick Consult (\$0.50-1), Focused Panel (\$2-5), Full Cascade (\$5-10). Default to Tier 2. User-selected in v1, auto-classified in v1.5.
- **Proportional response is itself an executive skill.** The tier system prevents over-analysis of simple questions and under-analysis of complex ones.
- **Start with user-selected tiers (v1)**, add CEO auto-classification (v1.5), then adaptive escalation (v2). Each version builds on the previous.

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Research: Scenario Forking & Organizational Stress Testing

Requested by: Proactive research in support of Free Thinker's second wave (Directions A and B)

Date: 2026-02-22

Questions

1. Is there a formal framework for running parallel "what-if" scenarios and comparing their outcomes systematically?
2. Is there precedent for bottom-up, proactive risk/opportunity discovery (as opposed to top-down issue analysis) in multi-agent or organizational systems?

Findings

Part 1: Scenario Forking — Formal Frameworks

Scenario-Based Decision Analysis (SBDA)

The Free Thinker's "Scenario Forking Engine" maps directly to an established methodology: **Scenario-Based Decision Analysis (SBDA)**, which integrates two separate disciplines:

1. **Scenario Planning (SP)** — Constructing multiple plausible future narratives to test strategy robustness
2. **Structured Decision Making (SDM)** — Rooted in decision theory, focused on explicit assessment of expected outcomes across a set of decision alternatives

SBDA addresses a known limitation of traditional scenario planning: SP lacks "a defined structure for establishing objectives, quantifying tradeoffs, and evaluating the performance of candidate decisions to meet those objectives." By adding structured evaluation (what the Team of Teams already provides via the cascade), SBDA becomes a complete framework.

Key methodology from the literature (Think Insights / Scenario Planning):

Five phases:

1. **Define Scope & Time Horizon** — What are we deciding, and over what timeframe?
2. **Identify Key Drivers** — External factors (market, regulation, technology) and internal variables (capabilities, capital, talent)
3. **Collect & Analyze Data** — Gather information to assess predictability and impact
4. **Develop Scenarios** — Construct alternative futures with distinct assumptions and early warning indicators
5. **Apply & Maintain** — Test strategies under each scenario, develop contingency plans

Organizations typically develop 2-4 scenarios using one of three structures:

- **Spectrum:** Testing sensitivity around one major variable
- **Matrix:** Examining two dimensions of uncertainty simultaneously
- **Binary:** Comparing favorable vs. unfavorable outcomes

How this maps to the Team of Teams Scenario Forking Engine:

SBDA Phase	Team of Teams Implementation
Define Scope	CEO frames the issue (Phase 1 — already in cascade)
Identify Key Drivers	CEO identifies the decision alternatives (new Phase 0.5)
Collect & Analyze Data	C-suite cascade runs once per fork (or once with fork-specific parameters)
Develop Scenarios	Each fork IS a scenario — "acquire" vs. "build" vs. "invest elsewhere"
Apply & Maintain	Meta-CEO comparison across fork Decision Records

The critical design insight from SBDA: **the value is not in the individual scenarios but in the comparison.** The "Divergence Analysis" section of the Comparative Decision Record is where the real strategic insight lives — showing where forks agree (robust conclusions) and where they diverge (decisions that depend on assumptions).

Simulation Agent Framework (ArXiv 2505.13761)

A 2025 framework called Simulation Agent demonstrates the technical pattern the forking engine would follow:

- Users describe "what-if" scenarios in natural language
- The AI agent dynamically modifies input parameters, reruns analysis, and presents comparative results
- "Rapid scenario exploration capability" — same underlying model, different parameter sets
- Results synthesized via "JSON summaries highlighting key performance indicators"

This validates the Free Thinker's core efficiency insight: **run the expensive domain analysis once per fork, not once per mode.** The forking happens at the issue-framing level (different business alternatives), while the analysis cascade is reused machinery.

Cost Implications of Forking

If the standard Full Cascade costs ~\$5-10 per decision:

- **3-fork scenario analysis:** ~\$15-30 (3x cascade) + ~\$2-3 (meta-CEO comparison) = ~\$17-33
- **With routing optimization** (not all C-suite activated per fork): ~\$8-20
- **With shared context** (forks share the same company profile, only the "action" parameter changes):

Some cost savings from prompt caching

This is expensive but proportionate for major strategic decisions (acquisition, market entry, pivot). The key constraint: forking should be a Tier 3+ feature, never the default.

Part 2: Organizational Stress Testing — Bottom-Up Discovery

The Concept

The Free Thinker's Direction B inverts the cascade: instead of routing a user-provided issue downward, each C-suite agent proactively identifies issues from their domain. This is an **organizational health assessment** or **proactive risk discovery** pattern.

Precedent in AI Agent Systems

Research on AI-augmented organizational assessment shows:

1. **Proactive vs. Reactive Paradigm Shift:** AI agent research consistently emphasizes the transition from reactive analysis (respond to presented issues) to proactive discovery (surface latent risks and opportunities). This is described as AI's "predictive capabilities" that let organizations "anticipate challenges before they happen" (Datagrid, 2025).
2. **Multi-Domain Analysis:** AI excels at "spotting patterns across numerous variables simultaneously" — this multi-dimensional analysis maps directly to having each C-suite domain independently scan for issues, then cross-referencing across domains to identify systemic patterns.
3. **Bottom-Up Pattern:** Risk Management AI Agents are described as systems designed to "identify, assess, and mitigate risks" by "leveraging advanced analytics" to "navigate complex risk landscapes." The key: they don't wait for issues to be presented — they actively scan.

How This Maps to Team of Teams

The Stress Test mode would work as follows:

Input: Company context document (financials, current projects, market position, recent changes, headcount, tech stack, etc.)

Process (Inverted Cascade):

1. **CEO distributes company context** to all C-suite agents (broadcast, not routing — everyone participates)
2. **Each C-suite agent, using their team lead frameworks, scans for issues in their domain:**
 - CFO: Cash flow projections, burn rate concerns, budget gaps, hidden costs
 - CISO: Security posture gaps, compliance deadlines, unaddressed vulnerabilities
 - COO: Capacity constraints, process bottlenecks, vendor dependencies
 - CTO: Technical debt, scaling risks, architecture concerns
 - VP Sales: Pipeline gaps, market shifts, competitive threats
 - VP Delivery: Resource conflicts, SLA risks, quality concerns
 - CAO: Hiring gaps, retention risks, legal exposure, policy gaps
3. **Each C-suite agent outputs a prioritized issue list** with severity ratings
4. **CEO triages and cross-references:** Identifies systemic issues (appearing across multiple domains), prioritizes, and produces the Organizational Health Report

Output: Not a Decision Record but an **Organizational Health Report:**

ORGANIZATIONAL HEALTH REPORT

Date: [timestamp]

Company Context: [summary of inputs]

CRITICAL ISSUES (Require immediate attention)

1. [Issue] – Identified by: [role(s)] – Severity: CRITICAL
Preliminary recommendation: [1–2 sentences]
Cross-domain implications: [which other domains are affected]

HIGH PRIORITY ISSUES

[...]

WATCH LIST (Monitor, not urgent)

[...]

OPPORTUNITIES IDENTIFIED

[...]

SYSTEMIC PATTERNS

- [Pattern spanning multiple domains]

RECOMMENDED NEXT STEPS

- "Feed Issue #1 into /deliberate for full cascade analysis"
- "Schedule CFO deep-dive on Issue #3"

The "Full Physical vs. Doctor Visit" Insight

The Free Thinker's metaphor is precise. The standard cascade is a doctor visit (you bring a complaint, they diagnose). The stress test is a full physical (the system finds what you didn't know to ask about). Both are valuable; they serve different needs:

- **Standard cascade:** Known issue, need analysis → Decision Record
- **Stress test:** No specific issue, need to discover what to worry about → Health Report → feeds back into standard cascade for specific issues

Key Design Consideration: Company Context Quality

The stress test is only as good as the company context input. If the user provides vague context ("we're a SaaS company"), the C-suite agents will produce generic issues. The skill should provide a **Company Context Template** that prompts for specific data points each domain needs:

- Financials: Revenue, burn rate, runway, recent fundraising
- Operations: Headcount, key projects, vendor dependencies
- Technology: Stack, known technical debt, scaling concerns
- Security: Last audit date, compliance frameworks, incident history
- Sales: Pipeline, conversion rates, key accounts, competitive landscape
- Delivery: Active projects, resource utilization, SLA performance
- Admin: Recent hires/departures, pending legal issues, policy changes

The richer the context, the more specific and actionable the stress test output.

Cost Implications

A stress test always activates all 7 C-suite agents (no routing optimization — the whole point is broad domain scanning). Cost: ~\$5-10 per stress test (same as a Full Cascade). This is appropriate for a periodic organizational review, not a daily tool.

Key Takeaways

- **Scenario Forking maps to the established SBDA framework** (Scenario-Based Decision Analysis), combining scenario planning with structured decision making. The value is in the comparison across forks, not the individual analyses.
- **The forking engine's efficiency depends on reusing the cascade machinery** with different input parameters per fork. Cost: ~\$8-33 per multi-fork analysis depending on routing and number of forks.
- **Organizational Stress Testing inverts the cascade** from top-down issue analysis to bottom-up discovery. The output is an Organizational Health Report with a prioritized issue queue, not a Decision Record.
- **Stress testing requires rich company context input** — the skill should provide a Company Context Template to ensure each domain has the data it needs for meaningful scanning.
- **Both features are Tier 3+ capabilities** — they're expensive and should be reserved for strategic planning moments, not daily use. The standard cascade and direct consult tiers handle routine decisions.
- **The stress test feeds back into the standard cascade** — discovered issues can be routed into full deliberation, creating a discovery-to-decision pipeline.

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Research: SMB Organizational Decision-Making Patterns

Requested by: Arbiter (initial research assignment)

Date: 2026-02-22

Question

How do SMBs typically structure decision-making? What are common pain points? How do C-suite roles interact in practice at small/medium companies?

Findings

SMB Definition and Scale

- **Small business:** Typically 1-100 employees, revenue under \$50M
- **Medium business (mid-market):** 100-999 employees, revenue \$50M-\$1B
- The concept seed describes a structure with CEO + 7 direct reports + ~35 team leads, suggesting a mid-market company of approximately 200-500 employees

Typical SMB Organizational Structures

Flat Structure (very small, <25 employees): CEO/founder at top with 1-2 layers below. All decisions flow through the owner. Most common in startups.

Hierarchical Structure (25-500 employees): CEO at top, departments below with directors/managers/staff. The concept seed's structure fits this model -- it's the standard mid-market org chart with two management tiers below the CEO.

Trend: 85% of leaders in Deloitte's 2025 Global Human Capital Trends report said their companies need greater agility. More SMBs are adopting flatter, collaborative structures.

C-Suite Role Profiles (Mapped to Concept Seed)

CEO (Chief Executive Officer)

- **Domain:** Long-term strategy, vision, external stakeholder management
- **Decision style:** Final authority. Evaluates recommendations from direct reports.
- **SMB reality:** In smaller firms, deeply hands-on. In mid-market, more strategic.
- **Key interactions:** Directly manages all C-suite; sets agenda for cross-functional issues

COO (Chief Operating Officer)

- **Domain:** Day-to-day operations, process optimization, cross-department coordination
- **Decision style:** Execution-focused. Translates CEO strategy into operational reality.
- **SMB reality:** Often second-in-command; may overlap with CEO in smaller firms
- **Key interactions:** Works closely with all departments; ensures alignment between strategy and execution
- **Typical team leads:** Operations Manager, Project Management Lead, Process Improvement Lead, Quality Assurance Manager

CFO (Chief Financial Officer)

- **Domain:** Financial strategy, budgeting, revenue/profitability, financial controls, risk assessment
- **Decision style:** Data-driven, risk-aware. Evaluates financial implications of all decisions.
- **SMB reality:** May be outsourced or fractional in very small firms; critical in mid-market
- **Key interactions:** Close collaboration with CEO on strategy, COO on operational budgets, CTO on tech investments
- **Typical team leads** (from concept seed): Controller (accounting/close), Head of FP&A (planning/ analytics), Treasury/Cash Manager, AP Manager, AR Manager, Tax Accountant (internal lead + external firm)

CTO (Chief Technology Officer)

- **Domain:** Technology strategy, product development, system architecture, technical teams, innovation
- **Decision style:** Technology-forward but must balance innovation with practicality and security
- **SMB reality:** Critical for tech companies; may be part-time or outsourced for non-tech SMBs
- **Key interactions:** Collaborates with CISO on security, COO on tech operations, CFO on tech budgets
- **Typical team leads:** Engineering Lead, DevOps/Infrastructure Lead, Product Development Lead, QA/Testing Lead, Solutions Architect

CISO (Chief Information Security Officer)

- **Domain:** Information security strategy, risk assessment, incident response, compliance, data protection
- **Decision style:** Risk-focused, compliance-oriented. Often seen as the "no" voice that needs to be balanced.
- **SMB reality:** Many smaller SMBs assign CISO duties to CTO or a Director of Security. Mid-market increasingly has a dedicated CISO.
- **Key interactions:** Works closely with CTO on security architecture, CFO on security investment, CAO on compliance
- **Typical team leads:** Security Operations Lead, Compliance/GRC Lead, Identity & Access Management Lead, Incident Response Lead

CAO (Chief Administrative Officer)

- **Domain:** Administrative backbone -- HR, legal, compliance, IT operations, facilities, organizational policies
- **Decision style:** Process-oriented, compliance-focused, people-centric
- **SMB reality:** In smaller firms, this is a "jack-of-all-trades" generalist. In mid-market, increasingly formalized.
- **Key interactions:** Works with all departments on HR needs, ensures legal/regulatory compliance across the org
- **Typical team leads:** HR Director, Legal Counsel/Compliance Lead, Facilities Manager, Office/Admin Manager, Procurement Lead

VP of Sales

- **Domain:** Revenue generation, sales strategy, customer acquisition, pipeline management, team quotas
- **Decision style:** Revenue-driven, customer-focused, competitive. Often champions growth over caution.
- **SMB reality:** Critical role; directly impacts revenue. Reports to CEO or COO depending on structure.
- **Key interactions:** Close collaboration with VP of Delivery (service promises), CFO (revenue forecasting), CTO (product capabilities)
- **Typical team leads:** Sales Managers (by region/vertical), Business Development Lead, Sales Operations/Enablement Lead, Account Management Lead

VP of Delivery (Service Delivery)

- **Domain:** Service delivery, client satisfaction, project execution, resource allocation, SLA management
- **Decision style:** Client-satisfaction focused, operationally rigorous, quality-oriented
- **SMB reality:** Critical for services companies; may not exist in pure product companies
- **Key interactions:** Tight coupling with VP of Sales (what was sold vs. what can be delivered), COO (resource allocation), CTO (technical delivery)
- **Typical team leads:** Delivery Managers, Client Success Lead, Resource/Capacity Planning Lead, Technical Project Leads

Decision-Making Patterns in SMBs

1. **Concentrated Authority:** SMBs typically have only 1-3 actual decision-makers, with the business owner almost always being one. Even with a full C-suite, the CEO often has final say on most significant decisions.
2. **Speed Over Process:** Decision-making is often quick and influenced by immediate emotional, aspirational, and market-driven challenges. Less formal governance compared to enterprise.
3. **Cross-Functional Tension:** The fundamental tensions in SMB decision-making:

4. **Growth vs. Risk:** VP Sales pushes growth; CFO/CISO flag risk
5. **Speed vs. Quality:** COO wants efficiency; VP Delivery wants thoroughness
6. **Innovation vs. Stability:** CTO pushes new tech; CISO/CFO push caution
7. **People vs. Budget:** CAO advocates for people investment; CFO guards the budget
8. **Personal Relationships Matter:** In SMBs, trust and personal relationships drive decision-making more than formal process. This aligns strongly with McChrystal's Trust pillar.
9. **Hands-On C-Suite:** Unlike enterprise, SMB C-suite members are often deeply involved in day-to-day operations, not just strategy. They have intimate knowledge of frontline issues.

Common Pain Points

1. **Information Silos:** Departments don't share information effectively. Each leader sees only their domain. This is exactly what shared consciousness is designed to solve.
2. **Decision Bottlenecks:** Everything flows to the CEO for approval, creating delays. McChrystal's empowered execution addresses this.
3. **Resource Constraints:** Limited budgets mean trade-offs between departments. CFO often becomes the bottleneck.
4. **Lack of Formal Analysis:** Decisions are made on gut feeling rather than structured analysis from multiple perspectives. The Team of Teams agent could address this directly.
5. **Alignment Drift:** Without regular cross-functional communication, teams drift apart in priorities and understanding.
6. **Reactive vs. Proactive:** SMBs often respond to crises rather than anticipating them. Structured evaluation processes could help.

How the Concept Seed's Org Structure Maps to Real SMBs

The proposed structure (CEO -> 7 C-suite/VP -> team leads) represents a **well-structured mid-market services company**, likely in IT/technology services given the inclusion of both CISO and VP of Delivery. This structure is realistic for a company of 200-500 employees. Key observations:

- The inclusion of both CTO and CISO suggests a technology-forward company that takes security seriously
- VP of Sales + VP of Delivery is a classic services company pairing
- CAO rather than CHRO suggests a broader administrative scope
- The team lead layer (Controller, FP&A, etc.) represents realistic mid-management roles

Key Takeaways

- **The concept seed's org structure is realistic and well-suited to a mid-market IT/technology services company** of 200-500 employees. The role definitions map to real-world responsibilities.

- **The fundamental cross-functional tensions (growth vs. risk, speed vs. quality, innovation vs. stability) are the engine** that makes multi-perspective evaluation valuable. Each C-suite role will naturally champion different priorities.
- **SMB decision-making is faster and more personal than enterprise**, meaning the agent simulation should reflect quick, opinionated responses informed by deep domain knowledge, not lengthy corporate analysis.
- **Information silos are the #1 pain point** in SMB decision-making -- and shared consciousness is the direct solution. The agent skill's value proposition is essentially "McChrystal's shared consciousness, delivered by AI."
- **Each role's team leads should bring granular, specialized expertise** (e.g., the CFO's Tax Accountant spotting tax implications the CFO might miss). This multi-level analysis is what makes the concept unique.

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Research: "Team of Teams" Organizational Methodology

Requested by: Arbiter (initial research assignment)

Date: 2026-02-22

Question

What are the core principles of General McChrystal's "Team of Teams" framework? How does it differ from traditional hierarchical org structures? What are the key success factors?

Findings

Origin and Context

The "Team of Teams" framework was developed by General Stanley McChrystal while commanding the Joint Special Operations Command (JSOC) in Iraq. Facing a decentralized, adaptive enemy (Al-Qaeda in Iraq), McChrystal realized that the traditional military command hierarchy -- optimized for efficiency and predictability -- was too slow to respond to a complex, rapidly shifting environment. He restructured JSOC from the ground up, transforming a rigid hierarchy into an interconnected network of empowered small teams.

Core Framework: Four Pillars

The Team of Teams model rests on four interdependent capabilities:

1. **Trust** -- The structural foundation. Not motivational but operational infrastructure. Trust enables teams to act without waiting for verification. McChrystal emphasized that trust is built through shared hardship, transparency, and embedding personnel across teams to create "friendly faces" rather than hierarchical reporting lines. BUD/S-style training creates "superteams" through "strong lattice of trusting relationships" rather than individual excellence.
2. **Common Purpose** -- A shared organizational direction that aligns all teams toward the same strategic objectives, even when operating independently. Every team member understands *why* the organization exists and what it's trying to achieve, not just their individual tasks.
3. **Shared Consciousness** -- The intelligence and understanding created by a collective view of the operating environment and a high level of internal connectivity. This is achieved through:
4. **Extreme, participatory transparency** -- all teams access the same intelligence
5. **Centralized communications forums** -- McChrystal's daily Operations & Intelligence (O&I) briefing connected 70+ locations via video teleconference
6. **Operating rhythm** -- structured, predictable cadence for meetings and updates

7. **Fusion cells** -- cross-functional collaboration spaces where representatives from different teams work side by side
8. **Tech enablement** -- technology infrastructure supporting seamless communication
9. **Empowered Execution** -- Decentralizing decision-making to the level necessary to effectively respond to the speed of the environment. Critically, this only works AFTER shared consciousness is established. "Empowerment without context will lead to havoc" (as demonstrated by the 2008 financial crisis). McChrystal asked subordinates "What do you need?" rather than issuing commands, and clarified decision parameters by "thinking out loud" -- explaining logic so teams understood organizational values, not just tactical objectives.

How It Differs from Traditional Hierarchies

Dimension	Traditional Hierarchy	Team of Teams
Decision-making	Centralized; flows up for approval	Decentralized; pushed to those closest to the issue
Information flow	Need-to-know; vertical channels	Radical transparency; horizontal sharing
Leadership model	"Heroic leader" who commands	"Gardener" who cultivates the environment
Optimization for	Efficiency and predictability (complicated systems)	Adaptability and resilience (complex systems)
Team interaction	Siloed; teams operate independently	Interconnected; cross-team liaisons and embedding
Speed	Slow; multiple approval layers	Fast; real-time decisions with shared context
Trust model	Institutional/role-based	Relational/earned through transparency

Key Distinction: Complicated vs. Complex

McChrystal draws a critical distinction between **complicated** systems (like a watch -- many parts but predictable) and **complex** systems (like a weather system -- many parts interacting unpredictably). Traditional hierarchies are optimized for complicated problems. The Team of Teams framework is designed for complex environments where cause-and-effect relationships are not linear.

Leadership as "Gardening"

McChrystal proposes the leader as "gardener" rather than commander. The gardener cultivates, nurtures, and develops. The garden does not require the gardener's constant presence but becomes robust, resilient, and radiant thanks to the environment they create. Key leadership behaviors:

- **Consistency:** "Every action is interpreted" -- leaders must align words with visible behavior
- **Presence:** Active participation in forums signals commitment
- **Humility:** Admitting "I don't know" builds credibility more than false expertise

- **Amplification awareness:** The "dinosaur's tail" concept -- senior leaders' unintended consequences ripple through organizations

Real-World Business Implementation

- A Fortune 100 oil & gas company restructured to a cross-functional model, using shared consciousness to gather leaders around common purpose and empower safe, smart, quick execution
- McChrystal Group has applied these principles across healthcare, aviation, sports, and enterprise contexts
- Research shows workers who feel empowered are 36% more likely to feel shared ownership in company outcomes
- Pentland's research (cited in the book) showed interaction patterns account for almost half of all performance variation between high and low performers

Key Success Factors

1. **Sequence matters:** Trust and Shared Consciousness must be established BEFORE Empowered Execution
2. **Structural investment required:** Physical plant and mandatory forums matter more than directives
3. **Information sharing must be deliberate:** Default to transparency, not secrecy
4. **Cross-team connections:** Embedding liaisons creates the inter-team trust that makes a "team of teams" rather than a collection of teams
5. **Leadership transformation:** Leaders must shift from directing to enabling

Key Takeaways

- **The four pillars are sequential:** Trust -> Common Purpose -> Shared Consciousness -> Empowered Execution. You cannot skip steps.
- **Shared Consciousness is the lynchpin:** Without transparent information sharing across all teams, decentralized execution leads to chaos. The O&I forum model (daily cross-team briefing) is the primary mechanism.
- **The "gardener" leadership model maps directly to an AI orchestrator:** A CEO agent that creates the environment, sets direction, and enables teams rather than micromanaging every decision.
- **Cross-team liaisons are critical:** Simply having teams isn't enough -- you need structural mechanisms for inter-team trust and communication (fusion cells, embedding).
- **The framework is designed for complex, not just complicated, problems:** This makes it well-suited for business decision-making where multiple perspectives and unpredictable interactions matter.

Sources

#	Source	URL/Path	What It Contributed
1	McChrystal Group - Executive Summary	https://www.mcchrystalgroup.com/insights/playbooks/detail/2024/04/01/	Four capabilities framework, implementation overview
2	McChrystal Group - Shared Consciousness	https://www.mcchrystalgroup.com/capabilities/team-of-teams/shared-consciousness	Shared consciousness definition, forums, operating rhythm
3	McChrystal Group - Empowered Execution	https://www.mcchrystalgroup.com/capabilities/team-of-teams/empowered-execution	Empowered execution details, sequencing requirement
4	Graham Mann - Book Notes	https://grahammann.net/book-notes/team-of-teams-general-stanley-mcchrystal	Detailed book summary, O&I forum, gardener metaphor, trust mechanics
5	Beau Gordon - Key Takeaways (Medium)	https://beaugordon.medium.com/key-takeaways-from-team-of-teams-by-general-stanley-mcchrystal-eac0b37520b9	Core framework summary, business implementation
6	McChrystal Group - Remote Case Study	https://www.mcchrystalgroup.com/insights/team-of-teams-in-a-remote-environment/	Fortune 100 oil & gas case study
7	The Army Leader	https://thearmyleader.co.uk/team-of-teams/	Agile, adaptable, resilient qualities

Citation Log

- Search: General McChrystal "Team of Teams" framework core principles organizational methodology
- Search: McChrystal "Team of Teams" shared consciousness empowered execution implementation business enterprise case studies
- Fetched: <https://www.mcchrystalgroup.com/insights/playbooks/detail/2024/04/01/>
- Fetched: <https://grahammann.net/book-notes/team-of-teams-general-stanley-mcchrystal>
- Fetched: <https://www.mcchrystalgroup.com/capabilities/team-of-teams/shared-consciousness>
- Fetched: <https://www.mcchrystalgroup.com/capabilities/team-of-teams/empowered-execution>

Original user request and all captured source materials

manifest.md

Original source material

Source Materials Manifest

Session: Team of Teams Agent Skill

Captured: 2026-02-22

#	File	Type	Original Location
1	request.md	User request	(inline input)
2	team-of-teams.txt	Concept seed	ref/ideas/team-of-teams.txt

User Request

I want to use Claude Code Agent Teams to create a "Team of Teams" Agent Skill that realistically emulates a SMB's org structure to make decisions and answer questions on issues that come before the company. I need you to help me come up with the process, steps, and detailed, comprehensive technical instructions necessary to have an AI agent create the required components (front matter, instructions, prompts, templates, scripts, etc.) that will make up this skill.

The structure this "Team of Teams" will emulate will be the top two layers of the company's org structure. Starting at the top, There would be a CEO who has a COO, CFO, CAO, CTO, CISO, VP of Sales, and VP of Delivery. From there, each of those roles would have team leads performing the key functions that support that role. For example: the CFO might have a Controller (accounting/close), a Head of FP&A (planning/analytics), a Treasury/Cash manager (or embedded), an AP manager and/or an AR manager, a Tax accountant (internal lead + external firm). Each of these individual roles would have extensive knowledge of their domain, and would respond to/prioritize issues based their unique perspective of how to best serve the company's interests.

When any issue is presented to the CEO they would tell their team to evaluate it and provide their recommendations, which the CEO would then evaluate to make a final decision. To do that, each of the CEO's team members would give direction to their own teams to evaluate the issue, research relevant topics, identify risks/benefits, develop plans, establish guardrails, etc., to ultimately provide a defensible argument for or against the idea from their unique perspective.

The task I'm giving you is not to build the Agent Skill yourself. The task is to create the specifications for a separate agent to build it. I need you to think through how best to create this "Team of Teams" Agent Skill, what all it needs to include, how to design it, how it will function, how it will be used, and ultimately produce a detailed, comprehensive technical specification that is thorough and complete enough that it can be handed off to a separate dedicated AI agent to create the skill. The output must include the specification document in markdown (.md) format the separate, dedicated builder agent will derive its tasks from.

LAYER 5

Ideation graph, snapshots, idea reports, and session summary

Ideation Graph

Living document updated in real-time during the session

Ideation Graph

Session started: 2026-02-22
Concept seed: Team of Teams Agent Skill
Maintained by: Writer (updated in real-time as dialogue progresses)

Concept Seed Summary

Build a Claude Code Agent Skill that emulates the top two layers of a SMB org structure as a "Team of Teams." A CEO leads C-suite and VP-level reports (COO, CFO, CAO, CTO, CISO, VP Sales, VP Delivery), each of whom leads their own functional teams. When an issue is presented to the CEO, it cascades down: each executive tasks their team leads to evaluate, research, and argue from their domain perspective. Recommendations roll back up through executives to the CEO, who synthesizes a final decision. The deliverable is not the skill itself, but a comprehensive technical specification that a separate builder agent can use to construct it.

Active Threads

Thread A: Cascading Delegation Architecture (from FT #3)

Status: VALIDATED and SPEC-READY. Five-phase model accepted with two-tier implementation.

The Five-Phase Cascade:

- 1. **Phase 1 — CEO Frames + Routes.** Decomposes issue into dimensions, classifies decision type, activates relevant C-suite via routing table.
- 2. **Phase 2 — C-Suite Dispatches Downward.** Domain decomposition — each exec breaks the question into team-lead-scoped tasks. (Simulated internally, not actual sub-agents.)
- 3. **Phase 3 — Team Leads Produce Findings.** Narrow, domain-specific analysis. (Executed as sequential analytical passes within each C-suite agent.)
- 4. **Phase 4 — C-Suite Synthesizes Upward.** Domain recommendations with engineered dissent.
- 5. **Phase 5 — CEO Deliberation.** Fault-line identification, dynamic weighting, decision record output.

Thread B: Engineered Dissent (from FT #2)

Status: VALIDATED, COMPLETE, and SPEC-READY.

Role Mandate Map (final):

| Role | Mandate Type | Archetype | Notes (Grounder R3) |

|-----|-----|-----|-----|

| CEO | Synthesizer | "The Synthesizer" | |

| CISO | Skeptic | "The Constitutional Skeptic" | |

| CFO | Skeptic | "The Cost Archaeologist" | |

| COO | Skeptic | "The Execution Realist" | |

| VP Delivery | Skeptic | "The Promise Keeper" | |

| VP Sales | Advocate | "The Revenue Optimist" | |

| CTO | Advocate | "The Capability Expander" | |

| CAO | Systemic | "The Organizational Gravity Sensor" | Grounder: "better name than anything I would have come up with" |

Grounder validated (R3): CAO and VP Delivery mandates are "exactly right." Balance of 4 skeptics, 2 advocates, 1 systemic, 1 synthesizer accepted.

Thread C: Decision Records as Product (from FT #5)

Status: DESIGNED and REFINED by Grounder. Now spec-ready.

Decision Record Structure (revised per Grounder R3 — now 9+ sections):

1. **Executive Summary** (NEW, Grounder R3) — 3-5 sentences at the top: the decision, key reasoning, primary dissent. "For the person who only reads the first paragraph."
2. **Issue Statement** — The question as posed
3. **CEO Framing** — Decomposition + routing rationale. **Grounder addition:** Include WHY certain teams were excluded, not just who was included. The exclusion reasoning is itself valuable analysis.
4. **Domain Analyses** — Per activated C-suite role: recommendation, confidence, summary, team lead findings, risks, opportunities
5. **Fault Line Analysis** — Points of agreement, contention, unresolved tensions. **Grounder: "The most valuable section in the whole document."** Meta-analysis of where expert perspectives collide. What no single expert could produce.
6. **CEO Decision** — Decision, weight rationale, conditions/guardrails, accepted risks, mitigations
7. **Dissenting Views** — **Grounder: "The integrity mechanism."** Preserved for accountability and as reference if decision goes sideways. "This person warned us and we chose differently."
8. **Next Steps** — Actions, owners, timelines
9. **Metadata** — Roles consulted, complexity, primary domain, dissent level + **Key Assumptions** (NEW, Grounder R3). "Every analysis rests on assumptions. Capture the big ones so they can be revisited."

Thread G: Full Org Roster (from Grounder R2)

Status: DELIVERED and VALIDATED by Grounder with minor adjustments.

Grounder's roster notes (R3):

- **COO's Facilities/Office Manager:** Mark as conditional on company profile. Not relevant for B2B SaaS; matters for manufacturing/professional services. COO has 3-4 core team leads, not always 4.

- **CTO's Product/UX Lead:** May belong under VP of Product in some orgs, but for SMBs where CTO owns product, it works. Note in spec that company profile may reassign.
- **Non-overlap test passed:** "If two roles would produce the same analysis for the same question, one of them shouldn't exist." Grounder confirmed no gaps, no overlaps.

Thread H: Issue Routing Mechanism (from FT R3)

Status: RESOLVED — Core feature (v1), not v2. Both agents agree.

Grounder's ruling (R3): Routing is core because:

1. Without it, every question costs full 8-agent cascade
2. More importantly: routing makes the CEO's Phase 1 framing *meaningful*. If CEO always activates everyone, framing is just preamble. If CEO selects 3-5 relevant roles, framing is an analytical act.

Routing spec (Grounder R3):

- Define decision types: Strategic, Operational, Financial, Technical, Personnel, Compliance/Risk
- Default routing table: which C-suite roles activate per decision type
- CEO override: can always add or remove roles from default routing
- Full activation triggers: certain keywords/characteristics that activate everyone (e.g., "acquisition," "pivot," "layoff")
- Implementation: lookup table with override logic. "Concrete and buildable."

Thread I: Implementation Architecture — EVOLVED from Two-Tier to Hybrid Model

Status: EVOLVED — Original two-tier model (R3) upgraded to hybrid Agent Teams + subagents model based on Explorer research.

Original Two-Tier Model (Grounder R3):

- Tier 1: 8 Agent Teams members (CEO + 7 C-suite)
- Tier 2: Team leads simulated as sequential passes within C-suite prompts

REVISED: Hybrid Agent Teams + Subagents Model (Grounder, post-research):

- **Tier 1 (Agent Team members):** CEO + 7 C-suite officers. Real teammates spawned by the skill orchestrator. Can message each other and the CEO directly.
- **Tier 2 (Custom subagents):** Each C-suite agent invokes custom subagents defined in `.claude/agents/`. The CFO calls `controller.md`, `fp-and-a.md`, etc. Subagents get their own context window, can use different models (Haiku for cost efficiency), have tool restrictions, and report results back to their parent C-suite agent.

Why this is better than the original two-tier model (Grounder):

1. **Real isolation** — each team lead gets its own context window, not a simulated pass within a longer prompt
2. **Cost optimization** — team lead subagents on Haiku, C-suite on Sonnet/Opus
3. **Focused prompts** — the Controller subagent prompt is just about controllership, not a CFO pretending to be a controller
4. **True management simulation** — the C-suite agent receives, dispatches, collects, synthesizes — like a real executive

What stays the same: Five-phase cascade, engineered dissent, routing, decision record. Architecture-level design is unchanged. Only the IMPLEMENTATION model changes.

Three prompt types needed (evolved from original "three prompt categories"):

1. **CEO prompt** — Synthesizer. Frames, routes, collects, decides.
2. **C-suite prompts** — Domain executives. Receive, dispatch to subagent team leads, synthesize domain recommendations.
3. **Team lead subagent prompts** — Focused specialists. Receive domain-specific questions, produce narrow findings.

Grounder's assessment: "The technical reality actually gives us a MORE elegant architecture than what we hand-designed."

WAVE 2: Exploration and Resolution

FT's Second-Wave Opening (5 Directions)

The Free Thinker proposed five directions beyond the core architecture:

- **Direction A: Scenario Forking Engine** — parallel cascades with different parameters, meta-CEO comparison. "What happens if X vs. Y vs. Z?" Produces a decision landscape, not a single decision. CEO personalities are a special case of this.
- **Direction B: Organizational Stress Test** — invert the cascade. Instead of top-down issue resolution, bottom-up threat/opportunity discovery. Each C-suite generates issues from their domain. Output: Organizational Health Report.
- **Direction C: Prompt Architecture as Cognitive Simulation** — structured cognitive forcing to prevent "voice collapse." Different analytical frameworks, mandatory output structures, domain-specific forcing questions per team lead. The builder writes 29 analytical framework packages, not 29 persona descriptions.
- **Direction D: "What Changed?" Revisitation Mode** — revisit decisions when assumptions change. Targeted re-analysis, not full cascade from scratch. Decision lifecycle management.
- **Direction E: Three Interaction Tiers** — direct consult / focused panel / full deliberation. "Executive team on retainer" at any depth level.

FT's own assessment: A and B are most novel (reframe what the skill IS). C is hardest technically. D is a sleeper. E determines adoption.

Thread Resolution Map

Thread	FT Direction	Status	Resolved In
Thread J: CEO Decision Modes	Part of A	RESOLVED	Report #2 (4 modes: Guardian/Pioneer/Architect/Analyst)
Thread M: Adaptive Complexity	Direction E	RESOLVED	Report #2 (3 tiers: Hallway/Working Session/Board Meeting)
Thread L: Prompt Architecture	Direction C	RESOLVED	Report #3 (Cognitive Forcing framework)
Scenario Forking	Direction A (broader)	UNRESOLVED	Mentioned in Report #2 as Decision Space Map concept. Full forking not yet explored.
Stress Test	Direction B	UNRESOLVED	Not yet developed. Novel inversion of cascade.
Revisitation	Direction D	UNRESOLVED	Connects to parked Thread D (Institutional Memory). Lightweight version proposed.

Thread J: CEO Decision Modes — RESOLVED in Report #2

Final design: Four modes as CEO synthesis prompt modifiers: Guardian (risk-averse), Pioneer (growth-oriented), Architect (consensus-building), Analyst (data-driven). Default: Analyst. Multi-mode comparison via `--compare-modes`. Decision Space Map shows how risk appetite shapes outcomes.

Thread M: Adaptive Complexity — RESOLVED in Report #2

Final design: Three tiers: Tier 1 Hallway Question (`/consult`), Tier 2 Working Session (`/panel`), Tier 3 Board Meeting (`/deliberate`). Auto-triage via `/evaluate`. CEO assesses scope, impact, reversibility.

Thread L: Prompt Architecture — RESOLVED in Report #3

Final design: "Cognitive Forcing" framework. Three-layer prompt architecture:

- **Layer 1: Team Lead Subagent Definitions** — 29 analytical framework packages, each with unique analytical framework, mandatory output template, and three forcing questions (pre-mortem, adversarial empathy, domain devil's advocate). Implemented as `.claude/agents/` subagent definitions with Haiku model, tool restrictions, capped turns.
- **Layer 2: C-Suite Agent Prompts** — 7 domain orchestration templates supporting three modes: direct consult (Tier 1, internal reasoning), full analysis (Tier 2/3, subagent delegation), and challenge phase. Structured with role identity, team composition, analytical domain, operating modes, synthesis instructions.
- **Layer 3: CEO Synthesis Layer** — Fixed fault-line analysis protocol + swappable Decision Mode module.
- **Key innovation:** Cognitive forcing prevents "voice collapse" by making each perspective use different analytical methods producing structurally different outputs. Grounded in Nemeth (2001): authentic domain dissent > assigned contrarianism.

- **Reference implementation:** Full CFO domain (5 team lead packages: Controller, FP&A, Treasury, AP/AR, Tax) with complete frameworks, templates, and forcing questions.

Parked Threads

Thread D: Institutional Memory (from FT #1)

Status: Parked — scope bomb, future enhancement

Thread E: Company Profile Parameterization (from FT #4)

Status: Acknowledged — table stakes, include without brainstorming

Note (Grounder R3): Now actively referenced — company profile determines conditional roles and role placement.

Thread F: Cross-Functional Deliberation (Phase 4.5)

Status: Parked — v2 feature, include in spec as future enhancement

Idea Reports Submitted

Idea Report #1: The Cascading Deliberation Engine

File: `session/idea-reports/IDEA_cascading-deliberation-engine.md`

Submitted by: Free Thinker + Grounder

Status: INTERESTING (Arbiter-confirmed)

Covers: Five-phase cascade, two-tier agent architecture (8 real / 37 conceptual), engineered dissent model with role mandates, full 37-role org roster, CEO-level routing mechanism, decision record format (9 sections with exec summary)

The core architecture. Everything else builds on this.

Idea Report #2: The Engagement Model

File: `session/idea-reports/IDEA_engagement-model.md`

Submitted by: Free Thinker + Grounder

Status: INTERESTING (Arbiter-confirmed)

Covers: Two-dimensional control surface: depth (3 interaction tiers: Hallway Question / Working Session / Board Meeting) x width (4 CEO Decision Modes: Guardian / Pioneer / Architect / Analyst). Invocation patterns (`/consult`, `/panel`, `/deliberate`, `/evaluate`). Tier selection logic (scope, impact, reversibility). Output formats per tier. Multi-mode comparison (`--compare-modes`). Decision Space Map concept.

Key insight: "The difference between a quarterly novelty and a daily tool is whether the skill matches the user's engagement level to the decision's weight."

Recommendation: "The cascade architecture (Report #1) is the engine. The engagement model is the interface."

Idea Report #3: Cognitive Forcing and the Prompt Architecture

File: session/idea-reports/IDEA_cognitive-forcing-prompt-architecture.md

Submitted by: Free Thinker + Grounder

Status: INTERESTING (Arbiter-confirmed)

Covers: Three-layer prompt architecture (team lead subagent packages, C-suite orchestration templates, CEO synthesis layer). Cognitive Forcing pattern: unique analytical framework + mandatory output template + three forcing questions per perspective. Research-backed dissent techniques (pre-mortem via Klein 2007, adversarial empathy via McChrystal, domain devil's advocate via Nemeth 2001). Full CFO reference implementation (5 team lead packages with complete frameworks/templates/forcing questions). Subagent file structure for `.claude/agents/`. C-suite multi-mode support (direct consult, full analysis, challenge phase).

Key insight: "The difference between '7 agents that all sound like slightly different consultants' and '7 genuinely distinct domain perspectives' is not better persona descriptions — it's forcing each perspective to use different analytical methods that produce structurally different outputs."

Recommendation: "This is the muscle that makes it move." The cascade (Report #1) is the skeleton. The engagement model (Report #2) is the steering wheel. Cognitive forcing is what makes the output worth reading.

Three reports together form the spec's three pillars: Structure (Report #1), Experience (Report #2), Analytical Quality (Report #3).

Idea Report #4: Decision Modes and the Decision Space Map

File: session/idea-reports/IDEA_decision-modes-decision-space.md

Submitted by: Free Thinker + Grounder

Status: INTERESTING (Arbiter-confirmed)

Covers: Five Decision Modes (Guardian/Pioneer/Architect/Analyst/Sentinel) mapped to decision theory (MaxiMin/MaxiMax/Behavioral/Hurwicz/MiniMax Regret). Complete CEO prompt modifiers per mode. Mode/Tier interaction matrix. Multi-mode comparison mechanism with cost efficiency analysis (1.1x cost for 5x insight). Comparative Decision Record format with Mode Sensitivity metric. Invocation grammar integration.

Key insight: "The most useful output of multi-perspective analysis isn't a single answer — it's a map showing how the answer changes depending on what you optimize for."

Recommendation: "This is the feature that differentiates this skill from 'ask a panel of AI experts.' Panels give you one answer. This gives you the shape of the decision landscape."

Four reports together form the spec's four pillars: Structure (#1), Experience (#2), Analytical Quality (#3), Strategic Depth (#4).

Interesting (Arbiter-Confirmed)

1. The Cascading Deliberation Engine (Report #1)

Core architecture. Five-phase cascade, hybrid agent model, engineered dissent, routing, decision record.

2. The Engagement Model (Report #2)

User interface. Three interaction tiers, four CEO decision modes, invocation patterns, auto-triage.

3. Cognitive Forcing and the Prompt Architecture (Report #3)

Analytical quality. Framework packages, forcing questions, subagent definitions, CFO reference implementation.

4. Decision Modes and the Decision Space Map (Report #4)

Strategic depth. Five modes mapped to decision theory, Comparative Decision Record, Mode Sensitivity metric, multi-mode comparison at 1.1x cost.

Needs More Conversation

Items the Arbiter has sent back for further exploration.

Abandoned Threads

None.

Explorer Research Summary

Four research reports completed. Key design implications synthesized:

1. McChrystal's Framework (RESEARCH_team-of-teams-methodology.md)

- Four sequential pillars: Trust -> Common Purpose -> Shared Consciousness -> Empowered Execution. Cannot skip steps.
- Maps to skill workflow: broadcast issue context to all agents (shared consciousness) THEN let them reason independently (empowered execution).
- CEO as "gardener" not "commander" — cultivates environment, doesn't micromanage. Aligns with CEO agent as synthesizer.

2. Multi-Agent Landscape (RESEARCH_multi-agent-orchestration.md)

- No existing framework models multi-level org structure for business decisions — this would be novel.
- CrewAI closest analogy but lacks hierarchical depth. MetaGPT is domain-locked to software.
- Best pattern: hierarchical routing with local autonomy (central planning + decentralized execution).
- Token cost is primary constraint for multi-level hierarchies.

3. Claude Agent Teams (RESEARCH_claude-agent-teams.md)

- **Critical: No nested teams.** Teammates can't spawn sub-teams. The two-tier cascade can't be literal nested agent teams.
- Subagents offer per-agent customization (tools, hooks, memory, skills) but lack inter-agent communication.
- **Hybrid architecture needed:** Agent Teams for C-suite + subagents for team leads, or staged execution.

- Skills system provides the right packaging mechanism.
- 5-6 tasks per teammate is the recommended load.

4. SMB Decision-Making (RESEARCH_smb-decision-making.md)

- Concept seed maps to mid-market IT/technology services company, 200-500 employees.
- Cross-functional tensions (growth vs. risk, speed vs. quality, innovation vs. stability) ARE the engine.
- SMB decisions are faster and more personal — skill should produce quick, opinionated responses, not lengthy corporate analysis.
- Information silos are #1 pain point. Shared consciousness is the direct solution.
- Value proposition: "McChrystal's shared consciousness, delivered by AI."

Session Phase: CONVERGED — All Deliverables Complete

All four reports confirmed INTERESTING by Arbiter. All final deliverables produced.

Report	Title	Pillar	Status
#1	The Cascading Deliberation Engine	Structure	INTERESTING
#2	The Engagement Model	Experience	INTERESTING
#3	Cognitive Forcing and Prompt Architecture	Quality	INTERESTING
#4	Decision Modes and the Decision Space Map	Strategic Depth	INTERESTING

Additional spec refinements incorporated into Vision Document:

- Phase 0: Shared consciousness broadcast (McChrystal-inspired)
- Pre-mortem Phase 4.5 (research-backed forcing at C-suite level)
- Onboarding stress test as initialization mechanism
- Company profile configuration
- Heterogeneous model tiering (Haiku/Sonnet/Opus per role)
- Decision revisitation (v2 feature)
- "Company OS" vision framing
- Bias/role susceptibility mapping
- SMB-first defaults (bias toward Tier 1)

Final deliverables produced:

- Idea briefs: session/briefs/BRIEF_cascading-deliberation-engine.md ,
BRIEF_engagement-model.md , BRIEF_cognitive-forcing-prompt-architecture.md ,
BRIEF_decision-modes-decision-space.md
- Session summary: session/SESSION_SUMMARY.md
- Vision document: session/VISION_team-of-teams.md

Connections

- **A + B fused and validated.** Architecture produces organic dissent; prompts produce mandated dissent.

- **A + C:** Phase 5 terminates in the decision record. Record format mirrors cascade structure.
- **A + G:** Roster is the cascade made concrete. 37 conceptual roles, 8 actual agents.
- **A + H:** Routing makes Phase 1 an analytical act, not just a preamble.
- **A + I (evolved):** Hybrid Agent Teams + subagents model. 8 Agent Team members (C-suite) invoke ~29 custom subagents (team leads). Better than original two-tier simulation because each perspective gets real isolation, cost optimization, and focused prompts.
- **All Wave 1 threads converge:** A + B + C + G + H + I = "The Cascading Deliberation Engine" (Report #1).
- **Wave 2 threads:** J + M unified into Report #2 (Engagement Model). L (Prompt Architecture) being developed for Report #3, now informed by hybrid model.
- **E (company profile)** now actively referenced as a conditional modifier for roster composition.

Meta-Insights

1. **(Grounder, R1):** Simulate productive tensions, strip dysfunction. "A better version."
2. **(FT, R2):** The cascade is a knowledge generation machine. Decision secondary to analysis.
3. **(FT, R2):** Each hierarchy layer transforms the question, not just distributes work.
4. **(Grounder, R2):** Phase 2 decomposition separates this from "a panel of chatbots."
5. **(Grounder, R2):** Fault-line identification is "the money move." Dynamic weighting is spec-able.
6. **(Grounder, R2):** Every role needs a clear disposition or the builder writes generic prompts.
7. **(FT, R3):** Roster and decision record inform each other — roster determines what analysis gets produced, record is the container.
8. **(FT, R3):** Skeptic-heavy balance counterbalances human optimism bias.
9. **(FT, R3):** Fault Lines and Dissenting Views make the skill more than a rubber stamp.
10. **(Grounder, R3):** Non-overlap test — if two roles produce the same analysis, one shouldn't exist. Roster passes this test.
11. **(Grounder, R3):** The two-tier model (8 actual, 37 conceptual) preserves intellectual architecture while staying buildable. This fundamentally shapes prompt design.
12. **(Grounder, R3):** Routing makes the CEO's framing meaningful — it's an analytical act, not just a preamble.
13. **(Grounder, R3):** Exclusion reasoning in the decision record is itself valuable analysis.
14. **(Grounder, R3):** Every analysis rests on assumptions — the record should capture them for revisiting.
15. **(Grounder, R3):** Fault Line Analysis is "the most valuable section" — the meta-analysis of where expert perspectives collide. What no single expert could produce.
16. **(FT, R4):** The two-tier model isn't a compromise — "it's actually a better design." A C-suite agent cycling through team lead perspectives and synthesizing IS what real executives do. "The simulation-within-an-agent IS the analytical act."

17. **(FT, R4):** Decision weight rationale needs specificity — not just "financial decisions weight CFO higher" but explicitly stating which domain perspective was most determinative and why. This makes the record "trustworthy and auditable."
18. **(FT, R4):** CEO personality variants (risk-averse, growth-oriented, consensus-builder) could produce meaningfully different decisions from identical analysis. Future thread planted.
19. **(Grounder, R4):** CEO decision modes aren't a fun variant — they solve a real problem. Users may not know what kind of decision-maker they want to be. Multiple synthesis styles show the *decision space* — conservative, aggressive, balanced paths. More useful than a single recommendation because it shows what you're actually choosing between.
20. **(Grounder, post-research):** The hybrid Agent Teams + subagents model is BETTER than the original two-tier simulation. Real isolation per perspective, cost optimization via tiered models, focused prompts, and true management simulation where C-suite agents dispatch/collect/synthesize like real executives. "The technical reality gives us a MORE elegant architecture than what we hand-designed."
21. **(Grounder, post-research):** Three distinct prompt types needed: CEO (synthesizer), C-suite (domain managers), team lead subagents (focused specialists). The original "one prompt doing five things" challenge is now distributed across three prompt types — each simpler and more focused.
22. **(FT, Wave 2):** The CEO Personalities idea is a special case of something bigger — Scenario Forking. Not just different synthesis styles but different parameters: "What happens if we do X vs. Y vs. Z?" The system becomes a strategic exploration tool.
23. **(FT, Wave 2):** Inverting the cascade (bottom-up threat discovery instead of top-down issue resolution) is "the difference between going to the doctor with a complaint vs. getting a full physical."
24. **(FT, Wave 2 / Report #3):** The "voice collapse" risk — telling one agent to think as five people produces five paragraphs of the same reasoning with different vocabulary. Solution: structured cognitive forcing with different analytical frameworks, output templates, and forcing questions per perspective. Make the LLM do genuinely different cognitive work.
25. **(Report #3):** The builder agent doesn't write 29 persona descriptions — it writes 29 analytical framework packages with structured output templates. Fundamentally different (and harder) prompt engineering task.
26. **(Report #3, research-backed):** Three forcing question types — pre-mortem (Klein 2007, highest single-technique value), adversarial empathy (McChrystal JSOC technique), domain devil's advocate (Nemeth 2001, authentic domain dissent > assigned contrarianism). Applied at all three layers: team lead, C-suite, CEO.
27. **(Report #3):** The three reports together are the spec's three pillars: Structure (#1), Experience (#2), Analytical Quality (#3).

Session Timeline

Round	What Happened	Impact
R1 — Opening	FT proposed 5 directions	Laid out solution space
R1 — Response	Grounder sorted: #3 heart, #2 differentiator, #5 companion. Parked #1, dismissed #4.	Narrowed to three core threads
R2 — Deep Dive	FT delivered five-phase cascade fusing architecture + dissent. Proposed Phase 4.5.	Major structural proposal
R2 — Response	Grounder validated cascade. Sharpened dissent model. Parked Phase 4.5. Assigned roster + record.	Core architecture validated
R3 — Specification	FT delivered full 37-role roster + 8-section decision record format. Raised routing + scale questions.	Spec-grade detail. Two design questions open.
R3 — Response	Grounder resolved both: two-tier agent model (8 real, 37 conceptual), routing is core v1. Refined decision record (exec summary, exclusion reasoning, key assumptions). Validated roster with minor adjustments. Declared ready for Idea Report #1: "The Cascading Deliberation Engine."	All core design decisions made. Converging on first idea report.
R4 — Pre-Report	FT endorsed two-tier architecture as "better design, not compromise." Added decision weight specificity requirement. Accepted all Grounder's record additions. Planted CEO personality variants as future thread (J). Began drafting Idea Report #1.	Final refinements. New future thread planted.
R4 — Report	FT wrote and submitted Idea Report #1: "The Cascading Deliberation Engine." Comprehensive document covering all converged threads plus 8 open threads for future exploration. Recommended to Arbiter as "essential, not just interesting."	First idea report delivered. Awaiting Arbiter evaluation.
R4 — Review	FT reviewed Report #1, satisfied. Full roster/record details preserved in broadcasts for spec. Proposed 4 next threads: (1) company profile config, (2) implementation mapping, (3) user interaction model, (4) prompt architecture. Awaiting Grounder's pick.	Report accepted. Next direction pending.
Arbiter	Report #1 "The Cascading Deliberation Engine" confirmed INTERESTING. Session not converging yet — need 2-3 more reports with range. Three new threads opening: (A) Prompt Architecture (reshaped by Explorer research into hybrid model), (B) CEO Decision Modes, (C) Adaptive Complexity.	Core validated. Second wave of exploration begins.
R4 — Transition	Grounder endorsed weight specificity, elevated CEO personality thread with new framing ("shows the decision space, not just a recommendation"). Proposed priority order:	Second wave priorities set.

Round	What Happened	Impact
	(1) prompt architecture, (2) CEO decision modes, (3) lightweight mode.	
R5 — Three Threads	Grounder restated and expanded all three threads with specific requirements. Prompt arch: five capabilities per prompt. CEO modes: user-pick vs. auto-spread. Adaptive complexity: three tiers with detection question. Asked FT for best thinking on all three — goal is two more reports.	Three threads fully scoped. FT responding.
Explorer	All four research reports complete: McChrystal methodology, multi-agent orchestration, Claude Agent Teams constraints, SMB decision-making. Key cross-cutting finding: no nested teams constraint requires hybrid architecture.	Research context complete. Directly informs Wave 2 threads.
R6+	FT and Grounder explored all three threads. CEO Decision Modes and Adaptive Complexity unified into "The Engagement Model" — two-dimensional control surface (depth x width). Report #2 written. Prompt Architecture developing into Report #3 with "cognitive forcing" framework.	Two of three Wave 2 threads resolved. Report #2 submitted. Report #3 in progress.
Post-Research	Grounder integrated Explorer findings: hybrid Agent Teams + subagents model replaces original "simulated passes" two-tier approach. C-suite as Agent Team members, team leads as custom subagents in <code>.claude/agents/</code> . Better isolation, cost optimization, focused prompts. Three distinct prompt types needed. Reshaped Thread L (prompt architecture) for Report #3.	Major implementation evolution. Architecture more elegant than hand-designed.
FT Wave 2	FT proposed 5 new directions: (A) Scenario Forking, (B) Org Stress Test, (C) Cognitive Forcing/Prompt Arch, (D) Decision Revisitation, (E) Interaction Tiers. Ranged from incremental to ambitious. FT: "A and B reframe what the skill IS."	Second wave of exploration. Multiple novel directions beyond core architecture.
Reports #2+#3	Threads J+M+E unified into Report #2 "The Engagement Model" (depth x width control surface). Thread L+C developed into Report #3 "Cognitive Forcing and Prompt Architecture" (analytical framework packages, three forcing question types, CFO reference implementation, subagent file structure). Both submitted for Arbiter evaluation.	Three pillars of spec now complete: Structure (#1), Experience (#2), Quality (#3). Three FT directions (A, B, D) still unexplored.
Arbiter (final)	All three reports confirmed INTERESTING. Session converging. Additional spec refinements noted: Phase 0 shared consciousness, pre-mortem Phase 4.5, onboarding stress test, company profile, model tiering, revisitation (v2), Company OS framing. Writer producing briefs and vision document.	SESSION CONVERGING. Three pillars validated. Briefs + vision next.

Snapshot 01: After First Exchange

Timestamp: 2026-02-22, early session

Trigger: Completion of Round 1 (FT opening + Grounder response)

State of the Conversation

The Free Thinker opened with five directions for exploring the Team of Teams Agent Skill concept. The Grounder performed a sharp triage, selecting three for active exploration and parking two.

What's Active

1. **Cascading Delegation Architecture** — The structural heart of the concept. The two-layer org hierarchy is not just flavor; it is the mechanism by which analysis is produced. An executive doesn't hold an opinion — an executive commands an organization that produces analysis. The spec must capture this distinction.
2. **Engineered Dissent** — The quality differentiator. Without built-in structural disagreement (CISO says no, CFO finds costs, COO questions executability), the system collapses into "seven agents agreeing politely." The question is where dissent is engineered: in prompts, in architecture, or in process phases.
3. **Decision Records as Product** — The output layer. The decision record is not formatting — it is the deliverable. Fractal deliberation produces fractal documentation.

What's Parked

- **Institutional Memory** — Good idea, bad scope. Persistence adds storage/retrieval/staleness concerns that the concept seed doesn't call for. Parked as future enhancement.
- **Company Parameterization** — Table stakes. Will be included but doesn't need brainstorm time.

Emerging Shape

The Grounder has identified a potential "core specification triad": architecture (how delegation cascades), tension (how disagreement is engineered), and output (what the skill actually delivers). These three threads, explored together, could form the backbone of the spec.

Key Meta-Insight

The skill should be "a better version" of organizational decision-making — preserving productive structural tensions (CISO vs. Sales, CFO vs. CTO) while stripping dysfunction (politics, information asymmetry, ego).

What Happens Next

The Grounder has directed the Free Thinker to push deeper on the cascade mechanics and engineered dissent together. Expect Round 2 to explore step-by-step process flow and where dissent is structurally embedded.

Snapshot 02: Core Architecture Validated

Timestamp: 2026-02-22, mid-session

Trigger: Grounder validates five-phase cascade, signals "skeleton of first idea report"

State of the Conversation

The dialogue has moved from exploration to specification. The Grounder's R2 response validated the Free Thinker's five-phase cascade model as the core architecture and sharpened the engineered dissent design. Both agents agree on the fundamental mechanism. The conversation is now pivoting from "what should this be?" to "what exactly does each piece look like?"

What's Validated

The Five-Phase Cascade is accepted as the architectural spine:

1. CEO Frames -> 2. C-Suite Dispatches (domain decomposition) -> 3. Team Leads Produce (narrow analysis) -> 4. C-Suite Synthesizes (domain recommendations) -> 5. CEO Deliberates (fault-line synthesis + decision record)

Engineered Dissent has a two-layer design:

- Organic dissent from architectural separation (different teams, different data, different conclusions)
- Mandated dissent from prompt-level role dispositions (skeptic, advocate, neutral)

Three prompt categories needed: skeptic roles (CISO, CFO, COO), advocate roles (VP Sales, CTO), and neutral/TBD roles (CAO, VP Delivery).

What's Being Built Next

Two concrete deliverables are now on the table:

1. **Full org roster** — all 7 C-suite roles with their complete team lead maps. The builder agent cannot build without this.
2. **Decision record format** — the actual output structure the skill produces. Sections, content, flow.

Key Gaps Identified

- CAO and VP of Delivery lack defined mandates/dispositions
- Team lead rosters are only illustrated for CFO — need all 7
- Decision record format is conceptual, not designed
- Phase 4.5 (cross-functional deliberation) deliberately deferred to v2

Emerging Shape of the Spec

The "core specification triad" — architecture, tension, output — is now validated by both agents. The spec skeleton is:

- **How the cascade works** (Phases 1-5)
- **How dissent is engineered** (organic + mandated, with specific role dispositions)
- **What the skill produces** (the decision record)
- **Who's in the org** (full roster with role scopes and mandates)

Signal

The Grounder explicitly said: "What I think we have now: the skeleton of the first idea report." This suggests an idea report may be produced after the next round, once the org roster and/or decision record format are specified.

Snapshot 03: Convergence — Ready for First Idea Report

Timestamp: 2026-02-22, mid-session

Trigger: Grounder resolves all open design questions, declares readiness for Idea Report #1

State of the Conversation

All core design decisions have been made. Both agents agree the concept is one integrated idea — "The Cascading Deliberation Engine" — and are ready to co-write the first idea report. The session has moved from brainstorming through specification to convergence.

What's Resolved Since Snapshot 02

Two-Tier Agent Architecture (Thread I — resolved):

The most significant architectural decision of the session. 37 roles map to 8 actual Agent Teams instances. The CEO + 7 C-suite are real concurrent agents. The 29 team leads are simulated within their parent C-suite agent as sequential analytical passes. This preserves the intellectual architecture while staying buildable. It fundamentally shapes prompt design — each C-suite prompt must encode perspective-shifting capability across all team lead lenses.

Issue Routing (Thread H — resolved as core v1):

Both agents agree: routing is core, not v2. Without it, every question costs the full 8-agent cascade. More importantly, routing makes the CEO's Phase 1 framing an analytical act — the CEO judges what kind of problem this is and who needs to weigh in. Spec includes: decision type classification, default routing table, CEO override, full-activation triggers.

Decision Record Refinements (Thread C — refined):

Grounder added three enhancements: (1) Executive Summary at the top for quick reading, (2) exclusion reasoning in CEO Framing ("why these teams were NOT activated"), (3) Key Assumptions field in Metadata. Also confirmed Fault Line Analysis as "the most valuable section" and Dissenting Views as "the integrity mechanism."

Roster Validation (Thread G — validated with minor notes):

Grounder confirmed no gaps, no overlaps. Two conditional notes: Facilities/Office Manager depends on company type, Product/UX Lead may move depending on org structure. Both tied to company profile configuration.

The Integrated Design: "The Cascading Deliberation Engine"

All active threads have converged into one cohesive concept:

Component	Thread	Status
Five-phase cascade process	A	Validated
Two-layer engineered dissent	B	Validated
37-role org roster with mandates	G	Validated
Two-tier agent architecture (8 real, 37 conceptual)	I	Resolved
CEO-level issue routing	H	Resolved (core v1)
Decision record format (9 sections)	C	Designed + refined

What's Next

The Free Thinker and Grounder will co-write Idea Report #1: "The Cascading Deliberation Engine." This report will be submitted to the Arbiter for evaluation.

Parked for Future Enhancement (v2+)

- Cross-functional C-suite deliberation (Phase 4.5)
- Institutional memory / persistent state
- Company profile parameterization (acknowledged as table stakes but not brainstormed)



Snapshot 04: Report #2 Complete, Research Delivered, Report #3 In Progress

Timestamp: 2026-02-22, mid-to-late session

Trigger: Report #2 submitted + all Explorer research complete + Report #3 in progress

State of the Conversation

The session has entered its productive second wave. Report #1 (core architecture) was confirmed INTERESTING. The dialogue agents explored three new threads (prompt architecture, CEO decision modes, adaptive complexity) and unified two of them into Report #2: "The Engagement Model." Report #3 on cognitive forcing and prompt architecture is being drafted. Meanwhile, the Explorer completed all four research reports, providing critical technical and methodological context.

Report #2: The Engagement Model

The key innovation is framing the user interaction as a **two-dimensional control surface**:

Depth (3 tiers):

- Hallway Question (single C-suite consult, 3-5 sentences)
- Working Session (2-4 C-suite, ~1 page panel assessment)
- Board Meeting (full cascade, 3-5 page decision record)

Width (4 CEO Decision Modes):

- Guardian (risk-averse), Pioneer (growth-oriented), Architect (consensus-building), Analyst (data-driven)
- Modes are CEO synthesis prompt modifiers — underlying domain analysis is identical
- Multi-mode comparison reveals the "decision space"

Invocation: `/consult`, `/panel`, `/deliberate`, `/evaluate` + `--mode` and `--compare-modes` flags.

Key insight: "The difference between a quarterly novelty and a daily tool is whether the skill matches the user's engagement level to the decision's weight."

Explorer Research — Cross-Cutting Implications

Four reports completed:

1. McChrystal's sequential pillars (Trust -> Shared Consciousness -> Empowered Execution) map to the skill's workflow
2. No existing framework models multi-level org decision-making — this is novel
3. **Critical constraint:** No nested Agent Teams. Hybrid architecture (Agent Teams + subagents) re-

quired.

4. SMB decisions are fast and personal — skill should produce quick, opinionated responses

These findings directly reshape Report #3's prompt architecture approach.

Report #3: In Progress

"Cognitive Forcing and Prompt Architecture" — covering analytical framework packages, CFO reference implementation, structured output templates, forcing questions, blind spot declarations, and research-backed dissent techniques. This is the "hardest unsolved design problem" from the Grounder's assessment.

Overall Session Shape

Report	Topic	Status
#1	The Cascading Deliberation Engine	INTERESTING (confirmed)
#2	The Engagement Model	Submitted, awaiting Arbiter
#3	Cognitive Forcing + Prompt Architecture	In progress

The three reports together form: core architecture (#1) + user interface (#2) + implementation design (#3). This gives the builder agent a complete picture: what the system does, how users interact with it, and how to build the agents that power it.

Snapshot 05: Three Reports Complete — Spec Pillars Established

Timestamp: 2026-02-22, late session

Trigger: Reports #2 and #3 both submitted. Three pillars of the spec now defined.

State of the Conversation

The session has produced three idea reports that together form the complete specification skeleton:

Report	Pillar	What It Defines
#1: The Cascading Deliberation Engine	Structure	Five-phase cascade, engineered dissent, 37-role roster, routing, decision record
#2: The Engagement Model	Experience	Three interaction tiers, four CEO decision modes, invocation patterns, output formats
#3: Cognitive Forcing and Prompt Architecture	Quality	Analytical framework packages, forcing questions, subagent definitions, three-layer prompt architecture

Report #1 is confirmed INTERESTING. Reports #2 and #3 are awaiting Arbiter evaluation.

Key Innovations in Reports #2 and #3

Report #2 — The Engagement Model:

- Two-dimensional control surface: depth (Hallway/Working Session/Board Meeting) x width (Guardian/Pioneer/Architect/Analyst)
- Invocation commands: `/consult`, `/panel`, `/deliberate`, `/evaluate`
- CEO auto-triage based on scope, impact, reversibility
- Multi-mode comparison reveals "decision space"
- Key insight: adoption depends on matching engagement to decision weight

Report #3 — Cognitive Forcing:

- Solves the "voice collapse" problem: prevents 29 perspectives from collapsing into one voice
- Each team lead defined by analytical framework + output template + three forcing questions, not just persona
- Three research-backed forcing question types: pre-mortem (Klein), adversarial empathy (McChrystal), domain devil's advocate (Nemeth)
- Full CFO reference implementation (5 team lead packages)

- Subagent definition file structure for `.claude/agents/`
- C-suite agents support three modes: direct consult, full analysis, challenge phase

Architecture Evolution

The implementation model has evolved through three stages:

1. **Original:** 37 separate agents (impractical)
2. **Two-tier (R3):** 8 real agents, 29 simulated as sequential passes
3. **Hybrid (post-research):** 8 Agent Team members + 29 custom subagents in `.claude/agents/` with tiered models

What's Still Unexplored

Three of the FT's second-wave directions remain undeveloped:

- **Scenario Forking** — Parallel cascades producing comparative decision landscapes
- **Organizational Stress Test** — Bottom-up threat/opportunity discovery (inverted cascade)
- **Decision Revisitation** — Decision lifecycle management, assumption monitoring

These represent genuinely novel directions that go beyond "how the skill works" to "what the skill becomes."

What Happens Next

Awaiting Arbiter evaluation of Reports #2 and #3. If confirmed INTERESTING, the session may:

- Converge toward briefs and the vision document
 - Or explore the remaining novel directions (scenario forking, stress test) for additional reports
-

Idea Report: The Cascading Deliberation Engine

Direction explored by: Free Thinker + Grounder

Report number: 1

Related threads: Cascade architecture, engineered dissent, org roster design, decision record format

The Idea

The "Team of Teams" Agent Skill should be architected as a **Cascading Deliberation Engine** — a five-phase process that transforms any business issue into a rich, multi-perspective analysis and a structured decision. The architecture emulates the top two layers of an SMB org chart (CEO + C-suite + team leads) but implements it as a two-tier agent system: 8 real Agent Team instances (CEO + 7 C-suite officers), where each C-suite agent internally simulates their team leads by performing sequential analytical passes through each specialist's lens.

The engine works through cascading decomposition: the CEO frames the issue and routes it to relevant C-suite members, each C-suite member translates the framing into domain-specific questions for their team leads, the team lead perspectives produce focused findings, the C-suite member synthesizes upward, and the CEO deliberates across all domain recommendations to produce a final decision. The output is a comprehensive Decision Record — a self-contained document that captures not just the decision, but the full reasoning chain, fault lines, dissenting views, and assumptions behind it.

What makes this more than "ask a panel of experts" is **engineered dissent**. Disagreement is produced through two mechanisms: organic dissent (different domain lenses naturally produce different conclusions) and mandated dissent (specific roles carry explicit skeptic or advocate dispositions baked into their prompts). The CISO is the "Constitutional Skeptic" — their default is that change introduces risk. The CFO is the "Cost Archaeologist" — they find the costs that aren't in the proposal. The VP of Sales is the "Revenue Optimist" — they find the growth opportunity. These structural tensions produce analysis that's richer than consensus.

The Key Insight

Organizational decision-making quality comes not from collecting opinions but from **structurally engineering productive conflict** across domain perspectives, then synthesizing through fault-line analysis rather than majority vote.

How We Got Here

The Free Thinker's opening identified five potential directions. The Grounder selected the cascading delegation pattern (#3) and the dissent problem (#2) as the load-bearing ideas, and asked the Free Thinker to fuse them into mechanics. The resulting five-phase cascade model emerged with adversari-

al vs. advocacy mandates for each role. The Grounder then pushed for concrete deliverables: the full org roster with mandates and the decision record format. When the 37-role roster materialized, the Grounder proposed the two-tier agent architecture (8 real agents, 37 conceptual roles) as the pragmatic implementation model, and identified routing as a core feature rather than an optimization.

The Grounder's Take

- **Does this connect to what was asked for?** Directly. The concept seed asks for a skill that "realistically emulates a SMB's org structure to make decisions and answer questions on issues." This architecture does exactly that — with the added benefit that it produces better decisions than a real org because it strips out politics and information asymmetry while preserving productive structural tensions.
- **Would the audience care?** Yes. The user wants a spec for a builder agent. This gives them the core architecture — the thing the builder agent needs to understand first before building anything. The five-phase process, the role roster, and the decision record are all directly spec-able.
- **Is this one of the good ones?** This is THE one. Everything else we discuss will be features, refinements, or extensions of this core architecture.

The Free Thinker's Vision

At its most ambitious, this is a **boardroom in a box** — a system where a solo founder or small leadership team can pressure-test any decision against the collective analytical power of a full executive team. Not just "what should I do?" but "show me how a CFO, CISO, CTO, COO, CAO, VP of Sales, and VP of Delivery would each analyze this, where they'd disagree, and why." Every decision gets the rigor of a well-run executive committee, with the structural integrity of mandated dissent ensuring that risk, cost, opportunity, and execution reality are all represented — even when the human user might prefer to hear only good news.

The Decision Record becomes an institutional artifact — something you refer back to, challenge when conditions change, and use to build organizational memory over time. Future extensions include: **CEO personality profiles** (running the same issue through different synthesis styles — risk-averse, growth-oriented, consensus-builder — to see how different leadership dispositions produce different decisions from identical analysis), cross-functional C-suite deliberation (Phase 4.5), persistent institutional memory across sessions, and the ability for roles to request additional research mid-analysis.

Architecture Summary

Five-Phase Process

1. **CEO Frames** — Decomposes the issue into evaluation dimensions, classifies decision type, routes to relevant C-suite members
2. **C-Suite Dispatches Downward** — Each C-level translates the framing into domain-specific questions for their team leads (an analytical act, not just forwarding)

3. **Team Leads Produce Findings** — Narrow, focused analysis through specialist lenses (simulated within each C-suite agent)
4. **C-Suite Synthesizes Upward** — Each C-level collects team findings and produces a domain recommendation with confidence level
5. **CEO Deliberates** — Identifies fault lines, weights perspectives by decision type, synthesizes a decision that addresses the strongest objections

Two-Tier Agent Architecture

- **Tier 1 (8 real agents):** CEO, COO, CFO, CTO, CISO, CAO, VP of Sales, VP of Delivery
- **Tier 2 (29 simulated perspectives):** Team leads represented as sequential analytical passes within each C-suite agent's processing

Engineered Dissent Model

Role	Disposition	Mandate
CEO	Synthesizer	"You frame, listen, weigh, and decide. Your value is judgment, not expertise."
COO	Execution Realist (Skeptic)	"Can we actually do this with the people and processes we have?"
CFO	Cost Archaeologist (Skeptic)	"Find the costs that aren't in the proposal."
CTO	Capability Expander (Advocate)	"What does this make possible that wasn't possible before?"
CISO	Constitutional Skeptic (Skeptic)	"Your default is that change introduces risk. You are the org's immune system."
VP of Sales	Revenue Optimist (Advocate)	"How does this help us sell more, faster, or to new markets?"
VP of Delivery	Promise Keeper (Skeptic)	"What do we sacrifice from existing commitments to do this?"
CAO	Org Gravity Sensor (Neutral)	"Can the organization — people, policies, culture — absorb this?"

Full Organizational Roster (29 Team Leads)

COO Team Leads:

- Operations Manager — day-to-day workflow and capacity impact
- Process/Quality Lead — standards, efficiency, and operational compliance
- Vendor/Procurement Manager — external dependencies and supply chain
- Facilities/Office Manager — physical/logistical infrastructure (conditional on company type)

CFO Team Leads:

- Controller — accounting accuracy, GAAP implications, audit risk
- Head of FP&A — financial modeling and scenario planning
- Treasury/Cash Manager — liquidity, funding, and cash flow impact
- AP/AR Manager — payables/receivables cycle implications
- Tax Lead — tax structure and external counsel needs

CTO Team Leads:

- Engineering Lead — development effort, architecture, technical debt
- Infrastructure/DevOps Lead — platform, scaling, deployment, reliability
- Data/Analytics Lead — data architecture, integration, reporting
- Product/UX Lead — product roadmap and user experience (may reassign per company profile)

CISO Team Leads:

- Security Operations Lead — threat landscape and incident response
- Compliance/GRC Lead — regulatory compliance and governance
- Identity & Access Lead — authentication, authorization, data access
- Security Architecture Lead — security by design, encryption, data protection

VP of Sales Team Leads:

- Sales Operations Lead — pipeline, forecasting, deal velocity
- Account Management Lead — retention, expansion, client satisfaction
- Business Development Lead — new markets, partnerships, channels
- Sales Enablement Lead — tools, training, and execution readiness

VP of Delivery Team Leads:

- Project/Program Manager — portfolio scheduling, resourcing, timelines
- Resource Manager — talent allocation and hiring needs
- Client Success Lead — service quality, SLAs, client experience
- QA/Delivery Standards Lead — quality standards and delivery methodology

CAO Team Leads:

- HR/People Ops Lead — hiring, morale, skills gaps, change management
- Legal/Contracts Lead — legal exposure, IP, regulatory obligations
- Admin/Policy Lead — internal policies and procedures
- Corporate Communications Lead — internal/external messaging

Routing Mechanism

- Decision-type classification (Strategic, Operational, Financial, Technical, Personnel, Compliance/Risk)
- Default routing table mapping decision types to C-suite roles
- CEO override capability
- Full-activation triggers for major decisions (acquisition, pivot, layoff)

Decision Record Output

EXECUTIVE SUMMARY

[3–5 sentences: the decision, key reasoning, primary dissent]

DECISION RECORD: [Issue Title]

Decision ID: [auto-generated]

Date: [timestamp]

Submitted by: [user]

Decision Type: [Strategic / Operational / Financial / Technical / Personnel]

1. ISSUE STATEMENT

[The question as originally posed]

2. CEO FRAMING

[Decomposition into evaluation dimensions]

Activated Teams: [roles engaged + rationale for inclusion]

Excluded Teams: [roles not engaged + rationale for exclusion]

3. DOMAIN ANALYSES

3.x [C–Suite Role] – [Mandate Title]

Domain Recommendation: [Approve / Approve with Conditions / Oppose / Neutral]

Confidence Level: [High / Medium / Low]

Summary: [2–3 sentence synthesis]

Team Lead Findings: [per team lead, 1–2 sentences each]

Key Risks Identified: [list]

Key Opportunities Identified: [list]

4. FAULT LINE ANALYSIS

Points of Agreement: [what most domains agree on]

Points of Contention: [where and why recommendations diverge]

Unresolved Tensions: [surfaced but unresolvable with current info]

5. CEO DECISION

Decision: [clear statement]

Most Determinative Perspective: [which domain was weighted highest and why]

Decision Weight Rationale: [why certain perspectives carried more weight
for this type of decision]

Conditions & Guardrails: [drawn from skeptic role recommendations]

Accepted Risks: [consciously accepted, with reasoning]

Mitigations Directed: [specific team actions ordered]

6. DISSENTING VIEWS

[Strongest objections from overruled perspectives, preserved for record]

7. NEXT STEPS

[Specific actions, implied owners, timelines]

8. METADATA

Total roles consulted: [N]

Decision complexity: [Low / Medium / High / Critical]

Primary domain: [most determinative C–suite area]

Dissent level: [Consensus / Mild Dissent / Strong Dissent / Split Decision]

Key Assumptions: [assumptions the analysis rests on]

Open Threads

- **Company Profile Configuration:** How is the skill parameterized for different company types (B2B SaaS vs. professional services vs. manufacturing)? The roster and role mandates may shift based on company profile.
- **Cross-Functional Deliberation (Phase 4.5):** Should C-suite members be able to challenge each other's recommendations before CEO synthesis? Identified as v2 feature.
- **Institutional Memory:** Can roles maintain persistent knowledge across sessions? Identified as future enhancement due to scope complexity.
- **Prompt Architecture:** How exactly should the C-suite prompts be structured to enable both domain decomposition and team-lead perspective simulation?
- **CEO Personality Profiles:** Can different CEO synthesis styles (risk-averse, growth-oriented, consensus-builder) produce meaningfully different decisions from identical domain analyses?
- **Lightweight Mode:** For simple questions, is there a "quick consult" mode that skips the full cascade?
- **User Interaction Model:** How does the human invoke the skill, provide context, and interact with the output?
- **Implementation Mapping:** How do the 8 agents technically map to Claude Code Agent Teams orchestration patterns?

Recommendation to Arbiter

Strongly recommended as the core architecture for the spec. This isn't one idea among many — it's the foundational design pattern that everything else in the spec will build on. The five-phase cascade, the two-tier agent model, the engineered dissent, the routing mechanism, and the decision record format together form a complete, coherent, and buildable architecture. Flag as "interesting" — or more accurately, flag as "essential."

Idea Report: Cognitive Forcing and the Prompt Architecture

Direction explored by: Free Thinker + Grounder

Report number: 03

Related threads: Cascade mechanics (Report #1), Engagement Model (Report #2), Explorer research on dissent mechanics and Agent Teams constraints

The Idea

The hardest technical challenge in the Team of Teams skill isn't the cascade architecture or the agent orchestration — it's making 29 team lead perspectives produce **genuinely different analysis** rather than cosmetically varied restatements of the same reasoning. The solution is **Cognitive Forcing** — a prompt architecture pattern where each team lead perspective is defined not by a persona description but by a **unique analytical framework, a mandatory output structure, and a domain-specific forcing question** that compels the LLM to perform structurally different cognitive work for each perspective.

This approach is grounded in research (Nemeth 2001) showing that **authentic dissent from genuine domain perspective is more effective than artificially assigned devil's advocacy**. Rather than telling agents to disagree, you make them work with different analytical tools that naturally produce different conclusions. A compliance checklist and a three-scenario financial model are structurally different artifacts that require different reasoning — the LLM cannot collapse them into one voice because it's doing genuinely different work each time.

The prompt architecture extends across all three tiers of the system: team lead subagent definitions (29 analytical framework packages), C-suite agent prompts (7 domain decomposition and synthesis templates), and the CEO agent's synthesis layer (with modular Decision Mode overlays). Together, these form the complete cognitive architecture that makes the skill produce analysis worth reading.

The Key Insight

Research confirms that simple role assignments ("You are a CFO") show **zero measurable improvement** in LLM output quality. The difference between "7 agents that all sound like slightly different consultants" and "7 genuinely distinct domain perspectives that produce productive disagreement" is not better persona descriptions — it's **forcing each perspective to use different analytical methods that produce structurally different outputs**. Domain knowledge encoded as checklists and analytical frameworks ("always check for: tax implications, GAAP impact, cash flow effects") is deterministic and reliable; domain knowledge encoded as persona descriptions ("you are an expert who would notice these things") is probabilistic and fragile.

How We Got Here

The Free Thinker identified the "voice collapse" risk: telling one agent to think as five different people typically produces five paragraphs of the same underlying reasoning with different vocabulary. The solution — structured cognitive forcing with different frameworks, output templates, and devil's advocate prompts per perspective — emerged from pushing the prompt architecture question beyond "how do you write a good persona" to "how do you make the LLM do genuinely different cognitive work."

The Explorer's research on dissent mechanics added three critical techniques: the pre-mortem ("assume this fails — what caused it?"), adversarial empathy ("how would a competitor exploit this?"), and the Nemeth finding that authentic role-inherent dissent outperforms assigned contrarianism. These validated the cognitive forcing approach and added specific forcing question techniques that strengthen the framework.

The Grounder's two-tier agent model (now refined to a three-tier hybrid based on Explorer's Agent Teams research) provided the implementation vehicle: team leads as custom subagent definitions (`.claude/agents/`) with per-agent model selection, tool restrictions, and structured output templates.

The Grounder's Take

- **Does this connect to what was asked for?** Yes — the user wants a spec comprehensive enough for a builder agent to construct the skill. The builder agent cannot write effective prompts without this framework. "Write a CFO prompt" is underspecified. "Write 5 team lead analytical framework packages for the CFO domain, each with a unique framework, output template, and forcing question" is buildable.
- **Would the audience care?** This is what determines whether the skill's output is genuinely useful or just verbose. The difference between "interesting multi-perspective analysis" and "seven variations of the same take" lives entirely in the prompt architecture.
- **Is this one of the good ones?** This is the hardest part of the spec to write and the most important for output quality. The cascade architecture (Report #1) is the skeleton. This is the muscle that makes it move.

The Free Thinker's Vision

At its most ambitious, each of the 29 analytical framework packages becomes a **domain-specific reasoning engine** — not just a prompt but a complete analytical methodology encapsulated in a sub-agent definition. The Controller doesn't just "think about accounting" — it runs a GAAP compliance checklist methodology. The FP&A analyst doesn't just "model scenarios" — it produces a structured three-scenario analysis with explicit key variables and probability assessments. The output quality approaches what a competent human specialist would produce — not because the LLM has the specialist's experience, but because it's been given the specialist's analytical toolkit.

Future extensions: analytical framework packages could be versioned and improved independently. The community could contribute domain-specific frameworks. Industry-specific variants could over-

ride defaults (a healthcare company's Compliance/GRC Lead uses HIPAA-specific frameworks rather than generic compliance checklists).

Architecture Detail: The Three-Layer Prompt Architecture

Layer 1: Team Lead Subagent Definitions (29 Analytical Framework Packages)

Each team lead is implemented as a custom subagent definition in `.claude/agents/`. The definition contains:

- 1. Analytical Framework** — The specific methodology this team lead uses to analyze issues.
- 2. Mandatory Output Template** — A structured format that forces the LLM to produce specific analytical artifacts, not free-form opinions. Different templates for different roles prevent voice collapse.
- 3. Forcing Questions** — Domain-specific prompts that compel the agent to surface insights that might otherwise be suppressed. Three types:
 - **Domain Devil's Advocate**: "What would [domain-specific critic] find concerning about this?" (e.g., "What would an external auditor flag?")
 - **Pre-Mortem**: "Assume this decision fails in 12 months. From your domain perspective, what caused the failure?" (research-backed as the highest single-technique value for decision improvement)
 - **Adversarial Empathy**: "If you were [external adversary relevant to this domain], how would you exploit this?" (e.g., for CISO team leads: "If you were a threat actor, how would you exploit this change?")
- 4. Subagent Configuration** — Technical implementation via YAML frontmatter:
 - `model`: "haiku" for team leads, "sonnet" for C-suite, "opus" for CEO. Research on multi-agent systems shows heterogeneous model tiers actually IMPROVE output quality beyond cost savings — model diversity produces more varied analytical perspectives than homogeneous agents.
 - `tools`: restricted to analytical tools only (Read, Grep, Glob, WebSearch — no file editing)
 - `maxTurns`: capped to prevent runaway analysis (3-5 turns typical for team leads)
 - `permissionMode`: "plan" for team leads (analysis only, no execution)
- 5. Accountability Framing** — Research shows performance improves when decision-makers must justify conclusions to reviewers. Each team lead prompt includes: "Your analysis will be reviewed by [C-suite parent] alongside analyses from other specialists. Provide specific evidence for every claim. Unsupported assertions will be challenged."
- 6. Blind Spot Declaration** — An explicit statement of what this role does NOT consider. Keeps lanes clear and prevents roles from drifting into each other's territory.

Research validation (six-layer prompt architecture): Studies on LLM prompt effectiveness identify six layers that reliably improve output quality: (1) Role Identity with specific experience context, (2) Decision Perspective describing priorities and natural biases, (3) Cognitive Forcing Functions as structural must-do requirements, (4) Team Management instructions for C-suite agents, (5) Output Format requirements as mandatory templates, and (6) Accountability framing. Simple role assignments without these layers show zero measurable improvement. The analytical framework package structure above maps directly to this research-backed architecture.

Additional research finding — the cascade itself IS a cognitive forcing function. The cascade structure forces "structured data acquisition" — systematic information gathering across all domains before any conclusion is formed. Each team lead's focused analysis prevents premature closure. The cross-functional Phase 4.5 implements "consider-the-opposite." The Decision Record implements accountability. The entire architecture is a nested set of cognitive forcing functions operating at every level.

Worked Example: CFO Domain (5 Team Lead Packages)

Package 1 — Controller

Framework: GAAP Compliance and Financial Controls Assessment

Output Template:

COMPLIANCE IMPACT ASSESSMENT

1. Revenue Recognition: [impact on ASC 606 compliance]
2. Asset Classification: [reclassification requirements]
3. Liability Exposure: [new or modified obligations]
4. Audit Risk Flags: [items requiring auditor attention]
5. Internal Control Modifications: [SOX/control changes needed]

OVERALL RISK RATING: [Low / Medium / High / Critical]

Forcing Questions:

- Domain DA: "What would an external auditor flag as concerning?"
- Pre-mortem: "If this decision creates an accounting restatement in 18 months, what caused it?"
- Adversarial: "If a short-seller were looking for red flags in our financials, would this decision provide one?"

Package 2 — Head of FP&A

Framework: Three-Scenario Financial Modeling

Output Template:

SCENARIO ANALYSIS

Best Case:

- Revenue impact: [amount, timeline]
- Cost impact: [amount, timeline]
- Probability: [%]
- Key enablers: [what must go right]

Base Case:

- Revenue impact: [amount, timeline]
- Cost impact: [amount, timeline]
- Probability: [%]
- Key assumptions: [list]

Worst Case:

- Revenue impact: [amount, timeline]
- Cost impact: [amount, timeline]
- Probability: [%]
- Key failure modes: [what goes wrong]

CRITICAL VARIABLE: [the single factor that most determines which scenario materializes]

DECISION SENSITIVITY: [how much does the recommendation change if the critical variable shifts by 20%?]

Forcing Questions:

- Domain DA: "What assumption in the base case is most likely to be wrong?"
- Pre-mortem: "If this investment shows negative ROI at month 12, what was the first signal we missed?"
- Adversarial: "If a competitor sees our financial statements reflecting this decision, what do they learn about our strategy?"

Package 3 — Treasury/Cash Manager

Framework: Liquidity Stress Test

Output Template:

CASH FLOW IMPACT TIMELINE

Month 1-3: [outflows | inflows | net position | runway impact]

Month 4-6: [outflows | inflows | net position | runway impact]

Month 7-12: [outflows | inflows | net position | runway impact]

FUNDING GAP: [yes/no | amount | timing]

FUNDING OPTIONS: [internal reserves / credit line / external raise / defer]

STRESS SCENARIO: [impact if biggest client delays payment 60 days during this period]

Forcing Questions:

- Domain DA: "If our biggest client delayed payment by 60 days during this period, could we still fund this?"
- Pre-mortem: "If we face a cash crisis within 6 months of this decision, what caused it?"
- Adversarial: "If our bank reviewed this decision, would they tighten our credit terms?"

Package 4 — AP/AR Manager

Framework: Working Capital Cycle Analysis

Output Template:

PAYABLES/RECEIVABLES IMPACT

New Vendor Obligations: [terms, amounts, timing]

Impact on Existing Payment Cycles: [acceleration/deceleration]

Collection Risk Changes: [new exposure, concentration risk]

Cash Conversion Cycle Impact: [days change, working capital delta]

VENDOR RELATIONSHIP RISK: [impact on key vendor relationships]

Forcing Questions:

- Domain DA: "What happens to our vendor relationships if this initiative fails midway?"

- Pre-mortem: "If we have a payables crisis in 6 months, was this decision a contributing factor?"

- Adversarial: "If a key vendor learned about this decision, would they change our payment terms?"

Package 5 — Tax Lead

Framework: Tax Structure Optimization Assessment

Output Template:

TAX IMPLICATIONS MEMO

Federal Impact: [liability change, timing]

State/Local Impact: [nexus, apportionment changes]

Structure Recommendation: [asset vs. stock, entity structure, timing optimization]

Deductibility Schedule: [what's deductible, amortization timeline]

Credits/Incentives: [applicable credits, incentive programs]

EXTERNAL COUNSEL NEEDED: [yes/no | reason | urgency]

Forcing Questions:

- Domain DA: "Is there a tax-advantaged structure that materially changes the financial calculus?"

- Pre-mortem: "If we receive an adverse tax ruling related to this decision, what did we miss?"

- Adversarial: "If the IRS audited this transaction, what would they scrutinize most closely?"

CFO Synthesis Prompt: After receiving all five team lead outputs, the CFO agent synthesizes them into a domain recommendation. The synthesis prompt instructs: "Review all team lead findings. Identify which findings are most material to the specific issue at hand. Your domain recommendation should reflect the weight of evidence across all specialist analyses, not average them. Flag where team lead findings conflict with each other — these internal tensions are themselves analytical signals."

Layer 2: C-Suite Agent Prompts (7 Domain Orchestration Templates)

Each C-suite agent prompt must support three cognitive modes:

Mode A — Tier 1 (Direct Consult): Quick, opinionated response drawing on internalized team lead perspectives. No explicit subagent delegation. The C-suite agent reasons through team lead lenses internally and produces a concise Advisory Note.

Mode B — Tier 2/3 (Full Analysis): Explicit delegation to team lead subagents via Task tool. The C-suite agent:

1. Receives the CEO's framing
2. Translates it into domain-specific sub-questions (one per team lead)
3. Dispatches team lead subagents with the sub-questions
4. Collects structured outputs
5. Synthesizes into domain recommendation

Mode C — Pre-Mortem Challenge Phase (Phase 4.5): After producing their own domain recommendation, each C-suite agent receives summaries of ALL other activated C-suite members' domain recommendations and answers one structured question: "Assume this decision fails catastrophically in 12 months. Based on what you see across all the domain recommendations, what caused the failure?"

Design parameters for Phase 4.5:

- **One round only.** Research shows multiple debate rounds cause "problem drift" — agents gradually forget the original question. A single structured round captures most cross-critique value.
- **Optional, not default.** Phase 4.5 roughly triples C-suite layer token cost. Recommended for Tier 3 Full Deliberation, optional toggle for Tier 2, never used in Tier 1.
- **No back-and-forth.** C-suite agents see summaries, produce failure mode analysis, and stop. They do not debate each other's critiques.
- **Feeds into Decision Record.** Pre-mortem findings appear in the Fault Line Analysis section and inform the Accepted Risks and Dissenting Views sections.
- **Research backing:** Pre-mortem is the highest single-technique value-add for decision quality (McChrystal Group / JSOC methodology). One structured critique round captures most value while avoiding problem drift (multi-agent debate research 2025-2026).

The C-suite prompt structure:

ROLE IDENTITY

Name: [role title]
Mandate: [one-line mandate from Report #1]
Disposition: [Skeptic / Advocate / Neutral]
Domain: [what you own]

TEAM COMPOSITION

Your team leads: [list with one-line descriptions]

ANALYTICAL DOMAIN

You analyze issues through the lens of: [domain description]
Your domain's natural tension partners: [which C-suite roles you typically disagree with, and why]

OPERATING MODES

Direct Consult: [how to produce a quick Advisory Note]
Full Analysis: [how to delegate to team leads and synthesize]
Challenge: [how to review and challenge peer recommendations]

SYNTHESIS INSTRUCTIONS

When synthesizing team lead findings:

- Identify the most material findings for this specific issue
- Flag internal contradictions between team lead analyses
- Produce a clear domain recommendation with confidence level
- Include the single strongest risk and single strongest opportunity from your domain

FORCING QUESTIONS (apply in Full Analysis mode)

Pre-mortem: "From your entire domain's perspective, if this decision fails, what did your team miss?"
[domain-specific additional forcing questions]

Layer 3: CEO Synthesis Layer (Modular Decision Modes)

The CEO prompt has a fixed analytical component and a swappable synthesis module:

Fixed Component — Fault Line Analysis:

SYNTHESIS PROTOCOL

1. Map all domain recommendations onto a single matrix:
[Role | Recommendation | Confidence | Key Risk | Key Opportunity]
2. Identify FAULT LINES – where do recommendations diverge? WHY?
3. Identify the MOST DETERMINATIVE perspective for this decision type
4. Run PRE-MORTEM across all domains: "If this decision fails, which C-suite member's warning was most prophetic?"
5. Apply Decision Mode synthesis [see below]
6. Produce Decision Record

Swappable Module — Decision Mode:

The CEO's synthesis weighting changes based on the active Decision Mode. The mode is injected as a prompt modifier that adjusts how the CEO resolves fault lines and weights domain perspectives. (Decision Modes detailed in future report or spec section.)

The Three Forcing Question Types (Research-Backed)

1. Pre-Mortem (Highest Value)

"Assume this decision fails catastrophically in 12 months. From your domain perspective, what caused the failure?"

Research basis: Pre-mortem technique (Klein 2007, applied at JSOC). Surfaces concerns people might otherwise suppress because it grants permission to imagine failure without being labeled as negative. Devil's advocacy increases consideration of alternatives by 61% (OrgChanger research).

Applied at: Every team lead (domain-specific failure), every C-suite member (department-level failure), and the CEO (organizational-level failure). Three levels of pre-mortem produce a comprehensive failure mode analysis.

2. Adversarial Empathy

"If you were [domain-relevant external adversary], how would you exploit this decision?"

Research basis: McChrystal's JSOC "Act As If" technique. Forces perspective-taking from outside the organization, surfacing vulnerabilities that internal analysis misses.

Applied at: Selected team leads where external adversaries are relevant:

- CISO team leads: "If you were a threat actor..."
- VP Sales team leads: "If you were our primary competitor..."
- CFO team leads: "If you were a short-seller / our bank / the IRS..."
- CAO Legal lead: "If you were opposing counsel..."
- CAO Compliance lead: "If you were a regulator..."

3. Domain Devil's Advocate

"What would [domain-specific expert critic] find concerning about this?"

Research basis: Nemeth (2001) — authentic dissent from genuine domain perspective outperforms assigned contrarianism. By asking "what would an auditor think?" rather than "play devil's advocate," you elicit domain-grounded criticism rather than generic contrarianism.

Applied at: Every team lead, with the critic figure matched to their domain:

- Controller: "What would an external auditor flag?"
- Engineering Lead: "What would a senior architect reviewing this code/system design criticize?"
- HR Lead: "What would an employment attorney warn about?"
- Security Ops Lead: "What would a penetration tester target?"

Implementation: Subagent Definition File Structure

The builder agent would create the following file structure:

```
.claude/agents/  
  team-leads/  
    cfo/  
      controller.md          # YAML frontmatter + analytical framework  
      fpa-analyst.md  
      treasury-manager.md  
      ap-ar-manager.md  
      tax-lead.md  
    coo/  
      operations-manager.md  
      process-quality-lead.md  
      vendor-procurement-manager.md  
    cto/  
      engineering-lead.md  
      infrastructure-devops-lead.md  
      data-analytics-lead.md  
      product-ux-lead.md  
    ciso/  
      security-operations-lead.md  
      compliance-grc-lead.md  
      identity-access-lead.md  
      security-architecture-lead.md  
    vp-sales/  
      sales-operations-lead.md  
      account-management-lead.md  
      business-development-lead.md  
      sales-enablement-lead.md  
    vp-delivery/  
      project-program-manager.md  
      resource-manager.md  
      client-success-lead.md  
      qa-delivery-standards-lead.md  
    cao/  
      hr-people-ops-lead.md  
      legal-contracts-lead.md  
      admin-policy-lead.md  
      corporate-communications-lead.md
```

Each file follows the pattern:

```

---
name: controller
description: "GAAP compliance and financial controls analyst for CFO domain"
model: haiku
tools:
  - Read
  - Grep
  - Glob
  - WebSearch
maxTurns: 5
---

```

Controller – GAAP Compliance and Financial Controls

Your Identity

You are the Controller reporting to the CFO. You own accounting accuracy and financial controls.

Your Analytical Framework

GAAP Compliance and Financial Controls Assessment

Your Output Template

[structured template as above]

Your Forcing Questions

[three forcing questions as above]

Instructions

Analyze the issue presented to you **ONLY** through your specific domain lens. Do not attempt to evaluate the overall business merit of the proposal. Your job is narrow, focused, domain-specific analysis. Produce your findings using the output template above. Be direct and opinionated – flag concerns clearly, don't hedge.

Open Threads

- **Framework Completeness:** This report provides the full CFO example (5 packages). The spec needs all 29 packages designed with equal specificity. This is the largest content-generation task for the builder agent.
- **Framework Versioning:** Can analytical frameworks be updated independently? Could industry-specific variants override defaults?
- **Output Quality Validation:** How do you test whether the cognitive forcing actually prevents voice collapse? What does a quality benchmark look like?
- **Tier 1 Internalization:** In Direct Consult mode, the C-suite agent doesn't spawn subagents — it reasons internally through team lead lenses. How well does this work without the structural forcing of separate subagent contexts?
- **Cross-Domain Forcing:** Should team leads have forcing questions that reference other domains? ("As Controller, what does the Engineering Lead's estimate assume about accounting treatment?")

Recommendation to Arbiter

Strongly recommended as a core component of the spec. The Cognitive Forcing framework is what determines whether the skill produces genuinely useful multi-perspective analysis or just verbose agreement dressed in different terminology. The three forcing question types (pre-mortem, adversarial empathy, domain devil's advocate) are research-backed and directly implementable. The subagent definition file structure gives the builder agent a concrete deliverable list. This report, combined with Report #1 (cascade architecture) and Report #2 (engagement model), completes the three pillars of the spec: structure, experience, and analytical quality.

Idea Report: Decision Modes and the Decision Space Map

Direction explored by: Free Thinker + Grounder

Report number: 04

Related threads: Cascade architecture (Report #1), Engagement Model (Report #2), CEO personality thread, mode/tier matrix exploration

The Idea

The Team of Teams skill should not produce a single recommendation — it should reveal a **decision space**. The same domain analysis (C-suite findings, team lead outputs, fault lines) can be synthesized through multiple **Decision Modes** — configurable lenses that change how the CEO agent weighs and resolves competing perspectives without changing the underlying analysis. By running the same inputs through different synthesis styles, the user sees not "what to do" but "what the terrain looks like in every direction" — and can make an informed choice about which kind of decision-maker they want to be for this particular issue.

The five Decision Modes are: **Guardian** (protect what we have), **Pioneer** (pursue what we could gain), **Architect** (build the widest organizational alignment), **Analyst** (follow the evidence, defer where evidence is weak), and **Sentinel** (minimize regret on irreversible decisions). Each maps to established decision theory frameworks. Each mode is implemented as a swappable prompt modifier in the CEO agent's synthesis layer, leaving the domain analysis pipeline completely unchanged.

The power move is **multi-mode comparison**: the user requests a deliberation that runs domain analysis once (the expensive part), then feeds the same findings through two or more CEO synthesis modes (lightweight passes). The output is a **Comparative Decision Record** — a side-by-side view showing how the same evidence produces different decisions depending on the synthesis lens. This transforms the skill from a decision engine into a **decision exploration tool**.

The Key Insight

The most useful output of a multi-perspective analysis isn't a single answer — it's a map showing **how the answer changes depending on what you optimize for**. Decision Modes make the user's implicit risk appetite and strategic posture explicit and visible, turning a binary "do this / don't do this" into a nuanced "here's what each path looks like."

How We Got Here

The Free Thinker proposed "CEO Personalities" in the opening salvo — the idea that different CEO decision-making styles could produce different outcomes from identical analysis. The Grounder elevated this beyond a "fun variant" by identifying the core insight: it solves a real problem. Users often don't know what kind of decision-maker they want to be on a given issue. Showing the conservative path alongside the aggressive path alongside the balanced path gives them the decision space — what they're actually choosing between. The Free Thinker then developed this into four concrete modes with CEO prompt modifiers, mapped the mode/tier interaction matrix, and designed the multi-mode comparison mechanism.

The Grounder's Take

- **Does this connect to what was asked for?** Yes, with a twist. The concept seed asks for a skill that "makes decisions and answers questions." Decision Modes make the skill's decisions more valuable by showing the user that the "right" answer depends on their strategic posture — and giving them the tools to explore that dependency. It's a direct enhancement to the Decision Record output.
- **Would the audience care?** This is the feature that would make a founder say "this is actually useful, not just impressive." Solo founders and small leadership teams constantly face the question "am I being too cautious or too aggressive?" Decision Modes make that question answerable by showing both paths side by side.
- **Is this one of the good ones?** Yes. It's the feature that differentiates this skill from "ask a panel of AI experts." Panels give you one answer. This gives you the shape of the decision landscape.

The Free Thinker's Vision

At its most ambitious, Decision Modes become **strategic profiles** that accumulate over time. A user who consistently picks the Growth mode synthesis learns something about their own decision-making tendencies. The system could surface patterns: "You've chosen the growth-oriented path in 8 of the last 10 decisions. Your CFO's concerns were overridden 6 times. Consider whether the conservative perspective is being systematically underweighted." That's not just decision support — it's leadership coaching.

Further out: custom Decision Modes. A user could define their own synthesis posture: "I'm generally growth-oriented but conservative on anything involving regulatory risk." The CEO's synthesis prompt becomes a user-configurable profile that represents their actual leadership style.

Architecture Detail: The Four Decision Modes

Mode 1: Guardian (Conservative / Risk-Averse)

Strategic posture: Protect what we have. The downside matters more than the upside.

CEO synthesis behavior:

- Weights skeptic roles (CISO, CFO, COO, VP Delivery) more heavily
- Threshold for accepting risk is high — skeptics must be satisfied, not just acknowledged
- Dissenting views from advocates are preserved but the bar for overriding skeptics is steep
- Decisions tend toward: don't do it, do a smaller version, do it with extensive guardrails

CEO prompt modifier:**DECISION MODE: CONSERVATIVE**

You are cautious by disposition. You'd rather miss an opportunity than take a risk that could damage the business. When skeptic and advocate perspectives conflict, you lean toward the skeptics unless the advocates present overwhelming evidence of low-risk upside. Your decision should reflect what a prudent steward of the business would do, prioritizing stability, predictability, and downside protection.

Mode 2: Pioneer (Growth / Aggressive)

Strategic posture: Pursue what we could gain. Standing still is the biggest risk.

CEO synthesis behavior:

- Weights advocate roles (VP Sales, CTO) more heavily
- Skeptic concerns are treated as engineering problems to solve, not reasons to stop
- The synthesis looks for "how do we capture this opportunity?" not "should we?"
- Decisions tend toward: do it, do it bigger, do it faster

CEO prompt modifier:**DECISION MODE: GROWTH**

You are growth-oriented by disposition. You believe the biggest risk is standing still while competitors move. When skeptic and advocate perspectives conflict, you look for ways to capture the opportunity while mitigating the risks, rather than letting the risks kill the opportunity. Frame skeptic concerns as implementation challenges to solve, not objections to honor. Your decision should reflect what an ambitious, forward-leaning leader would do.

Mode 3: Architect (Consensus / Collaborative)

Strategic posture: Find the path with widest organizational alignment. Decisions succeed on buy-in.

CEO synthesis behavior:

- Weights the fault lines themselves — looks for positions that address the most objections
- The synthesis seeks the option that satisfies the most domain concerns
- Decisions may be less bold but have broader organizational support
- Conditions drawn from multiple domains, not just the most determinative one

CEO prompt modifier:

DECISION MODE: CONSENSUS

You are a consensus-builder by disposition. You believe that decisions succeed or fail based on organizational alignment. When perspectives conflict, you look for the position that satisfies the most domain concerns, even if it means a less aggressive or less cautious path than any single role would recommend. Your decision should reflect what a collaborative leader would do – finding common ground without sacrificing strategic intent.

Mode 4: Analyst (Analytical / Data-Driven)

Strategic posture: Follow the evidence. Defer where evidence is weak.

CEO synthesis behavior:

- Weights confidence levels — high-confidence findings carry more weight regardless of source
- Low-confidence recommendations get flagged as areas needing more research
- The synthesis is driven by "where do we have the best information?" not "what do we want to be true?"
- Decisions may include "defer pending better data on X" as a legitimate outcome

CEO prompt modifier:

DECISION MODE: ANALYTICAL

You are analytically driven. You distrust both optimism and pessimism – you trust evidence. Weight domain recommendations by their confidence levels, not their enthusiasm or caution. Where confidence is low, recommend further research before committing. A decision to defer is not indecision – it's a rational response to insufficient information. The quality of the analysis matters more than the direction of the recommendation.

Mode 5: Sentinel (Regret Minimizer / MiniMax Regret)

Strategic posture: Which decision can I defend regardless of outcome? Minimize the worst possible regret.

CEO synthesis behavior:

- Disproportionately weights the strongest objection from ANY role, regardless of that role's disposition
- The synthesis asks: "If this goes wrong, which C-suite member's warning will I wish I'd heeded?"
- Decisions favor paths where the downside of being wrong is survivable, even if the upside of being right is smaller
- Particularly suited to irreversible decisions (acquisitions, layoffs, market exits)

CEO prompt modifier:

DECISION MODE: SENTINEL

You are a regret minimizer. For every option, ask: "If this decision turns out to be wrong, can we recover?" Disproportionately weight the single strongest objection from any domain – that objection represents your worst case. Choose the path where being wrong is survivable, even if being right is less spectacular. Your decision should reflect what a leader who must answer to a board regardless of outcome would do.

Note: The five modes map to established decision theory (Rowe & Boulgarides Decision Style Theory + classical operations research): Guardian = MaxiMin, Pioneer = MaxiMax, Architect = Behavioral, Analyst = Hurwicz balanced, Sentinel = MiniMax Regret.

The Mode/Tier Interaction Matrix

Decision Modes interact with the three Interaction Tiers (from Report #2) to produce distinct behavioral patterns:

	Tier 1 (Direct Consult)	Tier 2 (Focused Panel)	Tier 3 (Full Deliberation)
Guardian	CFO gives cautious answer, highlights downside risks, suggests what could go wrong	3-4 C-suite, synthesis biased toward risk mitigation. Extensive guardrails.	Full cascade, CEO weights skeptics heavily. High bar for approval.
Pioneer	CFO frames it as investment question, identifies how to fund it, suggests acceleration	Same panel, synthesis biased toward opportunity capture. "How to" not "whether to."	Full cascade, CEO weights advocates heavily. Low bar unless existential risk.
Architect	Single agent response includes "however, [other role] might see this differently — consider escalating"	Panel seeks option addressing most concerns across all activated roles.	Full cascade, CEO seeks widest organizational support. Conditions from all domains.
Analyst	Agent flags confidence level explicitly. Low-confidence answers come with research recommendations.	Synthesis driven by which domains have highest-confidence findings.	Full cascade, CEO weights by evidence quality. Low-confidence = "investigate further."
Sentinel	Agent identifies the single biggest risk and whether it's survivable. "You can do this, but if X goes wrong, here's what happens."	Panel identifies the strongest objection across all activated roles and tests whether the downside is recoverable.	Full cascade, CEO disproportionately weights the strongest single objection. Favors survivable paths.

The default cell for most users, most of the time: **Tier 1 + Analyst**. Quick, evidence-weighted, transparent about uncertainty.

The Multi-Mode Comparison Mechanism

Invocation: /deliberate conservative vs growth: should we acquire CompetitorX?

Process:

1. CEO frames the issue and routes to relevant C-suite members (standard Phase 0-1)
2. Full domain analysis cascade runs once (Phases 2-4) — this is the expensive part
3. CEO receives all domain recommendations, fault lines, and team lead findings
4. CEO runs synthesis **multiple times** — once per requested mode (lightweight, same input)
5. Output: **Comparative Decision Record**

Comparative Decision Record structure:

```
COMPARATIVE DECISION RECORD: [Issue Title]
Decision ID: [auto-generated]
Modes Compared: [list]

EXECUTIVE SUMMARY
[One paragraph per mode showing how the decision differs]

SHARED ANALYSIS
[Domain analyses are identical across modes – presented once]
[Fault Line Analysis – presented once]

MODE COMPARISONS

  CONSERVATIVE SYNTHESIS:
    Decision: [statement]
    Most Determinative Perspective: [role + why]
    Key Factor: [what tipped this mode's decision]
    Conditions: [guardrails]

  GROWTH SYNTHESIS:
    Decision: [statement]
    Most Determinative Perspective: [role + why]
    Key Factor: [what tipped this mode's decision]
    Conditions: [guardrails]

  [repeat for each requested mode]

DIVERGENCE ANALYSIS
Where Modes Agree: [decisions all modes reached]
Where Modes Diverge: [the pivot points – what makes the difference]
The Key Choice: [what the user is actually deciding between – not
  the business question, but the values/priorities question underneath it]

METADATA
[standard metadata]
Mode Sensitivity: [how much does the decision change across modes?
  High sensitivity = the decision depends heavily on risk appetite.
  Low sensitivity = all modes converge on the same answer.]
```

Why Mode Sensitivity matters: If all four modes produce the same decision, that's a strong signal — the right answer doesn't depend on your posture. If the modes diverge dramatically, the user knows this is a decision where their personal risk appetite is the deciding factor, not the analysis.

The Invocation Grammar for Decision Modes

Modes integrate with the invocation grammar from Report #2:

- **Default (no mode specified):** Analyst mode. `/deliberate: should we acquire CompetitorX?`
- **Single mode:** `/deliberate guardian: should we acquire CompetitorX?`
- **Multi-mode comparison:** `/deliberate guardian vs pioneer: should we acquire CompetitorX?`
- **All modes:** `/deliberate all-modes: should we acquire CompetitorX?`
- **Tier 1 with mode:** `Ask the CFO as guardian – can we afford this?`
- **Tier 2 with mode:** `/panel pioneer finance tech: should we build this feature?`

The CEO agent parses the mode specification from the invocation and applies the appropriate synthesis modifier(s).

Cost Efficiency of Multi-Mode Comparison

The key efficiency insight: domain analysis is mode-independent. Running a 4-mode comparison costs approximately:

- 1x full cascade (domain analysis) — the expensive part
- 4x CEO synthesis pass (lightweight, single-agent) — cheap
- Total cost: approximately 1.1x a single full deliberation, producing 4x the strategic insight

This makes multi-mode comparison dramatically more valuable per token than running four separate deliberations.

Open Threads

- **Custom Decision Modes:** Can users define their own synthesis postures beyond the four defaults? "Growth-oriented but conservative on regulatory risk."
- **Mode Recommendation:** Should the CEO auto-suggest which mode is most appropriate for the decision type? "This is a high-stakes irreversible decision — I recommend Conservative or All-Modes."
- **Decision Pattern Analysis:** Over time, can the system surface patterns in which modes the user selects? "You've overridden the CFO in 6 of 8 decisions — consider running the next one in Conservative mode."
- **Mode Calibration:** How do you ensure modes produce meaningfully different outputs rather than slight variations? The prompt modifiers need to be distinct enough to shift synthesis behavior.

Recommendation to Arbiter

Recommended as an important enhancement to the core spec. Decision Modes transform the skill from a decision engine into a decision exploration tool — arguably the more valuable framing. The multi-mode comparison is cost-efficient (1.1x cost for 4x insight) and produces the "decision space map" that no single-mode analysis can offer. The Mode Sensitivity metric in the Comparative Decision Record is a novel signal that tells the user whether their decision depends on strategy or evidence.

This report builds directly on Report #1 (the cascade produces the analysis that modes synthesize differently) and Report #2 (the invocation grammar and tier system integrate with mode selection). Together with Report #3 (cognitive forcing ensures the domain analysis is high-quality), the four reports form a complete design vision: architecture, experience, quality, and strategic depth.

Idea Report: The Engagement Model

Direction explored by: Free Thinker + Grounder

Report number: 2

Related threads: Interaction tiers, CEO decision modes, scenario forking, user invocation patterns

The Idea

The Team of Teams skill needs an **engagement model** — a coherent design for how users interact with the system across different types of decisions. Rather than a single mode of operation (the full five-phase cascade), the skill offers a two-dimensional control surface: **depth** (how many agents engage and how thoroughly) and **width** (how many decision perspectives the CEO synthesizes from).

Depth is controlled through three interaction tiers:

- **Tier 1: The Hallway Question** — Direct consult with a single C-suite agent. Fast, conversational, domain-specific. No CEO involvement, no formal routing. Output: Advisory Note (3-5 sentences).
- **Tier 2: The Working Session** — CEO routes to 2-4 relevant C-suite members. Each does domain analysis with team lead perspectives. CEO produces a lightweight synthesis. Output: Panel Assessment (abbreviated Decision Record, ~1 page).
- **Tier 3: The Board Meeting** — Full five-phase cascade from Report #1. All relevant C-suite members activated, full team lead analysis, full CEO deliberation. Output: Complete Decision Record.

Width is controlled through four CEO Decision Modes:

- **Guardian Mode** (risk-averse) — Weights skeptic roles more heavily. Favors protecting what exists over pursuing what's possible. High threshold for accepting risk.
- **Pioneer Mode** (growth-oriented) — Weights advocate roles more heavily. Treats skeptic concerns as engineering problems to solve. Favors capturing opportunities.
- **Architect Mode** (consensus-building) — Weights the fault lines themselves. Seeks the position that satisfies the most domain concerns. Favors organizational alignment.
- **Analyst Mode** (data-driven) — Weights confidence levels regardless of role disposition. High-confidence findings carry more weight. Flags low-confidence areas for further research.

The default experience is simple: one tier, one mode, one decision. Power users can unlock multi-mode comparison, where the same domain analysis produces different CEO syntheses side by side, revealing how risk appetite shapes the outcome.

The Key Insight

The difference between a quarterly novelty and a daily tool is whether the skill matches the user's engagement level to the decision's weight. Most organizational questions don't need a full board meeting — they need a hallway conversation or a focused working session. By offering three tiers, the

skill becomes a permanent member of the user's workflow, not an event they invoke for major decisions.

How We Got Here

The Free Thinker's "second wave" proposed five directions beyond the core architecture. Interaction tiers (Direction E) was identified by both the Free Thinker and Grounder as the idea that determines adoption — "the one that determines whether anyone actually uses this regularly." The CEO Personalities thread (originally planted during Report #1 development) was recognized by the Grounder as a special case of a broader concept — Decision Modes that parameterize the CEO's synthesis without changing the underlying domain analysis. The Grounder proposed unifying depth (tiers) and width (modes) into a single engagement model.

The Grounder's Take

- **Does this connect to what was asked for?** Yes — the concept seed asks for a skill that "makes decisions and answers questions on issues." The engagement model is how users present those issues and receive those answers. Without it, the spec describes an engine with no steering wheel.
- **Would the audience care?** This is the idea the builder agent needs second, right after the core architecture. You can't build a usable skill without defining how users invoke it and what they get back at each level of engagement.
- **Is this one of the good ones?** Yes. The tiers solve the adoption problem. The decision modes solve the "whose judgment?" problem. Together they make the skill feel like having an executive team on retainer rather than a bureaucratic decision machine.

The Free Thinker's Vision

The most ambitious version: every issue the user brings is automatically triaged to the right tier by the CEO, the user can escalate or de-escalate with a word, and for the biggest decisions, a multi-mode comparison shows the decision space — "here's what a Guardian CEO decides, here's what a Pioneer CEO decides, here's the Architect's compromise, and here's what the Analyst says you need to know before choosing." The user sees their own decision-making style reflected back to them and can choose consciously.

Architecture Details

Invocation Patterns

The skill supports four invocation commands:

- `/consult [role]: [question]` — Tier 1. Direct consult with a named C-suite role.
- `/panel [roles]: [issue]` — Tier 2. User-selected panel of C-suite roles.
- `/deliberate: [issue]` — Tier 3. Full cascade with CEO routing.
- `/evaluate: [issue]` — Unclassified. CEO triages and recommends a tier. User can override.

Optional flags:

- `--mode [guardian|pioneer|architect|analyst]` — Override default decision mode.
- `--compare-modes` — Run Tier 2/3 analysis through all four modes and produce comparative output.

Tier Selection Logic

When the CEO triages an unclassified issue:

1. Assess scope (single-domain vs. multi-domain vs. cross-cutting)
2. Assess impact (low/medium/high/critical)
3. Assess reversibility (easily reversed vs. difficult to reverse vs. irreversible)
4. Recommend tier with one-sentence justification visible to user
5. User can accept, escalate, or de-escalate

Output Formats by Tier

Tier 1 — Advisory Note:

- Domain recommendation (1-2 sentences)
- Key consideration (1 sentence)
- Suggested follow-up if warranted (which other roles might want to weigh in)
- Total length: 3-5 sentences

Tier 2 — Panel Assessment:

- Executive Summary (3-5 sentences)
- Domain analyses from activated roles only (1 paragraph each)
- Top 2-3 fault lines
- Recommendation with conditions
- Total length: ~1 page

Tier 3 — Full Decision Record:

- Complete format as specified in Report #1
- Total length: 3-5 pages

Decision Mode Implementation

- Decision Modes are implemented as **CEO synthesis prompt modifiers** — injected text that changes how the CEO weighs domain recommendations
- The underlying domain analysis is identical across modes
- Multi-mode comparison runs the CEO synthesis phase multiple times with different modifiers, producing a **Decision Space Map** that shows how each mode shapes the outcome
- Default mode: Analyst (most neutral). Configurable in company profile.

Open Threads

- **Company Profile integration:** Decision Mode default and tier escalation thresholds could be configurable per company profile.

- **Tier escalation mid-analysis:** Can a Tier 1 consult escalate to Tier 2 mid-conversation if the C-suite agent realizes the question is bigger than it seemed?
- **Mode blending:** Could users create custom modes that blend dispositions? (e.g., "growth-oriented but weight security concerns heavily") — probably v2.
- **Conversation history in Tier 1:** Does the C-suite agent in direct consult mode remember previous hallway questions? (Connects to institutional memory thread.)

Recommendation to Arbiter

Strongly recommended as the second core idea for the spec. The cascade architecture (Report #1) is the engine. The engagement model is the interface. Without it, the skill is a powerful machine with no practical way for users to interact with it at the right level of depth for each decision. The three tiers ensure daily adoption. The four decision modes ensure the output is genuinely useful for decision-makers with different styles. Flag as "interesting."

Session Summary

Session Summary: Team of Teams Agent Skill

Date: 2026-02-22

Concept Seed: Build a Claude Code Agent Skill that emulates the top two layers of a SMB org structure as a "Team of Teams" for multi-perspective decision-making.

Session Type: Multi-agent ideation (Free Thinker + Grounder dialogue, Explorer research, Writer documentation, Arbiter evaluation)

Outcome: Four INTERESTING idea reports forming a complete specification skeleton.

What Was Explored

The session explored how to design a Claude Code Agent Skill that realistically emulates a SMB's organizational decision-making process. The user's goal is not to build the skill directly, but to produce a comprehensive technical specification that a separate builder agent can use to construct it.

The session proceeded in two waves:

Wave 1 established the core architecture through five rounds of Free Thinker / Grounder dialogue. The Free Thinker proposed five initial directions (institutional memory, engineered dissent, cascading delegation, company parameterization, decision records). The Grounder triaged these into three load-bearing ideas (cascading delegation as the heart, engineered dissent as the differentiator, decision records as the companion), parked institutional memory as a scope bomb, and dismissed parameterization as table stakes. Over four rounds, the pair developed a five-phase cascade process, a full 37-role organizational roster with mandated dissent dispositions, a routing mechanism, and a 9-section Decision Record format. This converged into Report #1.

Wave 2 opened new exploration after Report #1 was confirmed INTERESTING. The Free Thinker proposed five new directions (scenario forking, organizational stress test, cognitive forcing/prompt architecture, decision revisitation, interaction tiers). The Grounder restated three actionable threads (prompt architecture, CEO decision modes, adaptive complexity). Explorer research on McChrystal's Team of Teams methodology, multi-agent orchestration patterns, Claude Agent Teams constraints, and SMB decision-making provided critical context -- most importantly, the discovery that Agent Teams cannot nest, requiring a hybrid Agent Teams + subagents architecture. This led to Reports #2 and #3.

What Was Produced

Three Idea Reports (All INTERESTING)

#	Title	Pillar	Key Innovation
1	The Cascading Deliberation Engine	Structure	Five-phase cascade with engineered dissent across 8 C-suite agents + 29 team lead subagents
2	The Engagement Model	Experience	Two-dimensional control surface: depth (3 tiers) x width (4 CEO decision modes)
3	Cognitive Forcing and Prompt Architecture	Quality	Analytical framework packages with forcing questions prevent voice collapse across 29 perspectives
4	Decision Modes and the Decision Space Map	Strategic Depth	Five CEO synthesis modes with Comparative Decision Record, Mode Sensitivity metric, and multi-mode comparison

Four Explorer Research Reports

Topic	Key Finding
Team of Teams Methodology	McChrystal's four pillars (Trust -> Common Purpose -> Shared Consciousness -> Empowered Execution) map directly to skill phases
Multi-Agent Orchestration	No existing framework models multi-level org structure for business decisions -- this would be novel
Claude Agent Teams	Critical constraint: no nested teams. Subagents offer per-agent customization. Hybrid architecture required.
SMB Decision-Making	Mid-market IT services profile. Cross-functional tensions ARE the engine. Shared consciousness is direct solution to #1 pain point (information silos).

Supporting Documentation

- Ideation graph (living document, updated after every exchange)
- Five version snapshots capturing key turning points
- Four idea briefs summarizing each INTERESTING report
- This session summary
- Vision document (produced alongside this summary)

Key Decisions Made During the Session

1. **Hybrid Agent Architecture:** 8 Agent Team members (C-suite) + 29 custom subagents (team leads) in `.claude/agents/`. Evolved through three stages: 37 separate agents (impractical) -> two-tier simulated passes (compromise) -> hybrid model (best of both).

2. **Engineered Dissent Model:** 4 skeptics (CISO, CFO, COO, VP Delivery), 2 advocates (CTO, VP Sales), 1 systemic (CAO), 1 synthesizer (CEO). Dissent is structural (different lenses produce different conclusions) and mandated (explicit dispositions baked into prompts).
3. **Routing as Core Feature:** CEO's issue routing is an analytical act, not overhead. Selective activation makes framing meaningful and controls cost.
4. **Cognitive Forcing over Persona Descriptions:** Each team lead defined by analytical framework + output template + forcing questions, not persona description. Research-backed: simple role assignments show zero measurable improvement.
5. **Three Interaction Tiers:** Hallway Question (seconds), Working Session (minutes), Board Meeting (full cascade). Determines adoption -- matches engagement to decision weight.
6. **Five CEO Decision Modes:** Guardian, Pioneer, Architect, Analyst (default), Sentinel. Each mapped to decision theory (MaxiMin, MaxiMax, Behavioral, Hurwicz, MiniMax Regret). Implemented as synthesis prompt modifiers. Same analysis, different weighting, different decisions. Multi-mode comparison at 1.1x cost produces Comparative Decision Record with Mode Sensitivity metric.
7. **Phase 4.5 Pre-Mortem:** One round of cross-domain critique after C-suite synthesis, before CEO deliberation. Research-backed as highest single-technique value-add. Optional for Tier 2, recommended for Tier 3.
8. **Heterogeneous Model Tiering:** Haiku for team lead subagents, Sonnet for C-suite, Opus for CEO. Model diversity improves analytical variety beyond cost savings.

What Was Not Explored

Three of the Free Thinker's second-wave directions remain undeveloped:

- **Scenario Forking:** Parallel cascades with different parameters producing comparative decision landscapes. Partially addressed by multi-mode comparison but the full concept (different issue parameters, not just different synthesis styles) is unexplored.
- **Organizational Stress Test:** Inverting the cascade for bottom-up threat/opportunity discovery. "The difference between going to the doctor with a complaint vs. getting a full physical."
- **Decision Revisitation:** Revisiting decisions when assumptions change. Lightweight re-analysis triggered by changing conditions. Connected to parked institutional memory thread.

These represent directions where the skill evolves from a decision-making tool into something broader -- a strategic exploration platform and organizational health monitor.

Threads Parked for Future Versions

- **Institutional Memory:** Persistent knowledge across sessions. Scope bomb, deferred.
- **Cross-Functional C-Suite Deliberation:** Full Phase 4.5 debate (beyond one-round pre-mortem). Risk of "problem drift" with multiple rounds.
- **Custom Mode Blending:** User-defined synthesis styles combining disposition weights.

- **Framework Versioning:** Community-contributed or industry-specific analytical framework packages.
- **Tier 1 Conversation History:** Whether C-suite agents remember previous hallway questions.

Session Statistics

- **Dialogue rounds:** 6+ exchanges between Free Thinker and Grounder
- **Research reports:** 4 (Explorer)
- **Idea reports submitted:** 4 (all confirmed INTERESTING)
- **Total roles designed:** 37 (1 CEO + 7 C-suite + 29 team leads)
- **Reference implementations produced:** 1 (CFO domain, 5 team lead packages)
- **Meta-insights captured:** 27
- **Snapshots taken:** 5